

THE EXPLICIT DEFINITION OF $\mathcal{W}$

with ε ignored

An appendix to *Primes in arithmetic progressions to large moduli and refinements of Harman's sieve*

RUNBO LI

$\mathcal{W} = \{(\theta_1, \theta_2, \theta_3) :$

- $(\theta_1 = 7/17 \text{ and } \theta_2 = 3/34 \text{ and } 0 < \theta_3 < 1/170),$
- or** $(\theta_1 = 7/17 \text{ and } 14/153 < \theta_2 < 19/204 \text{ and } 0 < \theta_3 < 1/170(19 - 204\theta_2)),$
- or** $(\theta_1 = 7/17 \text{ and } 3/34 < \theta_2 \leq 397/4403 \text{ and } 0 < \theta_3 < 1/170(19 - 204\theta_2)),$
- or** $(\theta_1 = 7/17 \text{ and } 397/4403 < \theta_2 \leq 185/2023 \text{ and } 0 < \theta_3 < 1/170(19 - 204\theta_2)),$
- or** $(\theta_1 = 7/17 \text{ and } 185/2023 < \theta_2 \leq 14/153 \text{ and } 0 < \theta_3 < 1/170(19 - 204\theta_2)),$
- or** $(\theta_1 = 7/17 \text{ and } 11/136 < \theta_2 < 3/34 \text{ and } 1/17(3 - 34\theta_2) \leq \theta_3 < 1/170(19 - 204\theta_2)),$
- or** $(\theta_1 = 107/280 \text{ and } \theta_2 = 19/280 \text{ and } 1/20 < \theta_3 < 143/2800),$
- or** $(25/58 < \theta_1 < 61/140 \text{ and } \theta_2 = 1/19(4 - 7\theta_1) \text{ and } 1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/4(1 - 2\theta_1)),$
- or** $(61/140 < \theta_1 < 43/98 \text{ and } \theta_2 = 1/19(4 - 7\theta_1) \text{ and } 1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/4(1 - 2\theta_1)),$
- or** $(43/98 < \theta_1 < 15/34 \text{ and } \theta_2 = 1/19(4 - 7\theta_1) \text{ and } 1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/4(1 - 2\theta_1)),$
- or** $(15/34 < \theta_1 < 4/9 \text{ and } \theta_2 = 1/19(4 - 7\theta_1) \text{ and } 1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/4(1 - 2\theta_1)),$
- or** $(\theta_1 = 11/24 \text{ and } 13/288 < \theta_2 < 11/240 \text{ and } 1/288(19 - 288\theta_2) \leq \theta_3 < 1/48),$
- or** $(37/90 < \theta_1 < 7/17 \text{ and } \theta_2 = 1/2(1 - 2\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
- or** $(\theta_1 = 13/32 \text{ and } \theta_2 = 3/32 \text{ and } 0 < \theta_3 < 1/320),$
- or** $(\theta_1 = 139/340 \text{ and } \theta_2 = 31/340 \text{ and } 0 < \theta_3 < 3/680),$
- or** $(\theta_1 = 37/90 \text{ and } \theta_2 = 4/45 \text{ and } 0 < \theta_3 < 1/180),$
- or** $(7/17 < \theta_1 \leq 12/29 \text{ and } \theta_2 = 1/2(1 - 2\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
- or** $(\theta_1 = 127/308 \text{ and } \theta_2 = 1/11 \text{ and } 0 < \theta_3 < 1/440),$
- or** $(\theta_1 = 5/12 \text{ and } \theta_2 = 1/12 \text{ and } 0 < \theta_3 < 1/120),$
- or** $(\theta_1 = 11/26 \text{ and } \theta_2 = 1/13 \text{ and } 0 < \theta_3 < 3/260),$
- or** $(\theta_1 = 61/140 \text{ and } \theta_2 = 1221/16660 \text{ and } 0 < \theta_3 < 47/6664),$
- or** $(\theta_1 = 15/34 \text{ and } \theta_2 = 9/119 \text{ and } 0 < \theta_3 < 1/2380),$
- or** $(\theta_1 = 43/98 \text{ and } \theta_2 = 435/5831 \text{ and } 0 < \theta_3 < 389/116620),$
- or** $(\theta_1 = 127/306 \text{ and } \theta_2 = 13/153 \text{ and } 0 < \theta_3 < 23/3060),$
- or** $(4/9 < \theta_1 < 29/65 \text{ and } \theta_2 = 1/19(4 - 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
- or** $(\theta_1 = 61/140 \text{ and } \theta_2 = 1/20 \text{ and } 1/35 \leq \theta_3 < 9/280),$
- or** $(\theta_1 = 15/34 \text{ and } \theta_2 = 31/646 \text{ and } 7/323 \leq \theta_3 < 1/34),$
- or** $(\theta_1 = 43/98 \text{ and } \theta_2 = 13/266 \text{ and } 23/931 \leq \theta_3 < 3/98),$
- or** $(\theta_1 = 4/9 \text{ and } \theta_2 = 8/171 \text{ and } 1/57 \leq \theta_3 < 1/36),$
- or** $(\theta_1 = 11/24 \text{ and } 11/240 \leq \theta_2 < 7/144 \text{ and } 1/288(19 - 288\theta_2) \leq \theta_3 < 1/48),$
- or** $(\theta_1 = 11/24 \text{ and } 7/144 \leq \theta_2 < 3/56 \text{ and } 1/24(1 - 12\theta_2) \leq \theta_3 < 1/240(19 - 288\theta_2)),$
- or** $(\theta_1 = 13/28 \text{ and } 5/112 < \theta_2 < 1/21 \text{ and } 1/16(1 - 16\theta_2) \leq \theta_3 < 1/56),$
- or** $(25/63 \leq \theta_1 < 2/5 \text{ and } \theta_2 = 1/112(29 - 56\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
- or** $(17/44 < \theta_1 < 7/18 \text{ and } \theta_2 = 1/112(29 - 56\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
- or** $(71/180 < \theta_1 \leq 36/91 \text{ and } \theta_2 = 1/112(29 - 56\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
- or** $(36/91 < \theta_1 < 25/63 \text{ and } \theta_2 = 1/112(29 - 56\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
- or** $(7/18 < \theta_1 \leq 39/100 \text{ and } \theta_2 = 1/112(29 - 56\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$

or $(39/100 < \theta_1 < 71/180$ and $\theta_2 = 1/112(29 - 56\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(\theta_1 = 2/5$ and $\theta_2 = 33/560$ and $23/560 < \theta_3 < 69/1400)$,
 or $(19/50 < \theta_1 \leq 807/2120$ and $\theta_2 = 1/112(29 - 56\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(807/2120 < \theta_1 < 55/144$ and $\theta_2 = 1/112(29 - 56\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(55/144 \leq \theta_1 < 107/280$ and $\theta_2 = 1/112(29 - 56\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(85/224 < \theta_1 < 19/50$ and $\theta_2 = 1/112(29 - 56\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(107/280 < \theta_1 < 17/44$ and $\theta_2 = 1/112(29 - 56\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(\theta_1 = 17/44$ and $\theta_2 = 81/1232$ and $59/1232 < \theta_3 < 39/770)$,
 or $(\theta_1 = 71/180$ and $\theta_2 = 311/5040$ and $221/5040 < \theta_3 < 157/3150)$,
 or $(\theta_1 = 19/50$ and $\theta_2 = 193/2800$ and $143/2800 < \theta_3 < 359/7000)$,
 or $(\theta_1 = 7/18$ and $\theta_2 = 65/1008$ and $47/1008 < \theta_3 < 127/2520)$,
 or $(25/63 \leq \theta_1 < 2/5$ and $1/4(1 - 2\theta_1) < \theta_2 < 1/20(9 - 20\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/4(1 - 2\theta_1)$),
 or $(25/63 \leq \theta_1 < 2/5$ and $\theta_2 = 1/20(9 - 20\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/4(1 - 2\theta_1)$),
 or $(\theta_1 = 2/5$ and $\theta_2 = 1/18$ and $2/45 < \theta_3 < 1/20)$,
 or $(7/18 < \theta_1 < 71/180$ and $1/4(1 - 2\theta_1) < \theta_2 < 1/18$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/4(1 - 2\theta_1)$),
 or $(\theta_1 = 71/180$ and $\theta_2 = 1/18$ and $1/20 < \theta_3 < 19/360)$,
 or $(71/180 < \theta_1 \leq 36/91$ and $1/4(1 - 2\theta_1) < \theta_2 < 1/20(9 - 20\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/4(1 - 2\theta_1)$),
 or $(36/91 < \theta_1 < 25/63$ and $1/4(1 - 2\theta_1) < \theta_2 < 1/20(9 - 20\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/4(1 - 2\theta_1)$),
 or $(71/180 < \theta_1 \leq 36/91$ and $\theta_2 = 1/20(9 - 20\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/4(1 - 2\theta_1)$),
 or $(36/91 < \theta_1 < 25/63$ and $\theta_2 = 1/20(9 - 20\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/4(1 - 2\theta_1)$),
 or $(39/100 \leq \theta_1 < 71/180$ and $\theta_2 = 1/20(9 - 20\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/4(1 - 2\theta_1)$),
 or $(3/8 < \theta_1 \leq 203/540$ and $1/4(1 - 2\theta_1) < \theta_2 \leq 1/24(3 - 4\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/4(1 - 2\theta_1)$),
 or $(203/540 < \theta_1 < 677/1800$ and $1/4(1 - 2\theta_1) < \theta_2 \leq 1/24(3 - 4\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/4(1 - 2\theta_1)$),
 or $(677/1800 < \theta_1 < 61/162$ and $1/4(1 - 2\theta_1) < \theta_2 \leq 1/24(3 - 4\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/4(1 - 2\theta_1)$),
 or $(61/162 \leq \theta_1 < 91/240$ and $1/4(1 - 2\theta_1) < \theta_2 \leq 1/24(3 - 4\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/4(1 - 2\theta_1)$),
 or $(91/240 < \theta_1 < 19/50$ and $1/4(1 - 2\theta_1) < \theta_2 \leq 1/24(3 - 4\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/4(1 - 2\theta_1)$),
 or $(19/50 < \theta_1 \leq 807/2120$ and $1/4(1 - 2\theta_1) < \theta_2 \leq 1/24(3 - 4\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/4(1 - 2\theta_1)$),
 or $(807/2120 < \theta_1 < 55/144$ and $1/4(1 - 2\theta_1) < \theta_2 \leq 1/24(3 - 4\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/4(1 - 2\theta_1)$),
 or $(55/144 \leq \theta_1 < 23/60$ and $1/4(1 - 2\theta_1) < \theta_2 \leq 1/24(3 - 4\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/4(1 - 2\theta_1)$),
 or $(23/60 \leq \theta_1 < 17/44$ and $1/4(1 - 2\theta_1) < \theta_2 < 1/9(4 - 9\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/4(1 - 2\theta_1)$),
 or $(23/60 \leq \theta_1 < 17/44$ and $\theta_2 = 1/9(4 - 9\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/4(1 - 2\theta_1)$),
 or $(17/44 < \theta_1 < 7/18$ and $1/4(1 - 2\theta_1) < \theta_2 < 1/9(4 - 9\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/4(1 - 2\theta_1)$),
 or $(17/44 < \theta_1 < 7/18$ and $\theta_2 = 1/9(4 - 9\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/4(1 - 2\theta_1)$),
 or $(107/280 < \theta_1 < 17/44$ and $\theta_2 = 1/20(9 - 20\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(17/44 < \theta_1 < 7/18$ and $\theta_2 = 1/20(9 - 20\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(7/18 < \theta_1 < 39/100$ and $\theta_2 = 1/20(9 - 20\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(17/45 < \theta_1 < 91/240$ and $\theta_2 = 1/9(4 - 9\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(91/240 < \theta_1 < 19/50$ and $\theta_2 = 1/9(4 - 9\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(19/50 < \theta_1 \leq 807/2120$ and $\theta_2 = 1/9(4 - 9\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(807/2120 < \theta_1 < 55/144$ and $\theta_2 = 1/9(4 - 9\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(55/144 \leq \theta_1 < 23/60$ and $\theta_2 = 1/9(4 - 9\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(\theta_1 = 17/44$ and $\theta_2 = 23/396$ and $1/18 < \theta_3 < 5/88)$,
 or $(\theta_1 = 17/44$ and $\theta_2 = 7/110$ and $1/20 < \theta_3 < 117/2200)$,
 or $(\theta_1 = 91/240$ and $\theta_2 = 47/720$ and $1/18 < \theta_3 < 9/160)$,
 or $(\theta_1 = 19/50$ and $\theta_2 = 29/450$ and $1/18 < \theta_3 < 17/300)$,
 or $(\theta_1 = 7/18$ and $\theta_2 = 11/180$ and $1/20 < \theta_3 < 49/900)$,
 or $(25/63 \leq \theta_1 < 2/5$ and $\theta_2 = 1/19(4 - 7\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(17/44 < \theta_1 < 7/18$ and $\theta_2 = 1/19(4 - 7\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(71/180 < \theta_1 \leq 36/91$ and $\theta_2 = 1/19(4 - 7\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),

or $(36/91 < \theta_1 < 25/63$ and $\theta_2 = 1/19(4 - 7\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(7/18 < \theta_1 < 71/180$ and $\theta_2 = 1/19(4 - 7\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(25/63 \leq \theta_1 < 2/5$ and $\theta_2 = 1/28(5 - 8\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(25/63 \leq \theta_1 < 2/5$ and $\theta_2 = 1/23(6 - 11\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(25/63 \leq \theta_1 < 2/5$ and $\theta_2 = 1/35(11 - 21\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(\theta_1 = 2/5$ and $\theta_2 = 6/95$ and $7/190 < \theta_3 < 21/475)$,
 or $(\theta_1 = 2/5$ and $\theta_2 = 9/140$ and $1/28 < \theta_3 < 3/70)$,
 or $(\theta_1 = 2/5$ and $\theta_2 = 8/115$ and $7/230 < \theta_3 < 21/575)$,
 or $(\theta_1 = 2/5$ and $\theta_2 = 13/175$ and $9/350 < \theta_3 < 27/875)$,
 or $(19/50 < \theta_1 \leq 807/2120$ and $\theta_2 = 1/20(9 - 20\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(807/2120 < \theta_1 < 55/144$ and $\theta_2 = 1/20(9 - 20\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(55/144 \leq \theta_1 < 107/280$ and $\theta_2 = 1/20(9 - 20\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(17/44 < \theta_1 < 7/18$ and $\theta_2 = 1/28(5 - 8\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(71/180 < \theta_1 \leq 36/91$ and $\theta_2 = 1/28(5 - 8\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(36/91 < \theta_1 < 25/63$ and $\theta_2 = 1/28(5 - 8\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(7/18 < \theta_1 < 71/180$ and $\theta_2 = 1/28(5 - 8\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(5/13 < \theta_1 < 17/44$ and $\theta_2 = 1/23(6 - 11\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(17/44 < \theta_1 < 7/18$ and $\theta_2 = 1/23(6 - 11\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(71/180 < \theta_1 \leq 36/91$ and $\theta_2 = 1/23(6 - 11\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(36/91 < \theta_1 < 25/63$ and $\theta_2 = 1/23(6 - 11\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(7/18 < \theta_1 < 71/180$ and $\theta_2 = 1/23(6 - 11\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(36/91 < \theta_1 < 25/63$ and $\theta_2 = 1/35(11 - 21\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(19/50 < \theta_1 \leq 807/2120$ and $\theta_2 = 1/140(-13 + 60\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(807/2120 < \theta_1 < 55/144$ and $\theta_2 = 1/140(-13 + 60\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(719/1820 < \theta_1 \leq 36/91$ and $\theta_2 = 1/91(43 - 91\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(\theta_1 = 17/44$ and $\theta_2 = 3/44$ and $1/22 < \theta_3 < 21/440)$,
 or $(\theta_1 = 17/44$ and $\theta_2 = 7/92$ and $19/506 < \theta_3 < 387/10120)$,
 or $(\theta_1 = 71/180$ and $\theta_2 = 223/3420$ and $23/570 < \theta_3 < 1561/34200)$,
 or $(\theta_1 = 71/180$ and $\theta_2 = 83/1260$ and $5/126 < \theta_3 < 113/2520)$,
 or $(\theta_1 = 71/180$ and $\theta_2 = 13/180$ and $1/30 < \theta_3 < 67/1800)$,
 or $(\theta_1 = 7/18$ and $\theta_2 = 23/342$ and $5/114 < \theta_3 < 161/3420)$,
 or $(\theta_1 = 7/18$ and $\theta_2 = 17/252$ and $11/252 < \theta_3 < 59/1260)$,
 or $(\theta_1 = 7/18$ and $\theta_2 = 31/414$ and $5/138 < \theta_3 < 157/4140)$,
 or $(71/180 < \theta_1 \leq 719/1820$ and $\theta_2 = 1/91(43 - 91\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(177/455 < \theta_1 < 71/180$ and $\theta_2 = 1/91(43 - 91\theta_1)$ and $1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(\theta_1 = 71/180$ and $\theta_2 = 1279/16380$ and $5/182 < \theta_3 < 989/32760)$,
 or $(2/5 < \theta_1 \leq 272/679$ and $\theta_2 = 1/112(29 - 56\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(\theta_1 = 7/17$ and $\theta_2 = 101/1904$ and $229/5712 \leq \theta_3 < 3/68)$,
 or $(\theta_1 = 5/12$ and $\theta_2 = 17/336$ and $5/126 \leq \theta_3 < 1/24)$,
 or $(\theta_1 = 139/340$ and $\theta_2 = 519/9520$ and $24/595 \leq \theta_3 < 31/680)$,
 or $(\theta_1 = 37/90$ and $\theta_2 = 269/5040$ and $607/15120 \leq \theta_3 < 2/45)$,
 or $(\theta_1 = 127/306$ and $\theta_2 = 881/17136$ and $2047/51408 \leq \theta_3 < 13/306)$,
 or $(\theta_1 = 205/504$ and $\theta_2 = 1/18$ and $35/864 \leq \theta_3 < 47/1008)$,
 or $(\theta_1 = 117/280$ and $\theta_2 = 1/20$ and $19/480 \leq \theta_3 < 23/560)$,
 or $(\theta_1 = 143/340$ and $\theta_2 = 463/9520$ and $281/7140 \leq \theta_3 < 27/680)$,
 or $(\theta_1 = 13/32$ and $\theta_2 = 25/448$ and $109/2688 \leq \theta_3 < 3/64)$,
 or $(\theta_1 = 53/126$ and $\theta_2 = 7/144$ and $17/432 \leq \theta_3 < 5/126)$,
 or $(2/5 < \theta_1 \leq 272/679$ and $\theta_2 = 1/19(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(2/5 < \theta_1 \leq 272/679$ and $\theta_2 = 1/28(5 - 8\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),

or $(2/5 < \theta_1 \leq 272/679 \text{ and } \theta_2 = 1/23(6 - 11\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(2/5 < \theta_1 \leq 272/679 \text{ and } \theta_2 = 1/35(11 - 21\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(\theta_1 = 31/70 \text{ and } \theta_2 = 9/190 \text{ and } 21/760 \leq \theta_3 < 1/35)$,
 or $(\theta_1 = 31/70 \text{ and } \theta_2 = 79/1610 \text{ and } 167/6440 \leq \theta_3 < 1/35)$,
 or $(\theta_1 = 31/70 \text{ and } \theta_2 = 17/350 \text{ and } 37/1400 \leq \theta_3 < 1/35)$,
 or $(61/140 < \theta_1 < 43/98 \text{ and } 1/8(3 - 6\theta_1) < \theta_2 < 1/19(4 - 7\theta_1) \text{ and } 1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(43/98 < \theta_1 < 15/34 \text{ and } 1/8(3 - 6\theta_1) < \theta_2 < 1/19(4 - 7\theta_1) \text{ and } 1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(15/34 < \theta_1 < 4/9 \text{ and } 1/8(3 - 6\theta_1) < \theta_2 < 1/19(4 - 7\theta_1) \text{ and } 1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(37/90 < \theta_1 < 7/17 \text{ and } 1/8(6 - 13\theta_1) < \theta_2 < 1/2(1 - 2\theta_1) \text{ and } 1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(7/17 < \theta_1 < 12/29 \text{ and } 1/8(6 - 13\theta_1) < \theta_2 < 1/2(1 - 2\theta_1) \text{ and } 1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(\theta_1 = 7/17 \text{ and } \theta_2 = 1/20 \text{ and } 11/255 \leq \theta_3 < 3/68)$,
 or $(\theta_1 = 5/12 \text{ and } \theta_2 = 1/20 \text{ and } 29/720 \leq \theta_3 < 1/24)$,
 or $(2/5 < \theta_1 \leq 272/679 \text{ and } \theta_2 = 1/18 \text{ and } 1/12(10/3 - 7\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(\theta_1 = 13/32 \text{ and } \theta_2 = 1/18 \text{ and } 47/1152 \leq \theta_3 < 3/64)$,
 or $(2/5 < \theta_1 \leq 272/679 \text{ and } \theta_2 = 1/20 \text{ and } 1/12(17/5 - 7\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(\theta_1 = 13/32 \text{ and } \theta_2 = 1/20 \text{ and } 89/1920 \leq \theta_3 < 3/64)$,
 or $(\theta_1 = 139/340 \text{ and } \theta_2 = 1/20 \text{ and } 61/1360 \leq \theta_3 < 31/680)$,
 or $(\theta_1 = 37/90 \text{ and } \theta_2 = 1/20 \text{ and } 47/1080 \leq \theta_3 < 2/45)$,
 or $(\theta_1 = 127/306 \text{ and } \theta_2 = 1/20 \text{ and } 757/18360 \leq \theta_3 < 13/306)$,
 or $(\theta_1 = 7/17 \text{ and } \theta_2 = 1/18 \text{ and } 23/612 \leq \theta_3 < 3/68)$,
 or $(\theta_1 = 7/17 \text{ and } \theta_2 = 1/17 \text{ and } 7/204 \leq \theta_3 < 7/170)$,
 or $(\theta_1 = 7/17 \text{ and } \theta_2 = 15/238 \text{ and } 43/1428 \leq \theta_3 < 43/1190)$,
 or $(\theta_1 = 7/17 \text{ and } \theta_2 = 25/391 \text{ and } 137/4692 \leq \theta_3 < 137/3910)$,
 or $(\theta_1 = 7/17 \text{ and } \theta_2 = 8/119 \text{ and } 37/1428 \leq \theta_3 < 37/1190)$,
 or $(\theta_1 = 37/84 \text{ and } \theta_2 = 1/20 \text{ and } 19/720 \leq \theta_3 < 5/168)$,
 or $(\theta_1 = 5/12 \text{ and } \theta_2 = 1/18 \text{ and } 5/144 \leq \theta_3 < 1/24)$,
 or $(\theta_1 = 5/12 \text{ and } \theta_2 = 11/204 \text{ and } 89/2448 \leq \theta_3 < 1/24)$,
 or $(\theta_1 = 11/26 \text{ and } \theta_2 = 1/18 \text{ and } 29/936 \leq \theta_3 < 29/780)$,
 or $(\theta_1 = 61/140 \text{ and } \theta_2 = 1/18 \text{ and } 17/720 \leq \theta_3 < 17/600)$,
 or $(\theta_1 = 43/98 \text{ and } \theta_2 = 1/18 \text{ and } 11/504 \leq \theta_3 < 11/420)$,
 or $(\theta_1 = 37/90 \text{ and } \theta_2 = 1/18 \text{ and } 41/1080 \leq \theta_3 < 2/45)$,
 or $(\theta_1 = 139/340 \text{ and } \theta_2 = 1/18 \text{ and } 481/12240 \leq \theta_3 < 31/680)$,
 or $(\theta_1 = 127/306 \text{ and } \theta_2 = 1/18 \text{ and } 131/3672 \leq \theta_3 < 13/306)$,
 or $(\theta_1 = 143/340 \text{ and } \theta_2 = 1/18 \text{ and } 397/12240 \leq \theta_3 < 397/10200)$,
 or $(\theta_1 = 53/126 \text{ and } \theta_2 = 1/18 \text{ and } 7/216 \leq \theta_3 < 7/180)$,
 or $(\theta_1 = 17/40 \text{ and } \theta_2 = 1/18 \text{ and } 43/1440 \leq \theta_3 < 43/1200)$,
 or $(\theta_1 = 85/198 \text{ and } \theta_2 = 1/18 \text{ and } 65/2376 \leq \theta_3 < 13/396)$,
 or $(\theta_1 = 163/378 \text{ and } \theta_2 = 1/18 \text{ and } 17/648 \leq \theta_3 < 17/540)$,
 or $(\theta_1 = 5/12 \text{ and } \theta_2 = 9/140 \text{ and } 131/5040 \leq \theta_3 < 131/4200)$,
 or $(\theta_1 = 5/12 \text{ and } \theta_2 = 31/476 \text{ and } 431/17136 \leq \theta_3 < 431/14280)$,
 or $(\theta_1 = 5/12 \text{ and } \theta_2 = 13/228 \text{ and } 91/2736 \leq \theta_3 < 91/2280)$,
 or $(\theta_1 = 5/12 \text{ and } \theta_2 = 17/276 \text{ and } 95/3312 \leq \theta_3 < 19/552)$,
 or $(\theta_1 = 11/26 \text{ and } \theta_2 = 1/20 \text{ and } 19/520 \leq \theta_3 < 1/26)$,
 or $(\theta_1 = 11/26 \text{ and } \theta_2 = 2/39 \text{ and } 11/312 \leq \theta_3 < 1/26)$,
 or $(\theta_1 = 11/26 \text{ and } \theta_2 = 27/494 \text{ and } 63/1976 \leq \theta_3 < 189/4940)$,
 or $(\theta_1 = 11/26 \text{ and } \theta_2 = 35/598 \text{ and } 67/2392 \leq \theta_3 < 201/5980)$,
 or $(\theta_1 = 11/26 \text{ and } \theta_2 = 11/182 \text{ and } 19/728 \leq \theta_3 < 57/1820)$,
 or $(\theta_1 = 61/140 \text{ and } \theta_2 = 1/20 \text{ and } 7/240 \leq \theta_3 < 9/280)$,
 or $(\theta_1 = 61/140 \text{ and } \theta_2 = 169/3220 \text{ and } 1031/38640 \leq \theta_3 < 1031/32200)$,

or $(\theta_1 = 61/140 \text{ and } \theta_2 = 37/700 \text{ and } 221/8400 \leq \theta_3 < 221/7000)$,
 or $(\theta_1 = 15/34 \text{ and } \theta_2 = 1/20 \text{ and } 53/2040 \leq \theta_3 < 1/34)$,
 or $(\theta_1 = 15/34 \text{ and } \theta_2 = 31/646 \text{ and } 217/7752 \leq \theta_3 < 1/34)$,
 or $(\theta_1 = 15/34 \text{ and } \theta_2 = 39/782 \text{ and } 245/9384 \leq \theta_3 < 1/34)$,
 or $(\theta_1 = 15/34 \text{ and } \theta_2 = 59/1190 \text{ and } 377/14280 \leq \theta_3 < 1/34)$,
 or $(\theta_1 = 43/98 \text{ and } \theta_2 = 1/20 \text{ and } 23/840 \leq \theta_3 < 3/98)$,
 or $(\theta_1 = 43/98 \text{ and } \theta_2 = 13/266 \text{ and } 13/456 \leq \theta_3 < 3/98)$,
 or $(\theta_1 = 43/98 \text{ and } \theta_2 = 5/98 \text{ and } 31/1176 \leq \theta_3 < 3/98)$,
 or $(\theta_1 = 143/340 \text{ and } \theta_2 = 1/20 \text{ and } 31/816 \leq \theta_3 < 27/680)$,
 or $(\theta_1 = 53/126 \text{ and } \theta_2 = 1/20 \text{ and } 41/1080 \leq \theta_3 < 5/126)$,
 or $(\theta_1 = 17/40 \text{ and } \theta_2 = 1/20 \text{ and } 17/480 \leq \theta_3 < 3/80)$,
 or $(\theta_1 = 97/220 \text{ and } \theta_2 = 1/20 \text{ and } 23/880 \leq \theta_3 < 13/440)$,
 or $(\theta_1 = 13/32 \text{ and } \theta_2 = 37/608 \text{ and } 259/7296 \leq \theta_3 < 259/6080)$,
 or $(\theta_1 = 13/32 \text{ and } \theta_2 = 1/16 \text{ and } 13/384 \leq \theta_3 < 13/320)$,
 or $(\theta_1 = 13/32 \text{ and } \theta_2 = 49/736 \text{ and } 263/8832 \leq \theta_3 < 263/7360)$,
 or $(\theta_1 = 13/32 \text{ and } \theta_2 = 79/1120 \text{ and } 347/13440 \leq \theta_3 < 347/11200)$,
 or $(\theta_1 = 139/340 \text{ and } \theta_2 = 21/340 \text{ and } 9/272 \leq \theta_3 < 27/680)$,
 or $(\theta_1 = 139/340 \text{ and } \theta_2 = 387/6460 \text{ and } 903/25840 \leq \theta_3 < 2709/64600)$,
 or $(\theta_1 = 139/340 \text{ and } \theta_2 = 31/510 \text{ and } 139/4080 \leq \theta_3 < 139/3400)$,
 or $(\theta_1 = 139/340 \text{ and } \theta_2 = 511/7820 \text{ and } 923/31280 \leq \theta_3 < 2769/78200)$,
 or $(\theta_1 = 139/340 \text{ and } \theta_2 = 821/11900 \text{ and } 1231/47600 \leq \theta_3 < 3693/119000)$,
 or $(\theta_1 = 37/90 \text{ and } \theta_2 = 91/1530 \text{ and } 125/3672 \leq \theta_3 < 25/612)$,
 or $(\theta_1 = 37/90 \text{ and } \theta_2 = 16/255 \text{ and } 113/3672 \leq \theta_3 < 113/3060)$,
 or $(\theta_1 = 37/90 \text{ and } \theta_2 = 101/1710 \text{ and } 707/20520 \leq \theta_3 < 707/17100)$,
 or $(\theta_1 = 37/90 \text{ and } \theta_2 = 8/135 \text{ and } 37/1080 \leq \theta_3 < 37/900)$,
 or $(\theta_1 = 37/90 \text{ and } \theta_2 = 133/2070 \text{ and } 727/24840 \leq \theta_3 < 727/20700)$,
 or $(\theta_1 = 37/90 \text{ and } \theta_2 = 71/1050 \text{ and } 979/37800 \leq \theta_3 < 979/31500)$,
 or $(\theta_1 = 127/306 \text{ and } \theta_2 = 23/357 \text{ and } 689/25704 \leq \theta_3 < 689/21420)$,
 or $(\theta_1 = 127/306 \text{ and } \theta_2 = 26/459 \text{ and } 127/3672 \leq \theta_3 < 127/3060)$,
 or $(\theta_1 = 127/306 \text{ and } \theta_2 = 335/5814 \text{ and } 2345/69768 \leq \theta_3 < 469/11628)$,
 or $(\theta_1 = 127/306 \text{ and } \theta_2 = 439/7038 \text{ and } 2437/84456 \leq \theta_3 < 2437/70380)$,
 or $(\theta_1 = 127/306 \text{ and } \theta_2 = 233/3570 \text{ and } 3337/128520 \leq \theta_3 < 3337/107100)$,
 or $(\theta_1 = 143/340 \text{ and } \theta_2 = 9/170 \text{ and } 143/4080 \leq \theta_3 < 27/680)$,
 or $(\theta_1 = 143/340 \text{ and } \theta_2 = 359/6460 \text{ and } 2513/77520 \leq \theta_3 < 2513/64600)$,
 or $(\theta_1 = 143/340 \text{ and } \theta_2 = 467/7820 \text{ and } 2653/93840 \leq \theta_3 < 2653/78200)$,
 or $(\theta_1 = 143/340 \text{ and } \theta_2 = 737/11900 \text{ and } 3721/142800 \leq \theta_3 < 3721/119000)$,
 or $(\theta_1 = 53/126 \text{ and } \theta_2 = 107/2142 \text{ and } 977/25704 \leq \theta_3 < 5/126)$,
 or $(\theta_1 = 53/126 \text{ and } \theta_2 = 10/189 \text{ and } 53/1512 \leq \theta_3 < 5/126)$,
 or $(\theta_1 = 53/126 \text{ and } \theta_2 = 173/2898 \text{ and } 983/34776 \leq \theta_3 < 983/28980)$,
 or $(\theta_1 = 53/126 \text{ and } \theta_2 = 13/210 \text{ and } 197/7560 \leq \theta_3 < 197/6300)$,
 or $(\theta_1 = 17/40 \text{ and } \theta_2 = 41/760 \text{ and } 287/9120 \leq \theta_3 < 3/80)$,
 or $(\theta_1 = 17/40 \text{ and } \theta_2 = 53/920 \text{ and } 307/11040 \leq \theta_3 < 307/9200)$,
 or $(\theta_1 = 17/40 \text{ and } \theta_2 = 83/1400 \text{ and } 439/16800 \leq \theta_3 < 439/14000)$,
 or $(\theta_1 = 2049/4964 \text{ and } \theta_2 = 315/4964 \text{ and } 1733/59568 \leq \theta_3 < 1733/49640)$,
 or $(\theta_1 = 509/1224 \text{ and } \theta_2 = 185/2856 \text{ and } 2671/102816 \leq \theta_3 < 2671/85680)$,
 or $(9/20 < \theta_1 < 73/162 \text{ and } \theta_2 = 1/2(1 - 2\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(4/9 < \theta_1 < 29/65 \text{ and } \theta_2 = 1/19(4 - 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(9/20 < \theta_1 < 73/162 \text{ and } \theta_2 = 1/2(1 - 2\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(4/9 < \theta_1 < 1868/4199 \text{ and } \theta_2 = 1/23(6 - 11\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,

or $(1868/4199 < \theta_1 < 116/259$ and $\theta_2 = 1/23(6 - 11\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(116/259 < \theta_1 < 185/413$ and $\theta_2 = 1/23(6 - 11\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(185/413 \leq \theta_1 < 49/109$ and $\theta_2 = 1/23(6 - 11\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(4/9 < \theta_1 < 97/217$ and $\theta_2 = 1/35(11 - 21\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(53/116 < \theta_1 \leq 16/35$ and $\theta_2 = 1/5(8 - 17\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(16/35 < \theta_1 < 91/199$ and $\theta_2 = 1/5(8 - 17\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(\theta_1 = 4/9$ and $\theta_2 = 1/20$ and $13/540 \leq \theta_3 < 1/36)$,
 or $(\theta_1 = 4/9$ and $\theta_2 = 1/21$ and $5/189 \leq \theta_3 < 1/36)$,
 or $(\theta_1 = 4/9$ and $\theta_2 = 8/171$ and $14/513 \leq \theta_3 < 1/36)$,
 or $(\theta_1 = 4/9$ and $\theta_2 = 10/207$ and $16/621 \leq \theta_3 < 1/36)$,
 or $(\theta_1 = 1868/4199$ and $\theta_2 = 202/4199$ and $108/4199 \leq \theta_3 < 463/16796)$,
 or $(\theta_1 = 116/259$ and $\theta_2 = 652/13209$ and $100/4403 \leq \theta_3 < 27/1036)$,
 or $(\theta_1 = 116/259$ and $\theta_2 = 278/5957$ and $454/17871 \leq \theta_3 < 27/1036)$,
 or $(9/20 < \theta_1 < 73/162$ and $1/2(1 - 2\theta_1) < \theta_2 < 1/40(11 - 20\theta_1)$ and $1/2(1 - 2\theta_1 - \theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(\theta_1 = 37/84$ and $\theta_2 = 1/20$ and $2/105 \leq \theta_3 < 5/168)$,
 or $(\theta_1 = 61/140$ and $\theta_2 = 1/18$ and $11/630 \leq \theta_3 < 17/600)$,
 or $(\theta_1 = 43/98$ and $\theta_2 = 1/18$ and $5/441 \leq \theta_3 < 11/420)$,
 or $(\theta_1 = 85/198$ and $\theta_2 = 1/18$ and $1/33 \leq \theta_3 < 13/396)$,
 or $(\theta_1 = 163/378$ and $\theta_2 = 1/18$ and $5/189 \leq \theta_3 < 17/540)$,
 or $(\theta_1 = 61/140$ and $\theta_2 = 169/3220$ and $19/805 \leq \theta_3 < 1031/32200)$,
 or $(\theta_1 = 61/140$ and $\theta_2 = 37/700$ and $4/175 \leq \theta_3 < 221/7000)$,
 or $(\theta_1 = 15/34$ and $\theta_2 = 1/20$ and $3/170 \leq \theta_3 < 1/34)$,
 or $(\theta_1 = 15/34$ and $\theta_2 = 39/782$ and $7/391 \leq \theta_3 < 1/34)$,
 or $(\theta_1 = 15/34$ and $\theta_2 = 59/1190$ and $11/595 \leq \theta_3 < 1/34)$,
 or $(\theta_1 = 43/98$ and $\theta_2 = 1/20$ and $11/490 \leq \theta_3 < 3/98)$,
 or $(\theta_1 = 43/98$ and $\theta_2 = 5/98$ and $1/49 \leq \theta_3 < 3/98)$,
 or $(\theta_1 = 4/9$ and $\theta_2 = 1/20$ and $1/90 \leq \theta_3 < 1/36)$,
 or $(\theta_1 = 97/220$ and $\theta_2 = 1/20$ and $1/55 \leq \theta_3 < 13/440)$,
 or $(\theta_1 = 4/9$ and $\theta_2 = 1/21$ and $1/63 \leq \theta_3 < 1/36)$,
 or $(\theta_1 = 4/9$ and $\theta_2 = 10/207$ and $1/69 \leq \theta_3 < 1/36)$,
 or $(\theta_1 = 1868/4199$ and $\theta_2 = 202/4199$ and $108/4199 \leq \theta_3 < 463/16796)$,
 or $(\theta_1 = 116/259$ and $\theta_2 = 652/13209$ and $100/4403 \leq \theta_3 < 27/1036)$,
 or $(\theta_1 = 116/259$ and $\theta_2 = 278/5957$ and $454/17871 \leq \theta_3 < 27/1036)$,
 or $(9/20 < \theta_1 < 73/162$ and $(1 - \theta_1)/12 < \theta_2 < 1/2(1 - 2\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(\theta_1 = 31/70$ and $\theta_2 = 1/20$ and $1/40 \leq \theta_3 < 1/35)$,
 or $(130/289 < \theta_1 < 9/20$ and $1/40(11 - 20\theta_1) \leq \theta_2 < 1/2(1 - 2\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(\theta_1 = 15/34$ and $\theta_2 = 1/18$ and $1/153 \leq \theta_3 < 5/204)$,
 or $(\theta_1 = 11/26$ and $\theta_2 = 15/221$ and $4/221 \leq \theta_3 < 99/4420)$,
 or $(\theta_1 = 143/340$ and $\theta_2 = 159/2380$ and $3/119 \leq \theta_3 < 121/4760)$,
 or $(\theta_1 = 53/126$ and $\theta_2 = 167/2499$ and $188/7497 \leq \theta_3 < 3803/149940)$,
 or $(\theta_1 = 17/40$ and $\theta_2 = 327/4760$ and $3/238 \leq \theta_3 < 191/9520)$,
 or $(9/20 < \theta_1 < 73/162$ and $\theta_2 = 1/5(8 - 17\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(9/20 < \theta_1 < 73/162$ and $\theta_2 = 1/204(-28 + 85\theta_1)$ and $1/17(8 - 17\theta_1) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(1772/3893 < \theta_1 < 11/24$ and $\theta_2 = 1/204(-28 + 85\theta_1)$ and $1/17(8 - 17\theta_1) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(\theta_1 = 1772/3893$ and $\theta_2 = 12/229$ and $60/3893 \leq \theta_3 < 72/3893)$,
 or $(\theta_1 = 11/24$ and $\theta_2 = 263/4896$ and $5/408 \leq \theta_3 < 1/68)$,
 or $(\theta_1 = 316/679$ and $\theta_2 = 654/11543$ and $60/11543 \leq \theta_3 < 72/11543)$,
 or $(\theta_1 = 316/679$ and $\theta_2 = 5/97$ and $1/97 \leq \theta_3 < 6/485)$,
 or $(\theta_1 = 226/481$ and $\theta_2 = 2/37$ and $5/962 \leq \theta_3 < 3/481)$,

or $(\theta_1 = 23/49 \text{ and } \theta_2 = 9/196 \text{ and } 2/147 \leq \theta_3 < 3/196)$,
 or $(2/5 < \theta_1 \leq 272/679 \text{ and } 1/2(1 - 2\theta_1) < \theta_2 < 1/12(4 - 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(\theta_1 = 7/17 \text{ and } \theta_2 = 3/34 \text{ and } 1/204 \leq \theta_3 < 1/170)$,
 or $(2/5 < \theta_1 \leq 272/679 \text{ and } \theta_2 = 1/2(1 - 2\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(\theta_1 = 127/308 \text{ and } \theta_2 = 1/11 \text{ and } 1/528 \leq \theta_3 < 1/440)$,
 or $(\theta_1 = 5/12 \text{ and } \theta_2 = 1/12 \text{ and } 1/144 \leq \theta_3 < 1/120)$,
 or $(\theta_1 = 11/26 \text{ and } \theta_2 = 1/13 \text{ and } 1/104 \leq \theta_3 < 3/260)$,
 or $(\theta_1 = 61/140 \text{ and } \theta_2 = 1221/16660 \text{ and } 235/39984 \leq \theta_3 < 47/6664)$,
 or $(\theta_1 = 15/34 \text{ and } \theta_2 = 9/119 \text{ and } 1/2856 \leq \theta_3 < 1/2380)$,
 or $(\theta_1 = 43/98 \text{ and } \theta_2 = 435/5831 \text{ and } 389/139944 \leq \theta_3 < 389/116620)$,
 or $(\theta_1 = 13/32 \text{ and } \theta_2 = 3/32 \text{ and } 1/384 \leq \theta_3 < 1/320)$,
 or $(\theta_1 = 139/340 \text{ and } \theta_2 = 31/340 \text{ and } 1/272 \leq \theta_3 < 3/680)$,
 or $(\theta_1 = 37/90 \text{ and } \theta_2 = 4/45 \text{ and } 1/216 \leq \theta_3 < 1/180)$,
 or $(\theta_1 = 127/306 \text{ and } \theta_2 = 13/153 \text{ and } 23/3672 \leq \theta_3 < 23/3060)$,
 or $(\theta_1 = 15/34 \text{ and } \theta_2 = 1/18 \text{ and } 25/1224 \leq \theta_3 < 5/204)$,
 or $(\theta_1 = 11/26 \text{ and } \theta_2 = 15/221 \text{ and } 33/1768 \leq \theta_3 < 99/4420)$,
 or $(\theta_1 = 127/306 \text{ and } \theta_2 = 23/306 \text{ and } 59/3672 \leq \theta_3 < 59/3060)$,
 or $(\theta_1 = 143/340 \text{ and } \theta_2 = 159/2380 \text{ and } 121/5712 \leq \theta_3 < 121/4760)$,
 or $(\theta_1 = 53/126 \text{ and } \theta_2 = 167/2499 \text{ and } 3803/179928 \leq \theta_3 < 3803/149940)$,
 or $(\theta_1 = 17/40 \text{ and } \theta_2 = 327/4760 \text{ and } 191/11424 \leq \theta_3 < 191/9520)$,
 or $(\theta_1 = 12/29 \text{ and } \theta_2 = 2/29 \text{ and } 2/87 \leq \theta_3 < 4/145)$,
 or $(451/1003 \leq \theta_1 < 9/20 \text{ and } 1/60(-7 + 25\theta_1) < \theta_2 < 1/12(4 - 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(9/20 < \theta_1 < 73/162 \text{ and } 1/5(8 - 17\theta_1) < \theta_2 < 1/12(4 - 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(9/20 < \theta_1 < 73/162 \text{ and } \theta_2 = 1/5(8 - 17\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(1772/3893 < \theta_1 < 11/24 \text{ and } 2/37 < \theta_2 < 1/12(4 - 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(11/24 < \theta_1 \leq 1444/3145 \text{ and } 2/37 < \theta_2 < 1/12(4 - 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(\theta_1 = 11/24 \text{ and } \theta_2 = 263/4896 \text{ and } 5/408 \leq \theta_3 < 1/68)$,
 or $(769/1673 \leq \theta_1 < 13/28 \text{ and } 1/204(-28 + 85\theta_1) < \theta_2 < 1/12(4 - 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(87/188 \leq \theta_1 < 13/28 \text{ and } \theta_2 = 1/5(-3 + 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(15/32 < \theta_1 < 8/17 \text{ and } 1/204(-28 + 85\theta_1) < \theta_2 < 1/12(4 - 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(15/32 < \theta_1 < 8/17 \text{ and } \theta_2 = 1/5(-3 + 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(177/388 < \theta_1 \leq 53/116 \text{ and } \theta_2 = 1/5(8 - 17\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(41/89 < \theta_1 < 87/188 \text{ and } \theta_2 = 1/5(-3 + 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(73/162 \leq \theta_1 < 5/11 \text{ and } 1/5(8 - 17\theta_1) < \theta_2 < 1/12(4 - 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(5/11 \leq \theta_1 \leq 286/629 \text{ and } 1/5(8 - 17\theta_1) < \theta_2 < 1/12(4 - 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(73/162 \leq \theta_1 < 5/11 \text{ and } \theta_2 = 1/5(8 - 17\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(1772/3893 < \theta_1 \leq 177/388 \text{ and } \theta_2 = 1/5(8 - 17\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(5/11 \leq \theta_1 < 286/629 \text{ and } \theta_2 = 1/5(8 - 17\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(286/629 \leq \theta_1 < 1772/3893 \text{ and } \theta_2 = 1/5(8 - 17\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(4/9 < \theta_1 < 116/259 \text{ and } 1/60(-7 + 25\theta_1) < \theta_2 < 1/12(4 - 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(116/259 < \theta_1 < 185/413 \text{ and } 1/60(-7 + 25\theta_1) < \theta_2 < 1/12(4 - 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(116/259 < \theta_1 < 185/413 \text{ and } \theta_2 = 1/70(32 - 63\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(185/413 \leq \theta_1 < 451/1003 \text{ and } 1/60(-7 + 25\theta_1) < \theta_2 < 1/12(4 - 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(13/28 \leq \theta_1 < 316/679 \text{ and } 1/204(-28 + 85\theta_1) < \theta_2 < 1/12(4 - 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(316/679 < \theta_1 \leq 15/32 \text{ and } 1/204(-28 + 85\theta_1) < \theta_2 < 1/12(4 - 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(1444/3145 < \theta_1 \leq 74/161 \text{ and } 1/204(-28 + 85\theta_1) < \theta_2 < 1/12(4 - 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(74/161 < \theta_1 < 769/1673 \text{ and } 1/204(-28 + 85\theta_1) < \theta_2 < 1/12(4 - 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(286/629 < \theta_1 < 1772/3893 \text{ and } 2/37 < \theta_2 < 1/12(4 - 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(13/28 \leq \theta_1 < 316/679 \text{ and } \theta_2 = 1/5(-3 + 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,

or $(316/679 < \theta_1 \leq 121/259 \text{ and } \theta_2 = 1/5(-3 + 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(121/259 < \theta_1 \leq 15/32 \text{ and } \theta_2 = 1/5(-3 + 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(\theta_1 = 116/259 \text{ and } \theta_2 = 2/37 \text{ and } 2/111 \leq \theta_3 < 4/185)$,
 or $(\theta_1 = 116/259 \text{ and } \theta_2 = 1087/15540 \text{ and } 11/5180 \leq \theta_3 < 33/12950)$,
 or $(\theta_1 = 1772/3893 \text{ and } \theta_2 = 12/229 \text{ and } 60/3893 \leq \theta_3 < 72/3893)$,
 or $(\theta_1 = 316/679 \text{ and } \theta_2 = 654/11543 \text{ and } 60/11543 \leq \theta_3 < 72/11543)$,
 or $(\theta_1 = 316/679 \text{ and } \theta_2 = 5/97 \text{ and } 1/97 \leq \theta_3 < 6/485)$,
 or $(\theta_1 = 226/481 \text{ and } \theta_2 = 2/37 \text{ and } 5/962 \leq \theta_3 < 3/481)$,
 or $(\theta_1 = 9/20 \text{ and } \theta_2 = 41/816 \text{ and } 7/340 \leq \theta_3 < 21/850)$,
 or $(\theta_1 = 23/49 \text{ and } \theta_2 = 9/196 \text{ and } 2/147 \leq \theta_3 < 3/196)$,
 or $(87/188 \leq \theta_1 < 13/28 \text{ and } \theta_2 = 1/5(-3 + 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(\theta_1 = 316/679 \text{ and } \theta_2 = 5/97 \text{ and } 1/97 \leq \theta_3 < 6/485)$,
 or $(\theta_1 = 226/481 \text{ and } \theta_2 = 2/37 \text{ and } 5/962 \leq \theta_3 < 3/481)$,
 or $(\theta_1 = 23/49 \text{ and } \theta_2 = 9/196 \text{ and } 2/147 \leq \theta_3 < 3/196)$,
 or $(2/5 < \theta_1 \leq 272/679 \text{ and } \theta_2 = 1/2(1 - 2\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(37/90 < \theta_1 < 7/17 \text{ and } 1/27(14 - 28\theta_1) < \theta_2 < 1/12(4 - 7\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(7/17 < \theta_1 \leq 127/308 \text{ and } 1/27(14 - 28\theta_1) < \theta_2 < 1/12(4 - 7\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(\theta_1 = 17/40 \text{ and } 3/40 \leq \theta_2 \leq 797/10360 \text{ and } 0 < \theta_3 < 1/400(41 - 480\theta_2))$,
 or $(37/90 < \theta_1 < 7/17 \text{ and } 1/2(1 - 2\theta_1) < \theta_2 \leq 1/259(130 - 259\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(7/17 < \theta_1 \leq 12/29 \text{ and } 1/2(1 - 2\theta_1) < \theta_2 \leq 1/259(130 - 259\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(272/679 < \theta_1 \leq 1041/2590 \text{ and } \theta_2 = 1/2(1 - 2\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(1041/2590 < \theta_1 < 13/32 \text{ and } \theta_2 = 1/2(1 - 2\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(13/32 < \theta_1 < 139/340 \text{ and } \theta_2 = 1/2(1 - 2\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(139/340 < \theta_1 < 37/90 \text{ and } \theta_2 = 1/2(1 - 2\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(37/90 < \theta_1 < 7/17 \text{ and } 1/259(130 - 259\theta_1) < \theta_2 \leq 1/119(57 - 112\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(37/90 < \theta_1 < 7/17 \text{ and } 1/119(57 - 112\theta_1) < \theta_2 \leq 1/27(14 - 28\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(\theta_1 = 13/32 \text{ and } 3/32 < \theta_2 \leq 793/8288 \text{ and } 0 < \theta_3 < 1/320(37 - 384\theta_2))$,
 or $(\theta_1 = 13/32 \text{ and } 793/8288 < \theta_2 < 37/384 \text{ and } 0 < \theta_3 < 1/320(37 - 384\theta_2))$,
 or $(\theta_1 = 139/340 \text{ and } 217/2295 < \theta_2 < 129/1360 \text{ and } 0 < \theta_3 < (387 - 4080\theta_2)/3400)$,
 or $(\theta_1 = 139/340 \text{ and } 31/340 < \theta_2 \leq 8199/88060 \text{ and } 0 < \theta_3 < (387 - 4080\theta_2)/3400)$,
 or $(\theta_1 = 139/340 \text{ and } 8199/88060 < \theta_2 \leq 953/10115 \text{ and } 0 < \theta_3 < (387 - 4080\theta_2)/3400)$,
 or $(\theta_1 = 139/340 \text{ and } 953/10115 < \theta_2 \leq 217/2295 \text{ and } 0 < \theta_3 < (387 - 4080\theta_2)/3400)$,
 or $(\theta_1 = 37/90 \text{ and } 112/1215 < \theta_2 < 101/1080 \text{ and } 0 < \theta_3 < 1/900(101 - 1080\theta_2))$,
 or $(\theta_1 = 37/90 \text{ and } 4/45 < \theta_2 \leq 2117/23310 \text{ and } 0 < \theta_3 < 1/900(101 - 1080\theta_2))$,
 or $(\theta_1 = 37/90 \text{ and } 2117/23310 < \theta_2 \leq 29/315 \text{ and } 0 < \theta_3 < 1/900(101 - 1080\theta_2))$,
 or $(\theta_1 = 37/90 \text{ and } 29/315 < \theta_2 \leq 112/1215 \text{ and } 0 < \theta_3 < 1/900(101 - 1080\theta_2))$,
 or $(\theta_1 = 15/34 \text{ and } 1/17 \leq \theta_2 \leq 535/8806 \text{ and } 0 < \theta_3 < 1/340(31 - 408\theta_2))$,
 or $(127/306 < \theta_1 < 5/12 \text{ and } \theta_2 = 1/2(1 - 2\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(12/29 < \theta_1 < 127/306 \text{ and } \theta_2 = 1/2(1 - 2\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(5/12 < \theta_1 \leq 108/259 \text{ and } \theta_2 = -2 + 5\theta_1 \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(108/259 < \theta_1 \leq 68/163 \text{ and } \theta_2 = -2 + 5\theta_1 \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(68/163 < \theta_1 < 28/67 \text{ and } \theta_2 = -2 + 5\theta_1 \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(73/170 \leq \theta_1 < 31/72 \text{ and } \theta_2 = 1/238(-27 + 102\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(31/72 \leq \theta_1 \leq 5419/12580 \text{ and } \theta_2 = 1/238(-27 + 102\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(4061/9418 < \theta_1 < 61/140 \text{ and } \theta_2 = 1/238(-27 + 102\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(61/140 < \theta_1 < 43/98 \text{ and } \theta_2 = 1/238(-27 + 102\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(43/98 < \theta_1 < 15/34 \text{ and } \theta_2 = 1/238(-27 + 102\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(15/34 < \theta_1 < 638/1445 \text{ and } \theta_2 = 1/238(-27 + 102\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(5419/12580 < \theta_1 \leq 4061/9418 \text{ and } \theta_2 = 1/238(-27 + 102\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,

or $(5/12 < \theta_1 \leq 108/259 \text{ and } 1/2(1 - 2\theta_1) \leq \theta_2 < -2 + 5\theta_1 \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(108/259 < \theta_1 < 21/50 \text{ and } 1/2(1 - 2\theta_1) \leq \theta_2 \leq 1/259(130 - 259\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(21/50 \leq \theta_1 < 143/340 \text{ and } 1/2(1 - 2\theta_1) \leq \theta_2 \leq 1/259(130 - 259\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(143/340 < \theta_1 < 53/126 \text{ and } 1/2(1 - 2\theta_1) \leq \theta_2 \leq 1/259(130 - 259\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(53/126 < \theta_1 < 11/26 \text{ and } 1/2(1 - 2\theta_1) \leq \theta_2 \leq 1/259(130 - 259\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(11/26 < \theta_1 \leq 204/481 \text{ and } 1/2(1 - 2\theta_1) \leq \theta_2 \leq 1/7(-2 + 6\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(204/481 < \theta_1 < 17/40 \text{ and } 1/2(1 - 2\theta_1) \leq \theta_2 \leq 1/259(130 - 259\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(17/40 < \theta_1 < 73/170 \text{ and } 1/2(1 - 2\theta_1) \leq \theta_2 \leq 1/259(130 - 259\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(73/170 < \theta_1 < 31/72 \text{ and } 1/2(1 - 2\theta_1) \leq \theta_2 < 1/238(-27 + 102\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(31/72 \leq \theta_1 \leq 5419/12580 \text{ and } 1/2(1 - 2\theta_1) \leq \theta_2 < 1/238(-27 + 102\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(5419/12580 < \theta_1 < 61/140 \text{ and } 1/2(1 - 2\theta_1) \leq \theta_2 \leq 1/259(130 - 259\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(61/140 < \theta_1 < 43/98 \text{ and } 1/2(1 - 2\theta_1) \leq \theta_2 \leq 1/259(130 - 259\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(43/98 < \theta_1 < 15/34 \text{ and } 1/2(1 - 2\theta_1) \leq \theta_2 \leq 1/259(130 - 259\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(15/34 < \theta_1 < 4/9 \text{ and } 1/2(1 - 2\theta_1) \leq \theta_2 \leq 1/259(130 - 259\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(7/17 < \theta_1 < 127/308 \text{ and } 1/259(130 - 259\theta_1) < \theta_2 \leq 1/119(57 - 112\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(7/17 < \theta_1 < 127/308 \text{ and } 1/119(57 - 112\theta_1) < \theta_2 \leq 1/27(14 - 28\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(\theta_1 = 127/308 \text{ and } 1021/11396 < \theta_2 < 1/11 \text{ and } 0 < \theta_3 < 1/440(49 - 528\theta_2))$,
 or $(\theta_1 = 5/12 \text{ and } 1/12 < \theta_2 \leq 265/3108 \text{ and } 0 < \theta_3 < 1/120(13 - 144\theta_2))$,
 or $(\theta_1 = 5/12 \text{ and } 265/3108 < \theta_2 \leq 7/81 \text{ and } 0 < \theta_3 < 1/120(13 - 144\theta_2))$,
 or $(\theta_1 = 5/12 \text{ and } 7/81 < \theta_2 < 13/144 \text{ and } 0 < \theta_3 < 1/120(13 - 144\theta_2))$,
 or $(\theta_1 = 11/26 \text{ and } 1/13 < \theta_2 \leq 531/6734 \text{ and } 0 < \theta_3 < 1/260(27 - 312\theta_2))$,
 or $(\theta_1 = 11/26 \text{ and } 531/6734 < \theta_2 \leq 28/351 \text{ and } 0 < \theta_3 < 1/260(27 - 312\theta_2))$,
 or $(\theta_1 = 11/26 \text{ and } 28/351 < \theta_2 < 9/104 \text{ and } 0 < \theta_3 < 1/260(27 - 312\theta_2))$,
 or $(\theta_1 = 61/140 \text{ and } 9/140 \leq \theta_2 \leq 49/740 \text{ and } 0 < \theta_3 < 1/200(19 - 240\theta_2))$,
 or $(\theta_1 = 61/140 \text{ and } 49/740 < \theta_2 \leq 1/15 \text{ and } 0 < \theta_3 < 1/200(19 - 240\theta_2))$,
 or $(\theta_1 = 61/140 \text{ and } 1/15 < \theta_2 < 1221/16660 \text{ and } 0 < \theta_3 < 1/200(19 - 240\theta_2))$,
 or $(\theta_1 = 61/140 \text{ and } 1221/16660 < \theta_2 < 19/240 \text{ and } 0 < \theta_3 < 1/200(19 - 240\theta_2))$,
 or $(\theta_1 = 15/34 \text{ and } 535/8806 < \theta_2 \leq 28/459 \text{ and } 0 < \theta_3 < 1/340(31 - 408\theta_2))$,
 or $(\theta_1 = 15/34 \text{ and } 28/459 < \theta_2 < 9/119 \text{ and } 0 < \theta_3 < 1/340(31 - 408\theta_2))$,
 or $(\theta_1 = 15/34 \text{ and } 9/119 < \theta_2 < 31/408 \text{ and } 0 < \theta_3 < 1/340(31 - 408\theta_2))$,
 or $(\theta_1 = 43/98 \text{ and } 3/49 \leq \theta_2 \leq 229/3626 \text{ and } 0 < \theta_3 < 1/140(13 - 168\theta_2))$,
 or $(\theta_1 = 43/98 \text{ and } 229/3626 < \theta_2 \leq 4/63 \text{ and } 0 < \theta_3 < 1/140(13 - 168\theta_2))$,
 or $(\theta_1 = 43/98 \text{ and } 4/63 < \theta_2 < 435/5831 \text{ and } 0 < \theta_3 < 1/140(13 - 168\theta_2))$,
 or $(\theta_1 = 43/98 \text{ and } 435/5831 < \theta_2 < 13/168 \text{ and } 0 < \theta_3 < 1/140(13 - 168\theta_2))$,
 or $(\theta_1 = 127/306 \text{ and } 364/4131 < \theta_2 \leq 1/11 \text{ and } 0 < \theta_3 < (335 - 3672\theta_2)/3060)$,
 or $(\theta_1 = 127/306 \text{ and } 1/11 < \theta_2 < 335/3672 \text{ and } 0 < \theta_3 < (335 - 3672\theta_2)/3060)$,
 or $(\theta_1 = 127/306 \text{ and } 13/153 < \theta_2 \leq 6887/79254 \text{ and } 0 < \theta_3 < (335 - 3672\theta_2)/3060)$,
 or $(\theta_1 = 127/306 \text{ and } 6887/79254 < \theta_2 \leq 364/4131 \text{ and } 0 < \theta_3 < (335 - 3672\theta_2)/3060)$,
 or $(\theta_1 = 143/340 \text{ and } 27/340 \leq \theta_2 \leq 7163/88060 \text{ and } 0 < \theta_3 < (359 - 4080\theta_2)/3400)$,
 or $(\theta_1 = 143/340 \text{ and } 7163/88060 < \theta_2 \leq 7/85 \text{ and } 0 < \theta_3 < (359 - 4080\theta_2)/3400)$,
 or $(\theta_1 = 143/340 \text{ and } 7/85 < \theta_2 < 359/4080 \text{ and } 0 < \theta_3 < (359 - 4080\theta_2)/3400)$,
 or $(\theta_1 = 53/126 \text{ and } 5/63 \leq \theta_2 \leq 379/4662 \text{ and } 0 < \theta_3 < 1/180(19 - 216\theta_2))$,
 or $(\theta_1 = 53/126 \text{ and } 379/4662 < \theta_2 \leq 20/243 \text{ and } 0 < \theta_3 < 1/180(19 - 216\theta_2))$,
 or $(\theta_1 = 53/126 \text{ and } 20/243 < \theta_2 < 19/216 \text{ and } 0 < \theta_3 < 1/180(19 - 216\theta_2))$,
 or $(\theta_1 = 17/40 \text{ and } 797/10360 < \theta_2 \leq 7/90 \text{ and } 0 < \theta_3 < 1/400(41 - 480\theta_2))$,
 or $(\theta_1 = 17/40 \text{ and } 7/90 < \theta_2 \leq 11/140 \text{ and } 0 < \theta_3 < 1/400(41 - 480\theta_2))$,
 or $(\theta_1 = 17/40 \text{ and } 11/140 < \theta_2 < 41/480 \text{ and } 0 < \theta_3 < 1/400(41 - 480\theta_2))$,
 or $(\theta_1 = 4/9 \text{ and } 1/18 \leq \theta_2 \leq 134/2331 \text{ and } 0 < \theta_3 < 1/45(4 - 54\theta_2))$,
 or $(\theta_1 = 4/9 \text{ and } 134/2331 < \theta_2 \leq 14/243 \text{ and } 0 < \theta_3 < 1/45(4 - 54\theta_2))$,

or $(\theta_1 = 4/9 \text{ and } 14/243 < \theta_2 < 2/27 \text{ and } 0 < \theta_3 < 1/45(4 - 54\theta_2))$,
 or $(\theta_1 = 116/259 \text{ and } \theta_2 = 2/37 \text{ and } 2/111 \leq \theta_3 < 4/185)$,
 or $(\theta_1 = 116/259 \text{ and } \theta_2 = 1087/15540 \text{ and } 11/5180 \leq \theta_3 < 33/12950)$,
 or $(\theta_1 = 73/170 \text{ and } 7/85 < \theta_2 < 169/2040 \text{ and } 0 < \theta_3 < (169 - 2040\theta_2)/1700)$,
 or $(\theta_1 = 9/20 \text{ and } \theta_2 = 41/816 \text{ and } 7/340 \leq \theta_3 < 21/850)$,
 or $(\theta_1 = 21/50 \text{ and } 56/675 < \theta_2 < 53/600 \text{ and } 0 < \theta_3 < 1/500(53 - 600\theta_2))$,
 or $(\theta_1 = 13/28 \text{ and } \theta_2 = 1/20 \text{ and } 1/80 \leq \theta_3 < 3/200)$,
 or $(\theta_1 = 31/70 \text{ and } \theta_2 = 1/18 \text{ and } 7/360 \leq \theta_3 < 7/300)$,
 or $(25/63 \leq \theta_1 < 2/5 \text{ and } 1/24(3 - 4\theta_1) < \theta_2 < 1/112(29 - 56\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(17/44 < \theta_1 < 7/18 \text{ and } 1/20(9 - 20\theta_1) < \theta_2 < 1/112(29 - 56\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(71/180 < \theta_1 \leq 36/91 \text{ and } 1/24(3 - 4\theta_1) < \theta_2 < 1/112(29 - 56\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(36/91 < \theta_1 < 25/63 \text{ and } 1/24(3 - 4\theta_1) < \theta_2 < 1/112(29 - 56\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(7/18 < \theta_1 \leq 39/100 \text{ and } 1/20(9 - 20\theta_1) < \theta_2 < 1/112(29 - 56\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(39/100 < \theta_1 < 71/180 \text{ and } 1/24(3 - 4\theta_1) < \theta_2 < 1/112(29 - 56\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(25/63 \leq \theta_1 < 2/5 \text{ and } 1/20(9 - 20\theta_1) < \theta_2 < 1/18 \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(25/63 \leq \theta_1 < 2/5 \text{ and } \theta_2 = 1/18 \text{ and } 1/9(4 - 9\theta_1) < \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(25/63 \leq \theta_1 < 2/5 \text{ and } 1/18 < \theta_2 \leq 1/24(3 - 4\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(\theta_1 = 2/5 \text{ and } 1/20 < \theta_2 < 1/18 \text{ and } 1/10(1 - 10\theta_2) < \theta_3 < 1/20)$,
 or $(\theta_1 = 2/5 \text{ and } 1/18 < \theta_2 < 7/120 \text{ and } 1/10(1 - 10\theta_2) < \theta_3 < 1/20)$,
 or $(\theta_1 = 2/5 \text{ and } 7/120 \leq \theta_2 < 33/560 \text{ and } 1/10(1 - 10\theta_2) < \theta_3 < 1/25(3 - 30\theta_2))$,
 or $(71/180 < \theta_1 \leq 36/91 \text{ and } 1/20(9 - 20\theta_1) < \theta_2 < 1/18 \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(36/91 < \theta_1 < 25/63 \text{ and } 1/20(9 - 20\theta_1) < \theta_2 < 1/18 \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(\theta_1 = 71/180 \text{ and } 19/360 < \theta_2 < 1/18 \text{ and } 1/180(19 - 180\theta_2) < \theta_3 < 19/360)$,
 or $(71/180 < \theta_1 \leq 36/91 \text{ and } \theta_2 = 1/18 \text{ and } 1/9(4 - 9\theta_1) < \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(36/91 < \theta_1 < 25/63 \text{ and } \theta_2 = 1/18 \text{ and } 1/9(4 - 9\theta_1) < \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(7/18 < \theta_1 < 71/180 \text{ and } \theta_2 = 1/18 \text{ and } 1/9(4 - 9\theta_1) < \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(39/100 \leq \theta_1 < 71/180 \text{ and } 1/18 < \theta_2 < 1/20(9 - 20\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(19/50 < \theta_1 \leq 807/2120 \text{ and } 1/9(4 - 9\theta_1) < \theta_2 < 1/112(29 - 56\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(807/2120 < \theta_1 < 55/144 \text{ and } 1/9(4 - 9\theta_1) < \theta_2 < 1/112(29 - 56\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(55/144 \leq \theta_1 \leq 107/280 \text{ and } 1/9(4 - 9\theta_1) < \theta_2 < 1/112(29 - 56\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(107/280 < \theta_1 \leq 23/60 \text{ and } 1/9(4 - 9\theta_1) < \theta_2 < 1/20(9 - 20\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(23/60 < \theta_1 < 17/44 \text{ and } 1/24(3 - 4\theta_1) < \theta_2 < 1/20(9 - 20\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(17/44 < \theta_1 < 7/18 \text{ and } 1/24(3 - 4\theta_1) < \theta_2 < 1/20(9 - 20\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(7/18 < \theta_1 < 39/100 \text{ and } 1/24(3 - 4\theta_1) < \theta_2 < 1/20(9 - 20\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(71/180 < \theta_1 \leq 36/91 \text{ and } 1/18 < \theta_2 \leq 1/24(3 - 4\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(36/91 < \theta_1 < 25/63 \text{ and } 1/18 < \theta_2 \leq 1/24(3 - 4\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(7/18 < \theta_1 < 39/100 \text{ and } 1/18 < \theta_2 \leq 1/24(3 - 4\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(23/6$

$\text{or } (19/50 < \theta_1 \leq 807/2120 \text{ and } 1/24(3 - 4\theta_1) < \theta_2 < 1/9(4 - 9\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (807/2120 < \theta_1 < 55/144 \text{ and } 1/24(3 - 4\theta_1) < \theta_2 < 1/9(4 - 9\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (55/144 \leq \theta_1 < 23/60 \text{ and } 1/24(3 - 4\theta_1) < \theta_2 < 1/9(4 - 9\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (107/280 < \theta_1 < 17/44 \text{ and } 1/20(9 - 20\theta_1) < \theta_2 < 1/112(29 - 56\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (\theta_1 = 71/180 \text{ and } 1/18 < \theta_2 \leq 8/135 \text{ and } 1/180(19 - 180\theta_2) < \theta_3 < 19/360),$
 $\text{or } (\theta_1 = 7/18 \text{ and } 1/18 < \theta_2 \leq 13/216 \text{ and } 1/9(1 - 9\theta_2) < \theta_3 < 1/18),$
 $\text{or } (\theta_1 = 17/44 \text{ and } 5/88 < \theta_2 < 23/396 \text{ and } 1/44(5 - 44\theta_2) < \theta_3 < 5/88),$
 $\text{or } (\theta_1 = 17/44 \text{ and } 23/396 < \theta_2 \leq 2/33 \text{ and } 1/44(5 - 44\theta_2) < \theta_3 < 5/88),$
 $\text{or } (\theta_1 = 17/44 \text{ and } 2/33 < \theta_2 < 7/110 \text{ and } 1/44(5 - 44\theta_2) < \theta_3 < 1/440(57 - 528\theta_2)),$
 $\text{or } (\theta_1 = 17/44 \text{ and } 7/110 < \theta_2 < 81/1232 \text{ and } 1/44(5 - 44\theta_2) < \theta_3 < 1/440(57 - 528\theta_2)),$
 $\text{or } (\theta_1 = 71/180 \text{ and } 8/135 < \theta_2 < 311/5040 \text{ and } 1/180(19 - 180\theta_2) < \theta_3 < (223 - 2160\theta_2)/1800),$
 $\text{or } (\theta_1 = 91/240 \text{ and } 29/480 < \theta_2 \leq 89/1440 \text{ and } 1/240(29 - 240\theta_2) < \theta_3 < 29/480),$
 $\text{or } (\theta_1 = 91/240 \text{ and } 89/1440 < \theta_2 < 47/720 \text{ and } 1/240(29 - 240\theta_2) < \theta_3 < (323 - 2880\theta_2)/2400),$
 $\text{or } (\theta_1 = 91/240 \text{ and } 47/720 < \theta_2 < 11/160 \text{ and } 1/240(29 - 240\theta_2) < \theta_3 < (323 - 2880\theta_2)/2400),$
 $\text{or } (\theta_1 = 19/50 \text{ and } 3/50 < \theta_2 \leq 37/600 \text{ and } 1/25(3 - 25\theta_2) < \theta_3 < 3/50),$
 $\text{or } (\theta_1 = 19/50 \text{ and } 37/600 < \theta_2 < 29/450 \text{ and } 1/25(3 - 25\theta_2) < \theta_3 < 1/500(67 - 600\theta_2)),$
 $\text{or } (\theta_1 = 19/50 \text{ and } 29/450 < \theta_2 < 193/2800 \text{ and } 1/25(3 - 25\theta_2) < \theta_3 < 1/500(67 - 600\theta_2)),$
 $\text{or } (\theta_1 = 7/18 \text{ and } 13/216 < \theta_2 < 11/180 \text{ and } 1/9(1 - 9\theta_2) < \theta_3 < 1/180(23 - 216\theta_2)),$
 $\text{or } (\theta_1 = 7/18 \text{ and } 11/180 < \theta_2 < 65/1008 \text{ and } 1/9(1 - 9\theta_2) < \theta_3 < 1/180(23 - 216\theta_2)),$
 $\text{or } (\theta_1 = 677/1800 \text{ and } 223/3600 < \theta_2 \leq 673/10800 \text{ and } (223 - 1800\theta_2)/1800 < \theta_3 < 223/3600),$
 $\text{or } (\theta_1 = 677/1800 \text{ and } 673/10800 < \theta_2 < 77/1200 \text{ and } (223 - 1800\theta_2)/1800 < \theta_3 < (2461 - 21600\theta_2)/18000),$
 $\text{or } (\theta_1 = 2/5 \text{ and } 13/175 < \theta_2 < 3/40 \text{ and } 1/10(1 - 10\theta_2) < \theta_3 < 1/25(3 - 30\theta_2)),$
 $\text{or } (25/63 \leq \theta_1 < 2/5 \text{ and } 1/19(4 - 7\theta_1) < \theta_2 < 1/28(5 - 8\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (137/360 < \theta_1 \leq 807/2120 \text{ and } 1/19(4 - 7\theta_1) < \theta_2 < 17/240 \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (807/2120 < \theta_1 < 55/144 \text{ and } 1/19(4 - 7\theta_1) < \theta_2 < 1/140(-13 + 60\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (55/144 \leq \theta_1 < 17/44 \text{ and } 1/19(4 - 7\theta_1) < \theta_2 < 1/10(3 - 6\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (17/44 < \theta_1 < 7/18 \text{ and } 1/19(4 - 7\theta_1) < \theta_2 < 1/28(5 - 8\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (71/180 < \theta_1 \leq 36/91 \text{ and } 1/19(4 - 7\theta_1) < \theta_2 < 1/28(5 - 8\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (36/91 < \theta_1 < 25/63 \text{ and } 1/19(4 - 7\theta_1) < \theta_2 < 1/28(5 - 8\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (7/18 < \theta_1 < 71/180 \text{ and } 1/19(4 - 7\theta_1) < \theta_2 < 1/28(5 - 8\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (27/71 < \theta_1 \leq 137/360 \text{ and } 1/19(4 - 7\theta_1) < \theta_2 < 1/2(-1 + 3\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (25/63 \leq \theta_1 < 2/5 \text{ and } 1/28(5 - 8\theta_1) < \theta_2 < 1/3(1 - 2\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (5/13 < \theta_1 < 17/44 \text{ and } 1/10(3 - 6\theta_1) \leq \theta_2 < 1/3(1 - 2\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (17/44 < \theta_1 < 7/18 \text{ and } 1/28(5 - 8\theta_1) < \theta_2 < 1/3(1 - 2\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (71/180 < \theta_1 \leq 36/91 \text{ and } 1/28(5 - 8\theta_1) < \theta_2 < 1/3(1$

[illegible]

$\text{or } (\theta_1 = 71/180 \text{ and } 223/3420 < \theta_2 < 83/1260 \text{ and } 1/180(19 - 180\theta_2) < \theta_3 < (223 - 2160\theta_2)/1800),$
 $\text{or } (\theta_1 = 71/180 \text{ and } 83/1260 < \theta_2 < 19/270 \text{ and } 1/180(19 - 180\theta_2) < \theta_3 < (223 - 2160\theta_2)/1800),$
 $\text{or } (\theta_1 = 71/180 \text{ and } 19/270 \leq \theta_2 < 13/180 \text{ and } 1/180(19 - 180\theta_2) < \theta_3 < (223 - 2160\theta_2)/1800),$
 $\text{or } (\theta_1 = 71/180 \text{ and } 13/180 < \theta_2 < 223/2880 \text{ and } 1/180(19 - 180\theta_2) < \theta_3 < (223 - 2160\theta_2)/1800),$
 $\text{or } (\theta_1 = 71/180 \text{ and } 223/2880 \leq \theta_2 < 7/90 \text{ and } 1/180(19 - 180\theta_2) < \theta_3 < (223 - 2160\theta_2)/1800),$
 $\text{or } (\theta_1 = 19/50 \text{ and } 193/2800 < \theta_2 < 7/100 \text{ and } 1/25(3 - 25\theta_2) < \theta_3 < 1/500(67 - 600\theta_2)),$
 $\text{or } (\theta_1 = 7/18 \text{ and } 17/252 < \theta_2 < 2/27 \text{ and } 1/9(1 - 9\theta_2) < \theta_3 < 1/180(23 - 216\theta_2)),$
 $\text{or } (\theta_1 = 7/18 \text{ and } 65/1008 < \theta_2 < 23/342 \text{ and } 1/9(1 - 9\theta_2) < \theta_3 < 1/180(23 - 216\theta_2)),$
 $\text{or } (\theta_1 = 7/18 \text{ and } 23/342 < \theta_2 < 17/252 \text{ and } 1/9(1 - 9\theta_2) < \theta_3 < 1/180(23 - 216\theta_2)),$
 $\text{or } (\theta_1 = 7/18 \text{ and } 2/27 \leq \theta_2 < 31/414 \text{ and } 1/9(1 - 9\theta_2) < \theta_3 < 1/180(23 - 216\theta_2)),$
 $\text{or } (\theta_1 = 7/18 \text{ and } 31/414 < \theta_2 < 23/288 \text{ and } 1/9(1 - 9\theta_2) < \theta_3 < 1/180(23 - 216\theta_2)),$
 $\text{or } (\theta_1 = 7/18 \text{ and } 23/288 \leq \theta_2 < 29/360 \text{ and } 1/9(1 - 9\theta_2) < \theta_3 < 1/180(23 - 216\theta_2)),$
 $\text{or } (\theta_1 = 2/5 \text{ and } 3/40 \leq \theta_2 < 1/10 \text{ and } 1/10(1 - 10\theta_2) < \theta_3 < 1/25(3 - 30\theta_2)),$
 $\text{or } (25/63 \leq \theta_1 < 2/5 \text{ and } 1/40(11 - 20\theta_1) \leq \theta_2 < 1/2(-1 + 3\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (\theta_1 = 71/180 \text{ and } 1279/16380 < \theta_2 < 11/120 \text{ and } 1/180(19 - 180\theta_2) < \theta_3 < (223 - 2160\theta_2)/1800),$
 $\text{or } (31/80 < \theta_1 < 7/18 \text{ and } 1/40(11 - 20\theta_1) \leq \theta_2 < 1/2(-1 + 3\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (71/180 < \theta_1 \leq 719/1820 \text{ and } 1/91(43 - 91\theta_1) < \theta_2 < 1/2(-1 + 3\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (719/1820 < \theta_1 \leq 36/91 \text{ and } 1/40(11 - 20\theta_1) \leq \theta_2 < 1/2(-1 + 3\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (71/180 < \theta_1 < 719/1820 \text{ and } 1/40(11 - 20\theta_1) \leq \theta_2 < 1/91(43 - 91\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (36/91 < \theta_1 < 25/63 \text{ and } 1/40(11 - 20\theta_1) \leq \theta_2 < 1/2(-1 + 3\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (7/18 < \theta_1 \leq 177/455 \text{ and } 1/40(11 - 20\theta_1) \leq \theta_2 < 1/2(-1 + 3\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (177/455 < \theta_1 < 71/180 \text{ and } 1/40(11 - 20\theta_1) \leq \theta_2 < 1/91(43 - 91\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (177/455 < \theta_1 < 71/180 \text{ and } 1/91(43 - 91\theta_1) < \theta_2 < 1/2(-1 + 3\theta_1) \text{ and } 1/2(1 - 2\theta_1 - 2\theta_2) < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (\theta_1 = 71/180 \text{ and } 7/90 \leq \theta_2 < 1279/16380 \text{ and } 1/180(19 - 180\theta_2) < \theta_3 < (223 - 2160\theta_2)/1800),$
 $\text{or } (\theta_1 = 7/18 \text{ and } 29/360 \leq \theta_2 < 1/12 \text{ and } 1/9(1 - 9\theta_2) < \theta_3 < 1/180(23 - 216\theta_2)),$
 $\text{or } (117/280 < \theta_1 < 21/50 \text{ and } \theta_2 = 1/112(29 - 56\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (21/50 \leq \theta_1 < 143/340 \text{ and } \theta_2 = 1/112(29 - 56\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (143/340 < \theta_1 < 53/126 \text{ and } \theta_2 = 1/112(29 - 56\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (53/126 < \theta_1 < 59/140 \text{ and } \theta_2 = 1/112(29 - 56\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (37/90 < \theta_1 < 7/17 \text{ and } \theta_2 = 1/112(29 - 56\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (7/17 < \theta_1 \leq 12/29 \text{ and } \theta_2 = 1/112(29 - 56\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (272/679 < \theta_1 \leq 45/112 \text{ and } \theta_2 = 1/112(29 - 56\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (205/504 < \theta_1 < 139/340 \text{ and } \theta_2 = 1/112(29 - 56\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (139/340 < \theta_1 < 37/90 \text{ and } \theta_2 = 1/112(29 - 56\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (12/29 < \theta_1 < 127/306 \text{ and } \theta_2 = 1/112(29 - 56\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \$

[illegible]

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or $(31/70 < \theta_1 < 4/9$ and $\theta_2 = 1/23(6 - 11\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(31/70 < \theta_1 < 4/9$ and $\theta_2 = 1/35(11 - 21\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(75/172 < \theta_1 < 43/98$ and $\theta_2 = 1/23(6 - 11\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(43/98 < \theta_1 < 97/220$ and $\theta_2 = 1/23(6 - 11\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(97/220 < \theta_1 < 15/34$ and $\theta_2 = 1/23(6 - 11\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(15/34 < \theta_1 < 4/9$ and $\theta_2 = 1/23(6 - 11\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(159/364 < \theta_1 < 43/98$ and $\theta_2 = 1/35(11 - 21\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(43/98 < \theta_1 < 37/84$ and $\theta_2 = 1/35(11 - 21\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(37/84 < \theta_1 < 15/34$ and $\theta_2 = 1/35(11 - 21\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(15/34 < \theta_1 < 4/9$ and $\theta_2 = 1/35(11 - 21\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(61/140 < \theta_1 < 43/98$ and $1/19(4 - 7\theta_1) < \theta_2 < 1/20$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(43/98 < \theta_1 \leq 37/84$ and $1/19(4 - 7\theta_1) < \theta_2 < 1/20$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(37/84 < \theta_1 < 15/34$ and $1/19(4 - 7\theta_1) < \theta_2 < 1/35(11 - 21\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(15/34 < \theta_1 < 4/9$ and $1/19(4 - 7\theta_1) < \theta_2 < 1/35(11 - 21\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(3/7 < \theta_1 \leq 85/198$ and $1/8(3 - 6\theta_1) < \theta_2 < 1/24(3 - 4\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(85/198 < \theta_1 < 73/170$ and $1/8(3 - 6\theta_1) < \theta_2 < 1/24(3 - 4\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(73/170 \leq \theta_1 < 31/72$ and $1/8(3 - 6\theta_1) < \theta_2 < 1/24(3 - 4\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(31/72 \leq \theta_1 \leq 25/58$ and $1/8(3 - 6\theta_1) < \theta_2 < 1/24(3 - 4\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(25/58 < \theta_1 < 61/140$ and $1/19(4 - 7\theta_1) < \theta_2 < 1/24(3 - 4\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(90/211 < \theta_1 \leq 53/124$ and $\theta_2 = 1/23(6 - 11\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(85/198 < \theta_1 < 73/170$ and $\theta_2 = 1/23(6 - 11\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(73/170 \leq \theta_1 < 31/72$ and $\theta_2 = 1/23(6 - 11\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(31/72 \leq \theta_1 < 61/140$ and $\theta_2 = 1/23(6 - 11\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(61/140 < \theta_1 \leq 75/172$ and $\theta_2 = 1/23(6 - 11\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(53/124 < \theta_1 < 85/198$ and $\theta_2 = 1/23(6 - 11\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(122/287 < \theta_1 < 73/170$ and $\theta_2 = 1/35(11 - 21\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(73/170 \leq \theta_1 < 31/72$ and $\theta_2 = 1/35(11 - 21\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(31/72 \leq \theta_1 < 163/378$ and $\theta_2 = 1/35(11 - 21\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(19/44 < \theta_1 < 61/140$ and $\theta_2 = 1/35(11 - 21\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(61/140 < \theta_1 \leq 159/364$ and $\theta_2 = 1/35(11 - 21\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(163/378 < \theta_1 \leq 19/44$ and $\theta_2 = 1/35(11 - 21\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(\theta_1 = 31/70$ and $9/190 < \theta_2 < 17/350$ and $1/40(3 - 40\theta_2) \leq \theta_3 < 1/35)$,
 or $(\theta_1 = 31/70$ and $13/280 < \theta_2 < 9/190$ and $1/40(3 - 40\theta_2) \leq \theta_3 < 1/35)$,
 or $(\theta_1 = 31/70$ and $1/20 < \theta_2 < 43/840$ and $1/40(3 - 40\theta_2) \leq \theta_3 < 1/35)$,
 or $(\theta_1 = 31/70$ and $51/980 \leq \theta_2 < 3/56$ and $1/40(3 - 40\theta_2) \leq \theta_3 < 1/10(9/10 - 12\theta_2))$,
 or $(\theta_1 = 31/70$ and $79/1610 < \theta_2 < 1/20$ and $1/40(3 - 40\theta_2) \leq \theta_3 < 1/35)$,
 or $(\theta_1 = 31/70$ and $17/350 < \theta_2 < 79/1610$ and $1/40(3 - 40\theta_2) \leq \theta$

or $(822/1955 < \theta_1 < 143/340$ and $\theta_2 = 1/238(-27 + 102\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(143/340 < \theta_1 < 53/126$ and $\theta_2 = 1/238(-27 + 102\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(53/126 < \theta_1 < 11/26$ and $\theta_2 = 1/238(-27 + 102\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(11/26 < \theta_1 < 17/40$ and $\theta_2 = 1/238(-27 + 102\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(17/40 < \theta_1 < 73/170$ and $\theta_2 = 1/238(-27 + 102\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(73/170 \leq \theta_1 < 31/72$ and $1/16(4 - 7\theta_1) \leq \theta_2 < 1/2(1 - 2\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(31/72 \leq \theta_1 < 61/140$ and $1/16(4 - 7\theta_1) \leq \theta_2 < 1/2(1 - 2\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(61/140 < \theta_1 < 43/98$ and $1/16(4 - 7\theta_1) \leq \theta_2 < 1/2(1 - 2\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(43/98 < \theta_1 < 15/34$ and $1/16(4 - 7\theta_1) \leq \theta_2 < 1/2(1 - 2\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(15/34 < \theta_1 < 4/9$ and $1/16(4 - 7\theta_1) \leq \theta_2 < 1/2(1 - 2\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(12/29 < \theta_1 < 127/306$ and $1/8(6 - 13\theta_1) < \theta_2 < 1/2(1 - 2\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(5/12 < \theta_1 \leq 58/139$ and $1/8(6 - 13\theta_1) < \theta_2 < 1/2(1 - 2\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(58/139 < \theta_1 < 21/50$ and $1/7(-2 + 6\theta_1) < \theta_2 < 1/2(1 - 2\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(21/50 \leq \theta_1 < 143/340$ and $1/7(-2 + 6\theta_1) < \theta_2 < 1/2(1 - 2\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(143/340 < \theta_1 < 53/126$ and $1/7(-2 + 6\theta_1) < \theta_2 < 1/2(1 - 2\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(53/126 < \theta_1 < 11/26$ and $1/7(-2 + 6\theta_1) < \theta_2 < 1/2(1 - 2\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(11/26 < \theta_1 < 17/40$ and $1/238(-27 + 102\theta_1) < \theta_2 < 1/2(1 - 2\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(17/40 < \theta_1 < 73/170$ and $1/238(-27 + 102\theta_1) < \theta_2 < 1/2(1 - 2\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(2/5 < \theta_1 \leq 272/679$ and $1/40(11 - 20\theta_1) \leq \theta_2 < 1/16(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(127/306 < \theta_1 < 5/12$ and $\theta_2 = -2 + 5\theta_1$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(12/29 < \theta_1 < 127/306$ and $\theta_2 = -2 + 5\theta_1$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(1849/4420 \leq \theta_1 < 692/1649$ and $\theta_2 = 1/238(-27 + 102\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(692/1649 \leq \theta_1 < 21/50$ and $\theta_2 = 1/238(-27 + 102\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(21/50 \leq \theta_1 < 143/340$ and $\theta_2 = 1/238(-27 + 102\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(143/340 < \theta_1 < 53/126$ and $\theta_2 = 1/238(-27 + 102\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(53/126 < \theta_1 < 11/26$ and $\theta_2 = 1/238(-27 + 102\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(11/26 < \theta_1 < 17/40$ and $\theta_2 = 1/238(-27 + 102\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(17/40 < \theta_1 < 73/170$ and $\theta_2 = 1/238(-27 + 102\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(\theta_1 = 31/70$ and $9/160 \leq \theta_2 < 2/35$ and $1/40(3 - 40\theta_2) \leq \theta_3 < 1/10(9/10 - 12\theta_2)$),
 or $(\theta_1 = 31/70$ and $3/56 \leq \theta_2 < 1/18$ and $1/40(3 - 40\theta_2) \leq \theta_3 < 1/10(9/10 - 12\theta_2)$),
 or $(\theta_1 = 31/70$ and $1/18 < \theta_2 < 9/160$ and $1/40(3 - 40\theta_2) \leq \theta_3 < 1/10(9/10 - 12\theta_2)$),
 or $(37/90 < \theta_1 < 7/17$ and $\theta_2 = 1/20$ and $1/12(17/5 - 7\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
 or $(\theta_1 = 7/17$ and $1/20 < \theta_2 < 101/1904$ and $1/12(19/17 - 12\theta_2) \leq \theta_3 < 3/68$),
 or $(\theta_1 = 7/17$ and $5/102 < \theta_2 < 1/20$ and $1/12(19/17 - 12\theta_2) \leq \theta_3 < 3/68$),
 or $(7/17 < \theta_1 \leq 12/29$ and $\theta_2 = 1/20$ and $1/12(17/5 - 7\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
 or $(\theta_1 = 5/12$ and $7/144 < \theta_2 < 1/20$ and $1/12(13/12 - 12\theta_2) \leq \theta_3 < 1/24$),
 or $(\theta_1 = 5/12$ and $1/20 < \theta_2 < 17/336$ and $1/12(13/12 - 12\theta_2) \leq \theta_3 < 1/24$),
 or $(\theta_1 = 13/32$ and $1/20 < \theta_2 < 1/18$ and $1/12(37/32 - 12\theta_2) \leq \theta_3 < 3/64$),
 or $(\theta_1 = 139/340$ and $1/20 < \theta_2 < 519/9520$ and $1/12(387/340 - 12\theta_2) \leq \theta_3 < 31/680$),
 or $(\theta_1 = 37/90$ and $1/20 < \theta_2 < 269/5040$ and $1/12(101/90 - 12\theta_2) \leq \theta_3 < 2/45$),
 or $(\theta_1 = 127/306$ and $1/20 < \theta_2 < 881/17136$ and $1/12(335/306 - 12\theta_2) \leq \theta_3 < 13/306$),
 or $(272/679 < \theta_1 < 13/32$ and $\theta_2 = 1/18$ and $1/12(10/3 - 7\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
 or $(13/32 < \theta_1 < 205/504$ and $\theta_2 = 1/18$ and $1/12(10/3 - 7\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
 or $(127/306 < \theta_1 < 5/12$ and $\theta_2 = 1/20$ and $1/12(17/5 - 7\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
 or $(272/679 < \theta_1 < 13/32$ and $\theta_2 = 1/20$ and $1/12(17/5 - 7\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
 or $(13/32 < \theta_1 < 139/340$ and $\theta_2 = 1/20$ and $1/12(17/5 - 7\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
 or $(139/340 < \theta_1 < 37/90$ and $\theta_2 = 1/20$ and $1/12(17/5 - 7\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
 or $(12/29 < \theta_1 < 127/306$ and $\theta_2 = 1/20$ and $1/12(17/5 - 7\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
 or $(5/12 < \theta_1 < 117/280$ and $\theta_2 = 1/20$ and $1/12(17/5 - 7\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1)$),

or $(\theta_1 = 13/32 \text{ and } 19/384 < \theta_2 < 1/20 \text{ and } 1/12(37/32 - 12\theta_2) \leq \theta_3 < 3/64)$,
 or $(\theta_1 = 139/340 \text{ and } 67/1360 < \theta_2 < 1/20 \text{ and } 1/12(387/340 - 12\theta_2) \leq \theta_3 < 31/680)$,
 or $(\theta_1 = 37/90 \text{ and } 53/1080 < \theta_2 < 1/20 \text{ and } 1/12(101/90 - 12\theta_2) \leq \theta_3 < 2/45)$,
 or $(\theta_1 = 127/306 \text{ and } 179/3672 < \theta_2 < 1/20 \text{ and } 1/12(335/306 - 12\theta_2) \leq \theta_3 < 13/306)$,
 or $(\theta_1 = 143/340 \text{ and } 197/4080 < \theta_2 < 463/9520 \text{ and } 1/12(359/340 - 12\theta_2) \leq \theta_3 < 27/680)$,
 or $(\theta_1 = 13/32 \text{ and } 1/18 < \theta_2 < 25/448 \text{ and } 1/12(37/32 - 12\theta_2) \leq \theta_3 < 3/64)$,
 or $(\theta_1 = 53/126 \text{ and } 73/1512 < \theta_2 < 7/144 \text{ and } 1/12(19/18 - 12\theta_2) \leq \theta_3 < 5/126)$,
 or $(37/90 < \theta_1 < 7/17 \text{ and } \theta_2 = 1/18 \text{ and } 1/12(10/3 - 7\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(\theta_1 = 7/17 \text{ and } 1/17 < \theta_2 < 29/476 \text{ and } 1/12(19/17 - 12\theta_2) \leq \theta_3 < 1/170(19 - 204\theta_2))$,
 or $(\theta_1 = 7/17 \text{ and } 23/408 \leq \theta_2 < 1/17 \text{ and } 1/12(19/17 - 12\theta_2) \leq \theta_3 < 1/170(19 - 204\theta_2))$,
 or $(\theta_1 = 7/17 \text{ and } 101/1904 < \theta_2 < 1/18 \text{ and } 1/12(19/17 - 12\theta_2) \leq \theta_3 < 3/68)$,
 or $(\theta_1 = 7/17 \text{ and } 1/18 < \theta_2 < 23/408 \text{ and } 1/12(19/17 - 12\theta_2) \leq \theta_3 < 3/68)$,
 or $(\theta_1 = 7/17 \text{ and } 8/119 < \theta_2 < 47/680 \text{ and } 1/12(19/17 - 12\theta_2) \leq \theta_3 < 1/170(19 - 204\theta_2))$,
 or $(\theta_1 = 7/17 \text{ and } 29/476 \leq \theta_2 < 15/238 \text{ and } 1/12(19/17 - 12\theta_2) \leq \theta_3 < 1/170(19 - 204\theta_2))$,
 or $(\theta_1 = 7/17 \text{ and } 15/238 < \theta_2 < 25/391 \text{ and } 1/12(19/17 - 12\theta_2) \leq \theta_3 < 1/170(19 - 204\theta_2))$,
 or $(\theta_1 = 7/17 \text{ and } 25/391 < \theta_2 < 8/119 \text{ and } 1/12(19/17 - 12\theta_2) \leq \theta_3 < 1/170(19 - 204\theta_2))$,
 or $(7/17 < \theta_1 \leq 12/29 \text{ and } \theta_2 = 1/18 \text{ and } 1/12(10/3 - 7\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(\theta_1 = 5/12 \text{ and } 17/336 < \theta_2 < 11/204 \text{ and } 1/12(13/12 - 12\theta_2) \leq \theta_3 < 1/24)$,
 or $(\theta_1 = 5/12 \text{ and } 11/204 < \theta_2 < 1/18 \text{ and } 1/12(13/12 - 12\theta_2) \leq \theta_3 < 1/24)$,
 or $(\theta_1 = 53/126 \text{ and } 13/210 < \theta_2 < 163/2520 \text{ and } 1/12(19/18 - 12\theta_2) \leq \theta_3 < 1/180(19 - 216\theta_2))$,
 or $(\theta_1 = 13/32 \text{ and } 37/608 < \theta_2 < 1/16 \text{ and } 1/12(37/32 - 12\theta_2) \leq \theta_3 < 1/320(37 - 384\theta_2))$,
 or $(\theta_1 = 43/98 \text{ and } 1/20 < \theta_2 < 5/98 \text{ and } 1/12(13/14 - 12\theta_2) \leq \theta_3 < 3/98)$,
 or $(\theta_1 = 139/340 \text{ and } 519/9520 < \theta_2 < 1/18 \text{ and } 1/12(387/340 - 12\theta_2) \leq \theta_3 < 31/680)$,
 or $(\theta_1 = 37/90 \text{ and } 269/5040 < \theta_2 < 1/18 \text{ and } 1/12(101/90 - 12\theta_2) \leq \theta_3 < 2/45)$,
 or $(\theta_1 = 127/306 \text{ and } 881/17136 < \theta_2 < 1/18 \text{ and } 1/12(335/306 - 12\theta_2) \leq \theta_3 < 13/306)$,
 or $(\theta_1 = 11/26 \text{ and } 1/20 < \theta_2 < 2/39 \text{ and } 1/12(27/26 - 12\theta_2) \leq \theta_3 < 1/26)$,
 or $(\theta_1 = 61/140 \text{ and } 1/20 < \theta_2 < 11/210 \text{ and } 1/12(19/20 - 12\theta_2) \leq \theta_3 < 9/280)$,
 or $(\theta_1 = 15/34 \text{ and } 1/20 < \theta_2 < 7/136 \text{ and } 1/12(31/34 - 12\theta_2) \leq \theta_3 < 1/34)$,
 or $(\theta_1 = 143/340 \text{ and } 1/20 < \theta_2 < 9/170 \text{ and } 1/12(359/340 - 12\theta_2) \leq \theta_3 < 27/680)$,
 or $(\theta_1 = 53/126 \text{ and } 1/20 < \theta_2 < 10/189 \text{ and } 1/12(19/18 - 12\theta_2) \leq \theta_3 < 5/126)$,
 or $(\theta_1 = 17/40 \text{ and } 1/20 < \theta_2 < 41/760 \text{ and } 1/12(41/40 - 12\theta_2) \leq \theta_3 < 3/80)$,
 or $(\theta_1 = 11/26 \text{ and } 27/494 < \theta_2 < 1/18 \text{ and } 1/12(27/26 - 12\theta_2) \leq \theta_3 < 1/260(27 - 312\theta_2))$,
 or $(\theta_1 = 61/140 \text{ and } 53/980 \leq \theta_2 < 1/18 \text{ and } 1/12(19/20 - 12\theta_2) \leq \theta_3 < 1/200(19 - 240\theta_2))$,
 or $(\theta_1 = 15/34 \text{ and } 25/476 \leq \theta_2 < 37/680 \text{ and } 1/12(31/34 - 12\theta_2) \leq \theta_3 < 1/340(31 - 408\theta_2))$,
 or $(\theta_1 = 43/98 \text{ and } 73/1372 \leq \theta_2 < 1/18 \text{ and } 1/12(13/14 - 12\theta_2) \leq \theta_3 < 1/140(13 - 168\theta_2))$,
 or $(\theta_1 = 143/340 \text{ and } 14/255 \leq \theta_2 < 1/18 \text{ and } 1/12(359/340 - 12\theta_2) \leq \theta_3 < (359 - 4080\theta_2)/3400)$,
 or $(\theta_1 = 53/126 \text{ and } 83/1512 \leq \theta_2 < 1/18 \text{ and } 1/12(19/18 - 12\theta_2) \leq \theta_3 < 1/180(19 - 216\theta_2))$,
 or $(\theta_1 = 17/40 \text{ and } 13/240 \leq \theta_2 < 1/18 \text{ and } 1/12(41/40 - 12\theta_2) \leq \theta_3 < 1/400(41 - 480\theta_2))$,
 or $(127/306 < \theta_1 < 5/12 \text{ and } \theta_2 = 1/18 \text{ and } 1/12(10/3 - 7\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(205/504 < \theta_1 < 139/340 \text{ and } \theta_2 = 1/18 \text{ and } 1/12(10/3 - 7\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(139/340 < \theta_1 < 37/90 \text{ and } \theta_2 = 1/18 \text{ and } 1/12(10/3 - 7\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(12/29 < \theta_1 < 127/306 \text{ and } \theta_2 = 1/18 \text{ and } 1/12(10/3 - 7\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(5/12 < \theta_1 < 21/50 \text{ and } \theta_2 = 1/18 \text{ and } 1/12(10/3 - 7\theta_1) \leq \theta_3 < 1/30(10 - 21\theta_1))$,
 or $(21/50 \leq \theta_1 < 143/340 \text{ and } \theta_2 = 1/18 \text{ and } 1/12(10/3 - 7\theta_1) \leq \theta_3 < 1/30(10 - 21\theta_1))$,
 or $(143/340 < \theta_1 < 53/126 \text{ and } \theta_2 = 1/18 \text{ and } 1/12(10/3 - 7\theta_1) \leq \theta_3 < 1/30(10 - 21\theta_1))$,
 or $(53/126 < \theta_1 < 11/26 \text{ and } \theta_2 = 1/18 \text{ and } 1/12(10/3 - 7\theta_1) \leq \theta_3 < 1/30(10 - 21\theta_1))$,
 or $(11/26 < \theta_1 < 17/40 \text{ and } \theta_2 = 1/18 \text{ and } 1/12(10/3 - 7\theta_1) \leq \theta_3 < 1/30(10 - 21\theta_1))$,
 or $(17/40 < \theta_1 < 85/198 \text{ and } \theta_2 = 1/18 \text{ and } 1/12(10/3 - 7\theta_1) \leq \theta_3 < 1/30(10 - 21\theta_1))$,
 or $(85/198 < \theta_1 < 73/170 \text{ and } \theta_2 = 1/18 \text{ and } 1/12(10/3 - 7\theta_1) \leq \theta_3 < 1/30(10 - 21\theta_1))$,

or $(73/170 \leq \theta_1 < 31/72$ and $\theta_2 = 1/18$ and $1/12(10/3 - 7\theta_1) \leq \theta_3 < 1/30(10 - 21\theta_1))$,
 or $(31/72 \leq \theta_1 < 163/378$ and $\theta_2 = 1/18$ and $1/12(10/3 - 7\theta_1) \leq \theta_3 < 1/30(10 - 21\theta_1))$,
 or $(163/378 < \theta_1 < 61/140$ and $\theta_2 = 1/18$ and $1/12(10/3 - 7\theta_1) \leq \theta_3 < 1/30(10 - 21\theta_1))$,
 or $(61/140 < \theta_1 < 43/98$ and $\theta_2 = 1/18$ and $1/12(10/3 - 7\theta_1) \leq \theta_3 < 1/30(10 - 21\theta_1))$,
 or $(43/98 < \theta_1 < 79/180$ and $\theta_2 = 1/18$ and $1/12(10/3 - 7\theta_1) \leq \theta_3 < 1/30(10 - 21\theta_1))$,
 or $(117/280 < \theta_1 < 21/50$ and $\theta_2 = 1/20$ and $1/12(17/5 - 7\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(21/50 \leq \theta_1 < 143/340$ and $\theta_2 = 1/20$ and $1/12(17/5 - 7\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(143/340 < \theta_1 < 53/126$ and $\theta_2 = 1/20$ and $1/12(17/5 - 7\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(53/126 < \theta_1 < 11/26$ and $\theta_2 = 1/20$ and $1/12(17/5 - 7\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(11/26 < \theta_1 < 17/40$ and $\theta_2 = 1/20$ and $1/12(17/5 - 7\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(17/40 < \theta_1 < 73/170$ and $\theta_2 = 1/20$ and $1/12(17/5 - 7\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(73/170 \leq \theta_1 < 31/72$ and $\theta_2 = 1/20$ and $1/12(17/5 - 7\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(31/72 \leq \theta_1 < 61/140$ and $\theta_2 = 1/20$ and $1/12(17/5 - 7\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(61/140 < \theta_1 < 43/98$ and $\theta_2 = 1/20$ and $1/12(17/5 - 7\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(43/98 < \theta_1 < 37/84$ and $\theta_2 = 1/20$ and $1/12(17/5 - 7\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(97/220 < \theta_1 < 15/34$ and $\theta_2 = 1/20$ and $1/12(17/5 - 7\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(15/34 < \theta_1 < 31/70$ and $\theta_2 = 1/20$ and $1/12(17/5 - 7\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(37/84 < \theta_1 < 97/220$ and $\theta_2 = 1/20$ and $1/12(17/5 - 7\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(\theta_1 = 11/26$ and $5/104 < \theta_2 < 1/20$ and $1/12(27/26 - 12\theta_2) \leq \theta_3 < 1/26)$,
 or $(\theta_1 = 61/140$ and $79/1680 < \theta_2 < 1/20$ and $1/12(19/20 - 12\theta_2) \leq \theta_3 < 9/280)$,
 or $(\theta_1 = 15/34$ and $39/782 < \theta_2 < 1/20$ and $1/12(31/34 - 12\theta_2) \leq \theta_3 < 1/34)$,
 or $(\theta_1 = 43/98$ and $13/266 < \theta_2 < 1/20$ and $1/12(13/14 - 12\theta_2) \leq \theta_3 < 3/98)$,
 or $(\theta_1 = 143/340$ and $463/9520 < \theta_2 < 1/20$ and $1/12(359/340 - 12\theta_2) \leq \theta_3 < 27/680)$,
 or $(\theta_1 = 53/126$ and $107/2142 < \theta_2 < 1/20$ and $1/12(19/18 - 12\theta_2) \leq \theta_3 < 5/126)$,
 or $(\theta_1 = 17/40$ and $23/480 < \theta_2 < 1/20$ and $1/12(41/40 - 12\theta_2) \leq \theta_3 < 3/80)$,
 or $(\theta_1 = 5/12$ and $1/18 < \theta_2 < 13/228$ and $1/12(13/12 - 12\theta_2) \leq \theta_3 < 1/120(13 - 144\theta_2))$,
 or $(\theta_1 = 5/12$ and $17/276 < \theta_2 < 9/140$ and $1/12(13/12 - 12\theta_2) \leq \theta_3 < 1/120(13 - 144\theta_2))$,
 or $(\theta_1 = 5/12$ and $9/140 < \theta_2 < 31/476$ and $1/12(13/12 - 12\theta_2) \leq \theta_3 < 1/120(13 - 144\theta_2))$,
 or $(\theta_1 = 5/12$ and $13/228 < \theta_2 < 5/84$ and $1/12(13/12 - 12\theta_2) \leq \theta_3 < 1/120(13 - 144\theta_2))$,
 or $(\theta_1 = 5/12$ and $5/84 \leq \theta_2 < 17/276$ and $1/12(13/12 - 12\theta_2) \leq \theta_3 < 1/120(13 - 144\theta_2))$,
 or $(\theta_1 = 5/12$ and $31/476 < \theta_2 < 1/15$ and $1/12(13/12 - 12\theta_2) \leq \theta_3 < 1/120(13 - 144\theta_2))$,
 or $(\theta_1 = 11/26$ and $1/18 < \theta_2 < 3/52$ and $1/12(27/26 - 12\theta_2) \leq \theta_3 < 1/260(27 - 312\theta_2))$,
 or $(\theta_1 = 11/26$ and $2/39 < \theta_2 < 17/312$ and $1/12(27/26 - 12\theta_2) \leq \theta_3 < 1/26)$,
 or $(\theta_1 = 11/26$ and $17/312 \leq \theta_2 < 27/494$ and $1/12(27/26 - 12\theta_2) \leq \theta_3 < 1/260(27 - 312\theta_2))$,
 or $(\theta_1 = 11/26$ and $3/52 \leq \theta_2 < 35/598$ and $1/12(27/26 - 12\theta_2) \leq \theta_3 < 1/260(27 - 312\theta_2))$,
 or $(\theta_1 = 11/26$ and $35/598 < \theta_2 < 11/182$ and $1/12(27/26 - 12\theta_2) \leq \theta_3 < 1/260(27 - 312\theta_2))$,
 or $(\theta_1 = 11/26$ and $11/182 < \theta_2 < 33/520$ and $1/12(27/26 - 12\theta_2) \leq \theta_3 < 1/260(27 - 312\theta_2))$,
 or $(\theta_1 = 61/140$ and $1/18 < \theta_2 < 2/35$ and $1/12(19/20 - 12\theta_2) \leq \theta_3 < 1/200(19 - 240\theta_2))$,
 or $(\theta_1 = 61/140$ and $11/210 \leq \theta_2 < 169/3220$ and $1/12(19/20 - 12\theta_2) \leq \theta_3 < 1/200(19 - 240\theta_2))$,
 or $(\theta_1 = 61/140$ and $169/3220 < \theta_2 < 37/700$ and $1/12(19/20 - 12\theta_2) \leq \theta_3 < 1/200(19 - 240\theta_2))$,
 or $(\theta_1 = 61/140$ and $37/700 < \theta_2 < 53/980$ and $1/12(19/20 - 12\theta_2) \leq \theta_3 < 1/200(19 - 240\theta_2))$,
 or $(\theta_1 = 15/34$ and $31/646 < \theta_2 < 59/1190$ and $1/12(31/34 - 12\theta_2) \leq \theta_3 < 1/34)$,
 or $(\theta_1 = 15/34$ and $19/408 < \theta_2 < 31/646$ and $1/12(31/34 - 12\theta_2) \leq \theta_3 < 1/34)$,
 or $(\theta_1 = 15/34$ and $59/1190 < \theta_2 < 39/782$ and $1/12(31/34 - 12\theta_2) \leq \theta_3 < 1/34)$,
 or $(\theta_1 = 15/34$ and $7/136 \leq \theta_2 < 25/476$ and $1/12(31/34 - 12\theta_2) \leq \theta_3 < 1/340(31 - 408\theta_2))$,
 or $(\theta_1 = 43/98$ and $1/18 < \theta_2 < 109/1960$ and $1/12(13/14 - 12\theta_2) \leq \theta_3 < 1/140(13 - 168\theta_2))$,
 or $(\theta_1 = 43/98$ and $55/1176 < \theta_2 < 13/266$ and $1/12(13/14 - 12\theta_2) \leq \theta_3 < 3/98)$,
 or $(\theta_1 = 43/98$ and $5/98 < \theta_2 < 61/1176$ and $1/12(13/14 - 12\theta_2) \leq \theta_3 < 3/98)$,
 or $(\theta_1 = 43/98$ and $61/1176 \leq \theta_2 < 73/1372$ and $1/12(13/14 - 12\theta_2) \leq \theta_3 < 1/140(13 - 168\theta_2))$,

or $(\theta_1 = 13/32 \text{ and } 25/448 < \theta_2 < 11/192 \text{ and } 1/12(37/32 - 12\theta_2) \leq \theta_3 < 3/64)$,
 or $(\theta_1 = 139/340 \text{ and } 1/18 < \theta_2 < 29/510 \text{ and } 1/12(387/340 - 12\theta_2) \leq \theta_3 < 31/680)$,
 or $(\theta_1 = 37/90 \text{ and } 1/18 < \theta_2 < 61/1080 \text{ and } 1/12(101/90 - 12\theta_2) \leq \theta_3 < 2/45)$,
 or $(\theta_1 = 127/306 \text{ and } 1/18 < \theta_2 < 205/3672 \text{ and } 1/12(335/306 - 12\theta_2) \leq \theta_3 < 13/306)$,
 or $(\theta_1 = 143/340 \text{ and } 1/18 < \theta_2 < 359/6460 \text{ and } 1/12(359/340 - 12\theta_2) \leq \theta_3 < (359 - 4080\theta_2)/3400)$,
 or $(\theta_1 = 53/126 \text{ and } 1/18 < \theta_2 < 103/1764 \text{ and } 1/12(19/18 - 12\theta_2) \leq \theta_3 < 1/180(19 - 216\theta_2))$,
 or $(\theta_1 = 17/40 \text{ and } 1/18 < \theta_2 < 2/35 \text{ and } 1/12(41/40 - 12\theta_2) \leq \theta_3 < 1/400(41 - 480\theta_2))$,
 or $(\theta_1 = 13/32 \text{ and } 11/192 \leq \theta_2 < 37/608 \text{ and } 1/12(37/32 - 12\theta_2) \leq \theta_3 < 1/320(37 - 384\theta_2))$,
 or $(\theta_1 = 13/32 \text{ and } 1/16 < \theta_2 < 49/736 \text{ and } 1/12(37/32 - 12\theta_2) \leq \theta_3 < 1/320(37 - 384\theta_2))$,
 or $(\theta_1 = 13/32 \text{ and } 49/736 < \theta_2 < 79/1120 \text{ and } 1/12(37/32 - 12\theta_2) \leq \theta_3 < 1/320(37 - 384\theta_2))$,
 or $(\theta_1 = 13/32 \text{ and } 79/1120 < \theta_2 < 23/320 \text{ and } 1/12(37/32 - 12\theta_2) \leq \theta_3 < 1/320(37 - 384\theta_2))$,
 or $(\theta_1 = 139/340 \text{ and } 29/510 \leq \theta_2 < 387/6460 \text{ and } 1/12(387/340 - 12\theta_2) \leq \theta_3 < (387 - 4080\theta_2)/3400)$,
 or $(\theta_1 = 139/340 \text{ and } 387/6460 < \theta_2 < 31/510 \text{ and } 1/12(387/340 - 12\theta_2) \leq \theta_3 < (387 - 4080\theta_2)/3400)$,
 or $(\theta_1 = 139/340 \text{ and } 31/510 < \theta_2 < 21/340 \text{ and } 1/12(387/340 - 12\theta_2) \leq \theta_3 < (387 - 4080\theta_2)/3400)$,
 or $(\theta_1 = 139/340 \text{ and } 11/170 < \theta_2 < 511/7820 \text{ and } 1/12(387/340 - 12\theta_2) \leq \theta_3 < (387 - 4080\theta_2)/3400)$,
 or $(\theta_1 = 139/340 \text{ and } 511/7820 < \theta_2 < 821/11900 \text{ and } 1/12(387/340 - 12\theta_2) \leq \theta_3 < (387 - 4080\theta_2)/3400)$,
 or $(\theta_1 = 139/340 \text{ and } 821/11900 < \theta_2 < 6/85 \text{ and } 1/12(387/340 - 12\theta_2) \leq \theta_3 < (387 - 4080\theta_2)/3400)$,
 or $(\theta_1 = 139/340 \text{ and } 21/340 < \theta_2 \leq 11/170 \text{ and } 1/12(387/340 - 12\theta_2) \leq \theta_3 < (387 - 4080\theta_2)/3400)$,
 or $(\theta_1 = 37/90 \text{ and } 133/2070 < \theta_2 \leq 1/15 \text{ and } 1/12(101/90 - 12\theta_2) \leq \theta_3 < 1/900(101 - 1080\theta_2))$,
 or $(\theta_1 = 37/90 \text{ and } 1/15 < \theta_2 < 71/1050 \text{ and } 1/12(101/90 - 12\theta_2) \leq \theta_3 < 1/900(101 - 1080\theta_2))$,
 or $(\theta_1 = 37/90 \text{ and } 91/1530 < \theta_2 < 11/180 \text{ and } 1/12(101/90 - 12\theta_2) \leq \theta_3 < 1/900(101 - 1080\theta_2))$,
 or $(\theta_1 = 37/90 \text{ and } 8/135 < \theta_2 < 91/1530 \text{ and } 1/12(101/90 - 12\theta_2) \leq \theta_3 < 1/900(101 - 1080\theta_2))$,
 or $(\theta_1 = 37/90 \text{ and } 61/1080 \leq \theta_2 < 101/1710 \text{ and } 1/12(101/90 - 12\theta_2) \leq \theta_3 < 1/900(101 - 1080\theta_2))$,
 or $(\theta_1 = 37/90 \text{ and } 101/1710 < \theta_2 < 8/135 \text{ and } 1/12(101/90 - 12\theta_2) \leq \theta_3 < 1/900(101 - 1080\theta_2))$,
 or $(\theta_1 = 37/90 \text{ and } 71/1050 < \theta_2 < 5/72 \text{ and } 1/12(101/90 - 12\theta_2) \leq \theta_3 < 1/900(101 - 1080\theta_2))$,
 or $(\theta_1 = 37/90 \text{ and } 11/180 \leq \theta_2 < 16/255 \text{ and } 1/12(101/90 - 12\theta_2) \leq \theta_3 < 1/900(101 - 1080\theta_2))$,
 or $(\theta_1 = 37/90 \text{ and } 16/255 < \theta_2 < 133/2070 \text{ and } 1/12(101/90 - 12\theta_2) \leq \theta_3 < 1/900(101 - 1080\theta_2))$,
 or $(\theta_1 = 127/306 \text{ and } 205/3672 \leq \theta_2 < 26/459 \text{ and } 1/12(335/306 - 12\theta_2) \leq \theta_3 < (335 - 3672\theta_2)/3060)$,
 or $(\theta_1 = 127/306 \text{ and } 26/459 < \theta_2 < 335/5814 \text{ and } 1/12(335/306 - 12\theta_2) \leq \theta_3 < (335 - 3672\theta_2)/3060)$,
 or $(\theta_1 = 127/306 \text{ and } 335/5814 < \theta_2 < 257/4284 \text{ and } 1/12(335/306 - 12\theta_2) \leq \theta_3 < (335 - 3672\theta_2)/3060)$,
 or $(\theta_1 = 127/306 \text{ and } 257/4284 \leq \theta_2 < 439/7038 \text{ and } 1/12(335/306 - 12\theta_2) \leq \theta_3 < (335 - 3672\theta_2)/3060)$,
 or $(\theta_1 = 127/306 \text{ and } 439/7038 < \theta_2 < 23/357 \text{ and } 1/12(335/306 - 12\theta_2) \leq \theta_3 < (335 - 3672\theta_2)/3060)$,
 or $(\theta_1 = 127/306 \text{ and } 23/357 < \theta_2 < 233/3570 \text{ and } 1/12(335/306 - 12\theta_2) \leq \theta_3 < (335 - 3672\theta_2)/3060)$,
 or $(\theta_1 = 127/306 \text{ and } 233/3570 < \theta_2 < 413/6120 \text{ and } 1/12(335/306 - 12\theta_2) \leq \theta_3 < (335 - 3672\theta_2)/3060)$,
 or $(\theta_1 = 143/340 \text{ and } 9/170 < \theta_2 < 14/255 \text{ and } 1/12(359/340 - 12\theta_2) \leq \theta_3 < 27/680)$,
 or $(\theta_1 = 143/340 \text{ and } 359/6460 < \theta_2 < 139/2380 \text{ and } 1/12(359/340 - 12\theta_2) \leq \theta_3 < (359 - 4080\theta_2)/3400)$,
 or $(\theta_1 = 143/340 \text{ and } 139/2380 \leq \theta_2 < 467/7820 \text{ and } 1/12(359/340 - 12\theta_2) \leq \theta_3 < (359 - 4080\theta_2)/3400)$,
 or $(\theta_1 = 143/340 \text{ and } 467/7820 < \theta_2 < 737/11900 \text{ and } 1/12(359/340 - 12\theta_2) \leq \theta_3 < (359 - 4080\theta_2)/3400)$,
 or $(\theta_1 = 143/340 \text{ and } 737/11900 < \theta_2 < 11/170 \text{ and } 1/12(359/340 - 12\theta_2) \leq \theta_3 < (359 - 4080\theta_2)/3400)$,
 or $(\theta_1 = 53/126 \text{ and } 10/189 < \theta_2 < 83/1512 \text{ and } 1/12(19/18 - 12\theta_2) \leq \theta_3 < 5/126)$,
 or $(\theta_1 = 53/126 \text{ and } 7/144 < \theta_2 < 107/2142 \text{ and } 1/12(19/18 - 12\theta_2) \leq \theta_3 < 5/126)$,
 or $(\theta_1 = 53/126 \text{ and } 103/1764 \leq \theta_2 < 173/2898 \text{ and } 1/12(19/18 - 12\theta_2) \leq \theta_3 < 1/180(19 - 216\theta_2))$,
 or $(\theta_1 = 53/126 \text{ and } 173/2898 < \theta_2 < 13/210 \text{ and } 1/12(19/18 - 12\theta_2) \leq \theta_3 < 1/180(19 - 216\theta_2))$,
 or $(\theta_1 = 17/40 \text{ and } 41/760 < \theta_2 < 13/240 \text{ and } 1/12(41/40 - 12\theta_2) \leq \theta_3 < 3/80)$,
 or $(\theta_1 = 17/40 \text{ and } 2/35 \leq \theta_2 < 53/920 \text{ and } 1/12(41/40 - 12\theta_2) \leq \theta_3 < 1/400(41 - 480\theta_2))$,
 or $(\theta_1 = 17/40 \text{ and } 53/920 < \theta_2 < 83/1400 \text{ and } 1/12(41/40 - 12\theta_2) \leq \theta_3 < 1/400(41 - 480\theta_2))$,
 or $(\theta_1 = 17/40 \text{ and } 83/1400 < \theta_2 < 1/16 \text{ and } 1/12(41/40 - 12\theta_2) \leq \theta_3 < 1/400(41 - 480\theta_2))$,
 or $(\theta_1 = 12/29 \text{ and } 67/1015 < \theta_2 < 79/1160 \text{ and } 1/12(32/29 - 12\theta_2) \leq \theta_3 < 1/145(16 - 174\theta_2))$,
 or $(130/289 < \theta_1 < 9/20 \text{ and } 1/28(5 - 8\theta_1) \leq \theta_2 < 1/40(11 - 20\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,

$\text{or } (9/20 < \theta_1 < 73/162 \text{ and } (1 - \theta_1)/12 < \theta_2 < 1/2(1 - 2\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (73/162 \leq \theta_1 < 5/11 \text{ and } (1 - \theta_1)/12 < \theta_2 < 1/2(1 - 2\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (4/9 < \theta_1 < 116/259 \text{ and } 1/28(5 - 8\theta_1) \leq \theta_2 < 1/40(11 - 20\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (116/259 < \theta_1 < 185/413 \text{ and } 1/28(5 - 8\theta_1) \leq \theta_2 < 1/40(11 - 20\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (185/413 \leq \theta_1 < 451/1003 \text{ and } 1/28(5 - 8\theta_1) \leq \theta_2 < 1/40(11 - 20\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (451/1003 \leq \theta_1 < 841/1870 \text{ and } 1/204(-28 + 85\theta_1) < \theta_2 < 1/40(11 - 20\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (9/20 < \theta_1 < 73/162 \text{ and } 1/2(1 - 2\theta_1) < \theta_2 < 1/40(11 - 20\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (157/343 \leq \theta_1 < 11/24 \text{ and } (1 - \theta_1)/12 < \theta_2 < 1/40(11 - 20\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (73/162 \leq \theta_1 \leq 4244/9401 \text{ and } 1/2(1 - 2\theta_1) \leq \theta_2 < 1/40(11 - 20\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (4244/9401 < \theta_1 \leq 279/616 \text{ and } 1/2(1 - 2\theta_1) \leq \theta_2 < 1/40(11 - 20\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (53/116 < \theta_1 \leq 16/35 \text{ and } \theta_2 = 1/5(8 - 17\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (16/35 < \theta_1 < 91/199 \text{ and } \theta_2 = 1/5(8 - 17\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (279/616 < \theta_1 < 5/11 \text{ and } 1/2(1 - 2\theta_1) \leq \theta_2 < 1/40(11 - 20\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (\theta_1 = 1772/3893 \text{ and } 707/15572 < \theta_2 < 7383/155720 \text{ and } (264 - 3893\theta_2)/3893 \leq \theta_3 < 349/15572),$
 $\text{or } (73/162 \leq \theta_1 \leq 4244/9401 \text{ and } 1/2(1 - 2\theta_1) \leq \theta_2 < 1/40(11 - 20\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (4244/9401 < \theta_1 \leq 279/616 \text{ and } 1/2(1 - 2\theta_1) \leq \theta_2 < 1/40(11 - 20\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (279/616 < \theta_1 < 5/11 \text{ and } 1/2(1 - 2\theta_1) \leq \theta_2 < 1/40(11 - 20\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (\theta_1 = 11/24 \text{ and } 13/288 < \theta_2 < 11/240 \text{ and } 1/288(19 - 288\theta_2) \leq \theta_3 < 1/48),$
 $\text{or } (\theta_1 = 74/161 \text{ and } 29/644 < \theta_2 < 291/6440 \text{ and } 1/46(3 - 46\theta_2) \leq \theta_3 < 13/644),$
 $\text{or } (9/20 < \theta_1 < 73/162 \text{ and } 1/2(1 - 2\theta_1) < \theta_2 < 1/40(11 - 20\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (4/9 < \theta_1 \leq 29/65 \text{ and } 1/19(4 - 7\theta_1) < \theta_2 < 1/35(11 - 21\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (4/9 < \theta_1 < 29/65 \text{ and } (1 - \theta_1)/12 < \theta_2 < 1/19(4 - 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (451/1003 \leq \theta_1 < 9/20 \text{ and } (1 - \theta_1)/12 < \theta_2 < 1/24(3 - 4\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (451/1003 \leq \theta_1 < 9/20 \text{ and } 1/24(3 - 4\theta_1) \leq \theta_2 < 1/28(5 - 8\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (157/343 \leq \theta_1 < 11/24 \text{ and } (1 - \theta_1)/12 < \theta_2 < 1/40(11 - 20\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (11/24 < \theta_1 \leq 1444/3145 \text{ and } (1 - \theta_1)/12 < \theta_2 < 1/40(11 - 20\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (\theta_1 = 11/24 \text{ and } 13/288 < \theta_2 < 11/240 \text{ and } 1/288(19 - 288\theta_2) \leq \theta_3 < 1/48),$
 $\text{or } (\theta_1 = 4/9 \text{ and } 1/20 < \theta_2 < 11/216 \text{ and } 1/12(8/9 - 12\theta_2) \leq \theta_3 < 1/36),$
 $\text{or } (\theta_1 = 4/9 \text{ and } 13/252 \leq \theta_2 < 19/360 \text{ and } 1/12(8/9 - 12\theta_2) \leq \theta_3 < 1/45(4 - 54\theta_2)),$
 $\text{or } (4/9 < \theta_1 \leq 1868/4199 \text{ and } 1/23(6 - 11\theta_1) < \theta_2 < 1/24(3 - 4\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (1868/4199 < \theta_1 < 116/259 \text{ and } 1/23(6 - 11\theta_1) < \theta_2 < 1/204(-28 + 85\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (116/259 < \theta_1 < 185/413 \text{ and } 1/23(6 - 11\theta_1) < \theta_2 < 1/204(-28 + 85\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (185/413 \le$

$\text{or } (185/413 \leq \theta_1 < 107/238 \text{ and } 1/204(-28 + 85\theta_1) < \theta_2 < 1/24(3 - 4\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (51/112 < \theta_1 \leq 177/388 \text{ and } 1/70(32 - 63\theta_1) \leq \theta_2 < 1/40(11 - 20\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (16/35 < \theta_1 < 157/343 \text{ and } 1/70(32 - 63\theta_1) \leq \theta_2 < 1/40(11 - 20\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (1772/3893 < \theta_1 \leq 51/112 \text{ and } (1 - \theta_1)/12 < \theta_2 < 1/40(11 - 20\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (51/112 < \theta_1 \leq 16/35 \text{ and } (1 - \theta_1)/12 < \theta_2 < 1/70(32 - 63\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (5/11 \leq \theta_1 < 1772/3893 \text{ and } (1 - \theta_1)/12 < \theta_2 < 1/40(11 - 20\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (91/199 < \theta_1 < 157/343 \text{ and } (1 - \theta_1)/12 < \theta_2 < 1/70(32 - 63\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (16/35 < \theta_1 < 91/199 \text{ and } (1 - \theta_1)/12 < \theta_2 < 1/5(8 - 17\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (49/109 < \theta_1 \leq 107/238 \text{ and } (1 - \theta_1)/12 < \theta_2 < 1/204(-28 + 85\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (4/9 < \theta_1 < 1868/4199 \text{ and } \theta_2 = 1/204(-28 + 85\theta_1) \text{ and } 1/17(8 - 17\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (1868/4199 < \theta_1 < 116/259 \text{ and } \theta_2 = 1/204(-28 + 85\theta_1) \text{ and } 1/17(8 - 17\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (116/259 < \theta_1 < 185/413 \text{ and } \theta_2 = 1/204(-28 + 85\theta_1) \text{ and } 1/17(8 - 17\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (185/413 \leq \theta_1 < 107/238 \text{ and } \theta_2 = 1/204(-28 + 85\theta_1) \text{ and } 1/17(8 - 17\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (16/35 < \theta_1 \leq 91/199 \text{ and } 1/5(8 - 17\theta_1) < \theta_2 < 1/70(32 - 63\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (177/388 < \theta_1 \leq 53/116 \text{ and } 1/70(32 - 63\theta_1) \leq \theta_2 < 1/40(11 - 20\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (53/116 < \theta_1 < 16/35 \text{ and } 1/70(32 - 63\theta_1) \leq \theta_2 < 1/5(8 - 17\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 $\text{or } (4/9 < \theta_1 \leq 1868/4199 \text{ and } 1/24(3 - 4\theta_1) \leq \theta_2 < 1/28(5 - 8\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (1868/4199 < \theta_1 < 116/259 \text{ and } 1/24(3 - 4\theta_1) \leq \theta_2 < 1/28(5 - 8\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (116/259 < \theta_1 < 185/413 \text{ and } 1/24(3 - 4\theta_1) \leq \theta_2 < 1/28(5 - 8\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (185/413 \leq \theta_1 < 107/238 \text{ and } 1/24(3 - 4\theta_1) \leq \theta_2 < 1/28(5 - 8\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (107/238 \leq \theta_1 < 451/1003 \text{ and } 1/204(-28 + 85\theta_1) < \theta_2 < 1/28(5 - 8\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (107/238 < \theta_1 < 451/1003 \text{ and } 1/24(3 - 4\theta_1) \leq \theta_2 < 1/204(-28 + 85\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (451/1003 < \theta_1 \leq 841/1870 \text{ and } 1/28(5 - 8\theta_1) \leq \theta_2 < 1/204(-28 + 85\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (841/1870 < \theta_1 \leq 130/289 \text{ and } 1/28(5 - 8\theta_1) \leq \theta_2 < 1/40(11 - 20\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (107/238 \leq \theta_1 < 451/1003 \text{ and } \theta_2 = 1/204(-28 + 85\theta_1) \text{ and } 1/17(8 - 17\theta_1) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (451/1003 \leq \theta_1 < 841/1870 \text{ and } \theta_2 = 1/204(-28 + 85\theta_1) \text{ and } 1/17(8 - 17\theta_1) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 $\text{or } (\theta_1 = 4/9 \text{ and } 10/207 < \theta_2 < 1/20 \text{ and } 1/12(8/9 - 12\theta_2) \leq \theta_3 < 1/36),$
 $\text{or } (\theta_1 = 9/20 \text{ and } 11/240 < \theta_2 < 1/20 \text{ and } 1/240(17 - 240\theta_2) \leq \theta_3 < 1/40),$
 $\text{or } (\theta_1 = 4/9 \text{ and } 8/171 < \theta_2 < 1/21 \text{ and } 1/12(8/9 - 12\theta_2) \leq \theta_3 < 1/36),$
 $\text{or } (\theta_1 = 4/9 \text{ and } 5/108 < \theta_2 < 8/171 \text{ and } 1/12(8/9 - 12\theta_2) \leq \theta_3 < 1/36),$
 $\text{or } (\theta_1 = 4/9 \text{ and } 1/21 < \theta_2 < 10/207 \text{ and } 1/12(8/9 - 12\theta_2) \leq \theta_3 < 1/36),$
 $\text{or } (\theta_1 = 4/9 \text{ and } 11/216 \leq \theta_2 < 13/252 \text{ and } 1/12(8/9 - 12\theta_2) \leq \theta_3 < 1/45(4 - 54\theta_2)),$
 $\text{or } (\theta_1 = 116/259 \text{ and } 143/3108 < \theta_2 < 278/5957 \text{ and } 1/111(8 - 1$

or $(\theta_1 = 1772/3893 \text{ and } ((707/15572 < \theta_2 \leq 179/3893 \text{ and } (264 - 3893\theta_2)/3893 \leq \theta_3 < 349/15572),$
 or $(179/3893 < \theta_2 < 7383/155720 \text{ and } 1/2(349/3893 - \theta_2) \leq \theta_3 < 349/15572))),$
 or $(\theta_1 = 43/98 \text{ and } 1/20 < \theta_2 < 5/98 \text{ and } 6/49 - 2\theta_2 \leq \theta_3 < 3/98),$
 or $(\theta_1 = 61/140 \text{ and } 1/20 < \theta_2 < 11/210 \text{ and } 9/70 - 2\theta_2 \leq \theta_3 < 9/280),$
 or $(\theta_1 = 15/34 \text{ and } 1/20 < \theta_2 < 7/136 \text{ and } 2/17 - 2\theta_2 \leq \theta_3 < 1/34),$
 or $(\theta_1 = 4/9 \text{ and } 1/20 < \theta_2 < 11/216 \text{ and } 1/9 - 2\theta_2 \leq \theta_3 < 1/36),$
 or $(\theta_1 = 61/140 \text{ and } 53/980 \leq \theta_2 < 1/18 \text{ and } 9/70 - 2\theta_2 \leq \theta_3 < 1/200(19 - 240\theta_2)),$
 or $(\theta_1 = 15/34 \text{ and } 25/476 \leq \theta_2 < 37/680 \text{ and } 2/17 - 2\theta_2 \leq \theta_3 < 1/340(31 - 408\theta_2)),$
 or $(\theta_1 = 43/98 \text{ and } 73/1372 \leq \theta_2 < 1/18 \text{ and } 6/49 - 2\theta_2 \leq \theta_3 < 1/140(13 - 168\theta_2)),$
 or $(\theta_1 = 4/9 \text{ and } 13/252 \leq \theta_2 < 19/360 \text{ and } 1/9 - 2\theta_2 \leq \theta_3 < 1/45(4 - 54\theta_2)),$
 or $(50/117 < \theta_1 < 85/198 \text{ and } \theta_2 = 1/18 \text{ and } 8/9 - 2\theta_1 \leq \theta_3 < 1/30(10 - 21\theta_1)),$
 or $(85/198 < \theta_1 < 73/170 \text{ and } \theta_2 = 1/18 \text{ and } 8/9 - 2\theta_1 \leq \theta_3 < 1/30(10 - 21\theta_1)),$
 or $(73/170 \leq \theta_1 < 31/72 \text{ and } \theta_2 = 1/18 \text{ and } 8/9 - 2\theta_1 \leq \theta_3 < 1/30(10 - 21\theta_1)),$
 or $(31/72 \leq \theta_1 < 163/378 \text{ and } \theta_2 = 1/18 \text{ and } 8/9 - 2\theta_1 \leq \theta_3 < 1/30(10 - 21\theta_1)),$
 or $(163/378 < \theta_1 < 61/140 \text{ and } \theta_2 = 1/18 \text{ and } 8/9 - 2\theta_1 \leq \theta_3 < 1/30(10 - 21\theta_1)),$
 or $(61/140 < \theta_1 < 43/98 \text{ and } \theta_2 = 1/18 \text{ and } 8/9 - 2\theta_1 \leq \theta_3 < 1/30(10 - 21\theta_1)),$
 or $(43/98 < \theta_1 < 79/180 \text{ and } \theta_2 = 1/18 \text{ and } 8/9 - 2\theta_1 \leq \theta_3 < 1/30(10 - 21\theta_1)),$
 or $(61/140 < \theta_1 < 43/98 \text{ and } \theta_2 = 1/20 \text{ and } 9/10 - 2\theta_1 \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 or $(43/98 < \theta_1 < 37/84 \text{ and } \theta_2 = 1/20 \text{ and } 9/10 - 2\theta_1 \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 or $(97/220 < \theta_1 < 15/34 \text{ and } \theta_2 = 1/20 \text{ and } 9/10 - 2\theta_1 \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 or $(15/34 < \theta_1 < 4/9 \text{ and } \theta_2 = 1/20 \text{ and } 9/10 - 2\theta_1 \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 or $(37/84 < \theta_1 < 97/220 \text{ and } \theta_2 = 1/20 \text{ and } 9/10 - 2\theta_1 \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 or $(4/9 < \theta_1 < 1868/4199 \text{ and } \theta_2 = 1/204(-28 + 85\theta_1) \text{ and } 1/17(8 - 17\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 or $(1868/4199 < \theta_1 < 116/259 \text{ and } \theta_2 = 1/204(-28 + 85\theta_1) \text{ and } 1/17(8 - 17\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 or $(116/259 < \theta_1 < 185/413 \text{ and } \theta_2 = 1/204(-28 + 85\theta_1) \text{ and } 1/17(8 - 17\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 or $(185/413 \leq \theta_1 < 107/238 \text{ and } \theta_2 = 1/204(-28 + 85\theta_1) \text{ and } 1/17(8 - 17\theta_1) \leq \theta_3 < 1/4(1 - 2\theta_1)),$
 or $(107/238 \leq \theta_1 < 451/1003 \text{ and } \theta_2 = 1/204(-28 + 85\theta_1) \text{ and } 1/17(8 - 17\theta_1) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 or $(451/1003 \leq \theta_1 < 841/1870 \text{ and } \theta_2 = 1/204(-28 + 85\theta_1) \text{ and } 1/17(8 - 17\theta_1) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)),$
 or $(\theta_1 = 15/34 \text{ and } 39/782 < \theta_2 < 1/20 \text{ and } 2/17 - 2\theta_2 \leq \theta_3 < 1/34),$
 or $(\theta_1 = 43/98 \text{ and } 13/266 < \theta_2 < 1/20 \text{ and } 6/49 - 2\theta_2 \leq \theta_3 < 3/98),$
 or $(\theta_1 = 4/9 \text{ and } 10/207 < \theta_2 < 1/20 \text{ and } 1/9 - 2\theta_2 \leq \theta_3 < 1/36),$
 or $(\theta_1 = 9/20 \text{ and } 11/240 < \theta_2 < 1/20 \text{ and } 1/240(17 - 240\theta_2) \leq \theta_3 < 1/40),$
 or $(\theta_1 = 11/26 \text{ and } 1/16 < \theta_2 < 33/520 \text{ and } 2/13 - 2\theta_2 \leq \theta_3 < 1/260(27 - 312\theta_2)),$
 or $(\theta_1 = 61/140 \text{ and } 1/18 < \theta_2 < 2/35 \text{ and } 9/70 - 2\theta_2 \leq \theta_3 < 1/200(19 - 240\theta_2)),$
 or $(\theta_1 = 61/140 \text{ and } 11/210 \leq \theta_2 < 169/3220 \text{ and } 9/70 - 2\theta_2 \leq \theta_3 < 1/200(19 - 240\theta_2)),$
 or $(\theta_1 = 61/140 \text{ and } 169/3220 < \theta_2 < 37/700 \text{ and } 9/70 - 2\theta_2 \leq \theta_3 < 1/200(19 - 240\theta_2)),$
 or $(\theta_1 = 61/140 \text{ and } 37/700 < \theta_2 < 53/980 \text{ and } 9/70 - 2\theta_2 \leq \theta_3 < 1/200(19 - 240\theta_2)),$
 or $(\theta_1 = 15/34 \text{ and } 31/646 < \theta_2 < 59/1190 \text{ and } 2/17 - 2\theta_2 \leq \theta_3 < 1/34),$
 or $(\theta_1 = 15/34 \text{ and } 59/1190 < \theta_2 < 39/782 \text{ and } 2/17 - 2\theta_2 \leq \theta_3 < 1/34),$
 or $(\theta_1 = 15/34 \text{ and } 7/136 \leq \theta_2 < 25/476 \text{ and } 2/17 - 2\theta_2 \leq \theta_3 < 1/340(31 - 408\theta_2)),$
 or $(\theta_1 = 43/98 \text{ and } 1/18 < \theta_2 < 109/1960 \text{ and } 6/49 - 2\theta_2 \leq \theta_3 < 1/140(13 - 168\theta_2)),$
 or $(\theta_1 = 43/98 \text{ and } 5/98 < \theta_2 < 61/1176 \text{ and } 6/49 - 2\theta_2 \leq \theta_3 < 3/98),$
 or $(\theta_1 = 43/98 \text{ and } 61/1176 \leq \theta_2 < 73/1372 \text{ and } 6/49 - 2\theta_2 \leq \theta_3 < 1/140(13 - 168\theta_2)),$
 or $(\theta_1 = 17/40 \text{ and } 19/320 < \theta_2 < 1/16 \text{ and } 3/20 - 2\theta_2 \leq \theta_3 < 1/400(41 - 480\theta_2)),$
 or $(\theta_1 = 4/9 \text{ and } 8/171 < \theta_2 < 1/21 \text{ and } 1/9 - 2\theta_2 \leq \theta_3 < 1/36),$
 or $(\theta_1 = 4/9 \text{ and } 1/21 < \theta_2 < 10/207 \text{ and } 1/9 - 2\theta_2 \leq \theta_3 < 1/36),$
 or $(\theta_1 = 4/9 \text{ and } 11/216 \leq \theta_2 < 13/252 \text{ and } 1/9 - 2\theta_2 \leq \theta_3 < 1/45(4 - 54\theta_2)),$
 or $(\theta_1 = 116/259 \text{ and } 143/3108 < \theta_2 < 278/5957 \text{ and } 1/111(8 - 111\theta_2) \leq \theta_3 < 27/1036),$
 or $(\theta_1 = 116/259 \text{ and } 652/13209 < \theta_2 < 313/6216 \text{ and } 1/111(8 - 111\theta_2) \leq \theta_3 < 27/1036),$

or $(\theta_1 = 116/259 \text{ and } 278/5957 < \theta_2 < 652/13209 \text{ and } 1/111(8 - 111\theta_2) \leq \theta_3 < 27/1036)$,
 or $(\theta_1 = 116/259 \text{ and } 313/6216 \leq \theta_2 < 367/7252 \text{ and } 1/111(8 - 111\theta_2) \leq \theta_3 < 1/185(16 - 222\theta_2))$,
 or $(\theta_1 = 116/259 \text{ and } 367/7252 \leq \theta_2 < 529/10360 \text{ and } 1/111(8 - 111\theta_2) \leq \theta_3 < 1/185(16 - 222\theta_2))$,
 or $(\theta_1 = 74/161 \text{ and } 29/644 < \theta_2 < 291/6440 \text{ and } 1/46(3 - 46\theta_2) \leq \theta_3 < 13/644)$,
 or $(73/162 \leq \theta_1 < 5/11 \text{ and } (1 - \theta_1)/12 < \theta_2 < 1/2(1 - 2\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(4/9 < \theta_1 < 29/65 \text{ and } (1 - \theta_1)/12 < \theta_2 < 1/19(4 - 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(13/30 < \theta_1 < 61/140 \text{ and } \theta_2 = 1/20 \text{ and } 9/10 - 2\theta_1 \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(\theta_1 = 61/140 \text{ and } 27/560 < \theta_2 < 1/20 \text{ and } 9/70 - 2\theta_2 \leq \theta_3 < 9/280)$,
 or $(\theta_1 = 15/34 \text{ and } 3/68 < \theta_2 < 31/646 \text{ and } 2/17 - 2\theta_2 \leq \theta_3 < 1/34)$,
 or $(\theta_1 = 43/98 \text{ and } 9/196 < \theta_2 < 13/266 \text{ and } 6/49 - 2\theta_2 \leq \theta_3 < 3/98)$,
 or $(\theta_1 = 4/9 \text{ and } 1/24 < \theta_2 < 8/171 \text{ and } 1/9 - 2\theta_2 \leq \theta_3 < 1/36)$,
 or $(130/289 < \theta_1 < 9/20 \text{ and } 1/40(11 - 20\theta_1) \leq \theta_2 < 1/2(1 - 2\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(4/9 < \theta_1 < 116/259 \text{ and } 1/40(11 - 20\theta_1) \leq \theta_2 < 1/2(1 - 2\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(116/259 < \theta_1 < 185/413 \text{ and } 1/40(11 - 20\theta_1) \leq \theta_2 < 1/2(1 - 2\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(185/413 \leq \theta_1 < 451/1003 \text{ and } 1/40(11 - 20\theta_1) \leq \theta_2 < 1/2(1 - 2\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(451/1003 \leq \theta_1 < 841/1870 \text{ and } 1/40(11 - 20\theta_1) \leq \theta_2 < 1/2(1 - 2\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(841/1870 \leq \theta_1 < 130/289 \text{ and } 1/204(-28 + 85\theta_1) < \theta_2 < 1/2(1 - 2\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(4/9 < \theta_1 < 116/259 \text{ and } 1/40(11 - 20\theta_1) \leq \theta_2 < 1/2(1 - 2\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(116/259 < \theta_1 < 185/413 \text{ and } 1/40(11 - 20\theta_1) \leq \theta_2 < 1/2(1 - 2\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(185/413 \leq \theta_1 < 451/1003 \text{ and } 1/40(11 - 20\theta_1) \leq \theta_2 < 1/2(1 - 2\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(451/1003 \leq \theta_1 < 841/1870 \text{ and } 1/40(11 - 20\theta_1) \leq \theta_2 < 1/2(1 - 2\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(841/1870 \leq \theta_1 < 130/289 \text{ and } 1/204(-28 + 85\theta_1) < \theta_2 < 1/2(1 - 2\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(\theta_1 = 17/40 \text{ and } 327/4760 < \theta_2 < 3/40 \text{ and } 3/20 - 2\theta_2 \leq \theta_3 < 1/400(41 - 480\theta_2))$,
 or $(\theta_1 = 143/340 \text{ and } 89/1190 < \theta_2 < 27/340 \text{ and } 27/170 - 2\theta_2 \leq \theta_3 < (359 - 4080\theta_2)/3400)$,
 or $(\theta_1 = 13/32 \text{ and } 23/256 < \theta_2 < 3/32 \text{ and } 3/16 - 2\theta_2 \leq \theta_3 < 1/320(37 - 384\theta_2))$,
 or $(\theta_1 = 139/340 \text{ and } 233/2720 < \theta_2 < 31/340 \text{ and } 31/170 - 2\theta_2 \leq \theta_3 < (387 - 4080\theta_2)/3400)$,
 or $(\theta_1 = 37/90 \text{ and } 59/720 < \theta_2 < 4/45 \text{ and } 8/45 - 2\theta_2 \leq \theta_3 < 1/900(101 - 1080\theta_2))$,
 or $(\theta_1 = 15/34 \text{ and } 31/544 \leq \theta_2 < 1/17 \text{ and } 2/17 - 2\theta_2 \leq \theta_3 < 1/340(31 - 408\theta_2))$,
 or $(\theta_1 = 53/126 \text{ and } 67/1008 < \theta_2 < 167/2499 \text{ and } 10/63 - 2\theta_2 \leq \theta_3 < 1/180(19 - 216\theta_2))$,
 or $(\theta_1 = 15/34 \text{ and } 37/680 \leq \theta_2 < 1/18 \text{ and } 2/17 - 2\theta_2 \leq \theta_3 < 1/340(31 - 408\theta_2))$,
 or $(\theta_1 = 4/9 \text{ and } 19/360 \leq \theta_2 < 1/18 \text{ and } 1/9 - 2\theta_2 \leq \theta_3 < 1/45(4 - 54\theta_2))$,
 or $(79/180 \leq \theta_1 < 15/34 \text{ and } \theta_2 = 1/18 \text{ and } 8/9 - 2\theta_1 \leq \theta_3 < 1/30(10 - 21\theta_1))$,
 or $(15/34 < \theta_1 < 4/9 \text{ and } \theta_2 = 1/18 \text{ and } 8/9 - 2\theta_1 \leq \theta_3 < 1/30(10 - 21\theta_1))$,
 or $(841/1870 \leq \theta_1 < 130/289 \text{ and } \theta_2 = 1/204(-28 + 85\theta_1) \text{ and } 1/17(8 - 17\theta_1) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(\theta_1 = 5/12 \text{ and } 7/96 < \theta_2 < 1/12 \text{ and } 1/6 - 2\theta_2 \leq \theta_3 < 1/120(13 - 144\theta_2))$,
 or $(\theta_1 = 11/26 \text{ and } 33/520 \leq \theta_2 < 27/416 \text{ and } 2/13 - 2\theta_2 \leq \theta_3 < 1/260(27 - 312\theta_2))$,
 or $(\theta_1 = 11/26 \text{ and } 27/416 \leq \theta_2 < 15/221 \text{ and } 2/13 - 2\theta_2 \leq \theta_3 < 1/260(27 - 312\theta_2))$,
 or $(\theta_1 = 11/26 \text{ and } 15/221 < \theta_2 < 1/13 \text{ and } 2/13 - 2\theta_2 \leq \theta_3 < 1/260(27 - 312\theta_2))$,
 or $(\theta_1 = 61/140 \text{ and } 2/35 \leq \theta_2 < 19/320 \text{ and } 9/70 - 2\theta_2 \leq \theta_3 < 1/200(19 - 240\theta_2))$,
 or $(\theta_1 = 61/140 \text{ and } 19/320 \leq \theta_2 < 9/140 \text{ and } 9/70 - 2\theta_2 \leq \theta_3 < 1/200(19 - 240\theta_2))$,
 or $(\theta_1 = 15/34 \text{ and } 1/18 < \theta_2 < 31/544 \text{ and } 2/17 - 2\theta_2 \leq \theta_3 < 1/340(31 - 408\theta_2))$,
 or $(\theta_1 = 43/98 \text{ and } 109/1960 \leq \theta_2 < 13/224 \text{ and } 6/49 - 2\theta_2 \leq \theta_3 < 1/140(13 - 168\theta_2))$,
 or $(\theta_1 = 43/98 \text{ and } 13/224 \leq \theta_2 < 3/49 \text{ and } 6/49 - 2\theta_2 \leq \theta_3 < 1/140(13 - 168\theta_2))$,
 or $(\theta_1 = 127/306 \text{ and } 185/2448 < \theta_2 < 13/153 \text{ and } 26/153 - 2\theta_2 \leq \theta_3 < (335 - 3672\theta_2)/3060)$,
 or $(\theta_1 = 143/340 \text{ and } 181/2720 < \theta_2 < 159/2380 \text{ and } 27/170 - 2\theta_2 \leq \theta_3 < (359 - 4080\theta_2)/3400)$,
 or $(\theta_1 = 143/340 \text{ and } 159/2380 < \theta_2 \leq 89/1190 \text{ and } 27/170 - 2\theta_2 \leq \theta_3 < (359 - 4080\theta_2)/3400)$,
 or $(\theta_1 = 53/126 \text{ and } 11/147 < \theta_2 < 5/63 \text{ and } 10/63 - 2\theta_2 \leq \theta_3 < 1/180(19 - 216\theta_2))$,
 or $(\theta_1 = 53/126 \text{ and } 167/2499 < \theta_2 \leq 11/147 \text{ and } 10/63 - 2\theta_2 \leq \theta_3 < 1/180(19 - 216\theta_2))$,
 or $(\theta_1 = 17/40 \text{ and } 1/16 \leq \theta_2 < 41/640 \text{ and } 3/20 - 2\theta_2 \leq \theta_3 < 1/400(41 - 480\theta_2))$,

or $(\theta_1 = 17/40 \text{ and } 41/640 \leq \theta_2 < 327/4760 \text{ and } 3/20 - 2\theta_2 \leq \theta_3 < 1/400(41 - 480\theta_2))$,
 or $(\theta_1 = 116/259 \text{ and } 529/10360 \leq \theta_2 < 27/518 \text{ and } 1/111(8 - 111\theta_2) \leq \theta_3 < 1/185(16 - 222\theta_2))$,
 or $(\theta_1 = 12/29 \text{ and } 9/116 < \theta_2 < 5/58 \text{ and } 5/29 - 2\theta_2 \leq \theta_3 < 1/145(16 - 174\theta_2))$,
 or $(130/289 < \theta_1 < 9/20 \text{ and } 1/2(1 - 2\theta_1) \leq \theta_2 < 1/204(-28 + 85\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(73/162 \leq \theta_1 \leq 4244/9401 \text{ and } 1/40(11 - 20\theta_1) \leq \theta_2 < 1/24(3 - 4\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(4244/9401 < \theta_1 \leq 279/616 \text{ and } 1/40(11 - 20\theta_1) \leq \theta_2 < 1/24(3 - 4\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(279/616 < \theta_1 < 5/11 \text{ and } 1/40(11 - 20\theta_1) \leq \theta_2 < 1/70(32 - 63\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(4/9 < \theta_1 < 116/259 \text{ and } 1/2(1 - 2\theta_1) \leq \theta_2 < 1/16(4 - 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(116/259 < \theta_1 < 185/413 \text{ and } 1/2(1 - 2\theta_1) \leq \theta_2 < 1/70(32 - 63\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(185/413 \leq \theta_1 < 451/1003 \text{ and } 1/2(1 - 2\theta_1) \leq \theta_2 < 1/70(32 - 63\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(451/1003 \leq \theta_1 < 130/289 \text{ and } 1/2(1 - 2\theta_1) \leq \theta_2 < 1/70(32 - 63\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(9/20 < \theta_1 < 73/162 \text{ and } 1/5(8 - 17\theta_1) < \theta_2 < 1/12(4 - 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(1772/3893 < \theta_1 < 11/24 \text{ and } 2/37 < \theta_2 < 1/12(4 - 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(9/20 < \theta_1 < 73/162 \text{ and } 1/40(11 - 20\theta_1) \leq \theta_2 < 1/24(3 - 4\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(157/343 \leq \theta_1 < 11/24 \text{ and } 1/40(11 - 20\theta_1) \leq \theta_2 < 1/24(3 - 4\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(177/388 < \theta_1 \leq 53/116 \text{ and } \theta_2 = 1/5(8 - 17\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(9/20 < \theta_1 < 73/162 \text{ and } 1/24(3 - 4\theta_1) \leq \theta_2 < 1/204(-28 + 85\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(9/20 < \theta_1 < 73/162 \text{ and } 1/204(-28 + 85\theta_1) < \theta_2 < 1/70(32 - 63\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(9/20 < \theta_1 < 73/162 \text{ and } 2/37 < \theta_2 < 1/5(8 - 17\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(9/20 < \theta_1 < 73/162 \text{ and } 1/70(32 - 63\theta_1) \leq \theta_2 \leq 2/37 \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(16/35 < \theta_1 < 11/24 \text{ and } 1/24(3 - 4\theta_1) \leq \theta_2 < 1/204(-28 + 85\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(73/162 \leq \theta_1 < 5/11 \text{ and } \theta_2 = 1/5(8 - 17\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(1772/3893 < \theta_1 \leq 177/388 \text{ and } \theta_2 = 1/5(8 - 17\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(5/11 \leq \theta_1 < 286/629 \text{ and } \theta_2 = 1/5(8 - 17\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(286/629 \leq \theta_1 < 1772/3893 \text{ and } \theta_2 = 1/5(8 - 17\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(1772/3893 < \theta_1 < 11/24 \text{ and } 1/204(-28 + 85\theta_1) < \theta_2 \leq 2/37 \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(87/188 \leq \theta_1 < 13/28 \text{ and } \theta_2 = 1/5(-3 + 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(15/32 < \theta_1 < 8/17 \text{ and } \theta_2 = 1/5(-3 + 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(41/89 < \theta_1 < 87/188 \text{ and } \theta_2 = 1/5(-3 + 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(13/28 \leq \theta_1 < 316/679 \text{ and } \theta_2 = 1/5(-3 + 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(316/679 < \theta_1 \leq 121/259 \text{ and } \theta_2 = 1/5(-3 + 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(121/259 < \theta_1 \leq 15/32 \text{ and } \theta_2 = 1/5(-3 + 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(73/162 \leq \theta_1 < 4244/9401 \text{ and } \theta_2 = 1/204(-28 + 85\theta_1) \text{ and } 1/17(8 - 17\theta_1) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(4244/9401 \leq \theta_1 < 5/11 \text{ and } \theta_2 = 1/204(-28 + 85\theta_1) \text{ and } 1/17(8 - 17\theta_1) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(5/11 \leq \theta_1 < 1772/3893 \text{ and } \theta_2 = 1/204(-28 + 85\theta_1) \text{ and } 1/17(8 - 17\theta_1) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(\theta_1 = 1772/3893 \text{ and } 7383/155720 \leq \theta_2 < 1294/27251 \text{ and } (264 - 3893\theta_2)/3893 \leq \theta_3 < 349/15572)$,
 or $(\theta_1 = 1772/3893 \text{ and } 1294/27251 \leq \theta_2 < 4591/93432 \text{ and } (264 - 3893\theta_2)/3893 \leq \theta_3 < 349/15572)$,
 or $(\theta_1 = 1772/3893 \text{ and } 2/37 < \theta_2 < 264/3893 \text{ and } (264 - 3893\theta_2)/3893 \leq \theta_3 < (1584 - 23358\theta_2)/19465)$,
 or $(\theta_1 = 1772/3893 \text{ and } 12/229 < \theta_2 \leq 2/37 \text{ and } (264 - 3893\theta_2)/3893 \leq \theta_3 < (1584 - 23358\theta_2)/19465)$,
 or $(\theta_1 = 1772/3893 \text{ and } 4591/93432 \leq \theta_2 < 12/229 \text{ and } (264 - 3893\theta_2)/3893 \leq \theta_3 < (1584 - 23358\theta_2)/19465)$,
 or $(11/24 < \theta_1 \leq 1444/3145 \text{ and } \theta_2 = 1/204(-28 + 85\theta_1) \text{ and } 1/17(8 - 17\theta_1) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(\theta_1 = 11/24 \text{ and } 2/37 < \theta_2 < 19/288 \text{ and } 1/288(19 - 288\theta_2) \leq \theta_3 < 1/240(19 - 288\theta_2))$,
 or $(\theta_1 = 11/24 \text{ and } 11/240 \leq \theta_2 < 7/144 \text{ and } 1/288(19 - 288\theta_2) \leq \theta_3 < 1/48)$,
 or $(\theta_1 = 11/24 \text{ and } 263/4896 < \theta_2 \leq 2/37 \text{ and } 1/288(19 - 288\theta_2) \leq \theta_3 < 1/240(19 - 288\theta_2))$,
 or $(\theta_1 = 11/24 \text{ and } 7/144 \leq \theta_2 < 263/4896 \text{ and } 1/288(19 - 288\theta_2) \leq \theta_3 < 1/240(19 - 288\theta_2))$,
 or $(769/1673 \leq \theta_1 < 13/28 \text{ and } \theta_2 = 1/204(-28 + 85\theta_1) \text{ and } 1/17(8 - 17\theta_1) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(\theta_1 = 13/28 \text{ and } 5/112 < \theta_2 < 1/21 \text{ and } 1/16(1 - 16\theta_2) \leq \theta_3 < 1/56)$,
 or $(15/32 < \theta_1 < 8/17 \text{ and } \theta_2 = 1/204(-28 + 85\theta_1) \text{ and } 1/17(8 - 17\theta_1) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(13/28 \leq \theta_1 < 316/679 \text{ and } \theta_2 = 1/204(-28 + 85\theta_1) \text{ and } 1/17(8 - 17\theta_1) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,

or $(316/679 < \theta_1 \leq 15/32$ and $\theta_2 = 1/204(-28 + 85\theta_1)$ and $1/17(8 - 17\theta_1) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(1444/3145 < \theta_1 \leq 74/161$ and $\theta_2 = 1/204(-28 + 85\theta_1)$ and $1/17(8 - 17\theta_1) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(74/161 < \theta_1 < 769/1673$ and $\theta_2 = 1/204(-28 + 85\theta_1)$ and $1/17(8 - 17\theta_1) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(\theta_1 = 316/679$ and $61/1358 \leq \theta_2 < 773/16296$ and $1/97(6 - 97\theta_2) \leq \theta_3 < 47/2716)$,
 or $(\theta_1 = 316/679$ and $121/2716 < \theta_2 < 61/1358$ and $1/97(6 - 97\theta_2) \leq \theta_3 < 47/2716)$,
 or $(\theta_1 = 316/679$ and $2/37 < \theta_2 < 654/11543$ and $1/97(6 - 97\theta_2) \leq \theta_3 < 1/485(36 - 582\theta_2))$,
 or $(\theta_1 = 316/679$ and $5/97 < \theta_2 \leq 2/37$ and $1/97(6 - 97\theta_2) \leq \theta_3 < 1/485(36 - 582\theta_2))$,
 or $(\theta_1 = 316/679$ and $773/16296 \leq \theta_2 < 5/97$ and $1/97(6 - 97\theta_2) \leq \theta_3 < 1/485(36 - 582\theta_2))$,
 or $(\theta_1 = 316/679$ and $654/11543 < \theta_2 < 6/97$ and $1/97(6 - 97\theta_2) \leq \theta_3 < 1/485(36 - 582\theta_2))$,
 or $(\theta_1 = 74/161$ and $291/6440 \leq \theta_2 < 187/3864$ and $1/46(3 - 46\theta_2) \leq \theta_3 < 13/644)$,
 or $(\theta_1 = 74/161$ and $187/3864 \leq \theta_2 \leq 2/37$ and $1/46(3 - 46\theta_2) \leq \theta_3 < 1/115(9 - 138\theta_2))$,
 or $(\theta_1 = 17/40$ and $3/40 \leq \theta_2 \leq 797/10360$ and $1/12(41/40 - 12\theta_2) \leq \theta_3 < 1/400(41 - 480\theta_2))$,
 or $(272/679 < \theta_1 \leq 1041/2590$ and $1/2(1 - 2\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(1041/2590 < \theta_1 \leq 524/1295$ and $1/2(1 - 2\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(37/90 < \theta_1 < 7/17$ and $\theta_2 = 1/2(1 - 2\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(37/90 < \theta_1 < 7/17$ and $1/27(14 - 28\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(\theta_1 = 7/17$ and $14/153 < \theta_2 < 19/204$ and $1/12(19/17 - 12\theta_2) \leq \theta_3 < 1/170(19 - 204\theta_2))$,
 or $(\theta_1 = 7/17$ and $3/34 < \theta_2 \leq 397/4403$ and $1/12(19/17 - 12\theta_2) \leq \theta_3 < 1/170(19 - 204\theta_2))$,
 or $(\theta_1 = 7/17$ and $397/4403 < \theta_2 \leq 185/2023$ and $1/12(19/17 - 12\theta_2) \leq \theta_3 < 1/170(19 - 204\theta_2))$,
 or $(\theta_1 = 7/17$ and $185/2023 < \theta_2 \leq 14/153$ and $1/12(19/17 - 12\theta_2) \leq \theta_3 < 1/170(19 - 204\theta_2))$,
 or $(7/17 < \theta_1 \leq 12/29$ and $\theta_2 = 1/2(1 - 2\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(7/17 < \theta_1 \leq 127/308$ and $1/27(14 - 28\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(\theta_1 = 15/34$ and $1/17 \leq \theta_2 \leq 535/8806$ and $1/12(31/34 - 12\theta_2) \leq \theta_3 < 1/340(31 - 408\theta_2))$,
 or $(127/306 < \theta_1 < 5/12$ and $\theta_2 = 1/2(1 - 2\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(272/679 < \theta_1 \leq 1041/2590$ and $\theta_2 = 1/2(1 - 2\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(1041/2590 < \theta_1 < 13/32$ and $\theta_2 = 1/2(1 - 2\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(13/32 < \theta_1 < 139/340$ and $\theta_2 = 1/2(1 - 2\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(139/340 < \theta_1 < 37/90$ and $\theta_2 = 1/2(1 - 2\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(12/29 < \theta_1 < 127/306$ and $\theta_2 = 1/2(1 - 2\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(127/306 < \theta_1 < 32/77$ and $1/11 < \theta_2 < 1/12(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(32/77 \leq \theta_1 < 5/12$ and $1/27(14 - 28\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(5/12 < \theta_1 \leq 108/259$ and $\theta_2 = -2 + 5\theta_1$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(108/259 < \theta_1 \leq 68/163$ and $\theta_2 = -2 + 5\theta_1$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(68/163 < \theta_1 < 28/67$ and $\theta_2 = -2 + 5\theta_1$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(73/170 \leq \theta_1 < 31/72$ and $\theta_2 = 1/238(-27 + 102\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(31/72 \leq \theta_1 \leq 5419/12580$ and $\theta_2 = 1/238(-27 + 102\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(4061/9418 < \theta_1 < 61/140$ and $\theta_2 = 1/238(-27 + 102\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(61/140 < \theta_1 < 43/98$ and $\theta_2 = 1/238(-27 + 102\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(43/98 < \theta_1 < 15/34$ and $\theta_2 = 1/238(-27 + 102\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(15/34 < \theta_1 < 638/1445$ and $\theta_2 = 1/238(-27 + 102\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(5419/12580 < \theta_1 \leq 4061/9418$ and $\theta_2 = 1/238(-27 + 102\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(524/1295 < \theta_1 < 13/32$ and $1/259(130 - 259\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(13/32 < \theta_1 \leq 208/511$ and $1/259(130 - 259\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(20/49 < \theta_1 < 139/340$ and $1/27(14 - 28\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(139/340 < \theta_1 < 37/90$ and $1/27(14 - 28\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(127/308 < \theta_1 \leq 12/29$ and $1/11 \leq \theta_2 < 1/12(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(12/29 < \theta_1 < 29/70$ and $1/11 < \theta_2 < 1/12(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(29/70 \leq \theta_1 < 127/306$ and $1/11 \leq \theta_2 < 1/12(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(5/12 < \theta_1 < 68/163$ and $1/27(14 - 28\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,

or $(34/81 \leq \theta_1 < 21/50$ and $1/27(14 - 28\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(21/50 < \theta_1 < 143/340$ and $1/27(14 - 28\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(143/340 < \theta_1 < 53/126$ and $1/27(14 - 28\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(53/126 < \theta_1 < 11/26$ and $1/27(14 - 28\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(11/26 < \theta_1 < 76/179$ and $1/27(14 - 28\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(76/179 \leq \theta_1 < 17/40$ and $1/7(-2 + 6\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(17/40 < \theta_1 < 73/170$ and $1/7(-2 + 6\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(52/121 \leq \theta_1 < 31/72$ and $1/27(14 - 28\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(31/72 \leq \theta_1 \leq 4061/9418$ and $1/27(14 - 28\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(4061/9418 < \theta_1 < 61/140$ and $1/238(-27 + 102\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(61/140 < \theta_1 < 43/98$ and $1/238(-27 + 102\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(43/98 < \theta_1 \leq 29/66$ and $1/238(-27 + 102\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(29/66 < \theta_1 < 15/34$ and $1/238(-27 + 102\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(15/34 < \theta_1 < 638/1445$ and $1/238(-27 + 102\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(638/1445 < \theta_1 < 31/70$ and $1/27(14 - 28\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(68/163 \leq \theta_1 < 28/67$ and $-2 + 5\theta_1 < \theta_2 < 1/12(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(73/170 < \theta_1 < 52/121$ and $1/7(-2 + 6\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(23/55 < \theta_1 < 34/81$ and $1/27(14 - 28\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(28/67 < \theta_1 \leq 23/55$ and $1/27(14 - 28\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(208/511 < \theta_1 < 257/630$ and $1/119(57 - 112\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(257/630 \leq \theta_1 \leq 20/49$ and $1/119(57 - 112\theta_1) \leq \theta_2 < 1/12(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
 or $(\theta_1 = 127/308$ and $1021/11396 < \theta_2 < 1/11$ and $1/12(49/44 - 12\theta_2) \leq \theta_3 < 1/440(49 - 528\theta_2)$),
 or $(\theta_1 = 5/12$ and $1/12 < \theta_2 \leq 265/3108$ and $1/12(13/12 - 12\theta_2) \leq \theta_3 < 1/120(13 - 144\theta_2)$),
 or $(\theta_1 = 5/12$ and $265/3108 < \theta_2 \leq 7/81$ and $1/12(13/12 - 12\theta_2) \leq \theta_3 < 1/120(13 - 144\theta_2)$),
 or $(\theta_1 = 5/12$ and $7/81 < \theta_2 < 13/144$ and $1/12(13/12 - 12\theta_2) \leq \theta_3 < 1/120(13 - 144\theta_2)$),
 or $(\theta_1 = 11/26$ and $1/13 < \theta_2 \leq 531/6734$ and $1/12(27/26 - 12\theta_2) \leq \theta_3 < 1/260(27 - 312\theta_2)$),
 or $(\theta_1 = 11/26$ and $531/6734 < \theta_2 \leq 28/351$ and $1/12(27/26 - 12\theta_2) \leq \theta_3 < 1/260(27 - 312\theta_2)$),
 or $(\theta_1 = 11/26$ and $28/351 < \theta_2 < 9/104$ and $1/12(27/26 - 12\theta_2) \leq \theta_3 < 1/260(27 - 312\theta_2)$),
 or $(\theta_1 = 61/140$ and $9/140 \leq \theta_2 \leq 49/740$ and $1/12(19/20 - 12\theta_2) \leq \theta_3 < 1/200(19 - 240\theta_2)$),
 or $(\theta_1 = 61/140$ and $49/740 < \theta_2 \leq 1/15$ and $1/12(19/20 - 12\theta_2) \leq \theta_3 < 1/200(19 - 240\theta_2)$),
 or $(\theta_1 = 61/140$ and $1/15 < \theta_2 < 1221/16660$ and $1/12(19/20 - 12\theta_2) \leq \theta_3 < 1/200(19 - 240\theta_2)$),
 or $(\theta_1 = 61/140$ and $1221/16660 < \theta_2 < 19/240$ and $1/12(19/20 - 12\theta_2) \leq \theta_3 < 1/200(19 - 240\theta_2)$),
 or $(\theta_1 = 15/34$ and $535/8806 < \theta_2 \leq 28/459$ and $1/12(31/34 - 12\theta_2) \leq \theta_3 < 1/340(31 - 408\theta_2)$),
 or $(\theta_1 = 15/34$ and $28/459 < \theta_2 < 9/119$ and $1/12(31/34 - 12\theta_2) \leq \theta_3 < 1/340(31 - 408\theta_2)$),
 or $(\theta_1 = 15/34$ and $9/119 < \theta_2 < 31/408$ and $1/12(31/34 - 12\theta_2) \leq \theta_3 < 1/340(31 - 408\theta_2)$),
 or $(\theta_1 = 43/98$ and $3/49 \leq \theta_2 \leq 229/3626$ and $1/12(13/14 - 12\theta_2) \leq \theta_3 < 1/140(13 - 168\theta_2)$),
 or $(\theta_1 = 43/98$ and $229/3626 < \theta_2 \leq 4/63$ and $1/12(13/14 - 12\theta_2) \leq \theta_3 < 1/140(13 - 168\theta_2)$),
 or $(\theta_1 = 43/98$ and $4/63 < \theta_2 < 435/5831$ and $1/12(13/14 - 12\theta_2) \leq \theta_3 < 1/140(13 - 168\theta_2)$),
 or $(\theta_1 = 43/98$ and $435/5831 < \theta_2 < 13/168$ and $1/12(13/14 - 12\theta_2) \leq \theta_3 < 1/140(13 - 168\theta_2)$),
 or $(\theta_1 = 13/32$ and $3/32 < \theta_2 \leq 793/8288$ and $1/12(37/32 - 12\theta_2) \leq \theta_3 < 1/320(37 - 384\theta_2)$),
 or $(\theta_1 = 13/32$ and $793/8288 < \theta_2 < 37/384$ and $1/12(37/32 - 12\theta_2) \leq \theta_3 < 1/320(37 - 384\theta_2)$),
 or $(\theta_1 = 139/340$ and $217/2295 < \theta_2 < 129/1360$ and $1/12(387/340 - 12\theta_2) \leq \theta_3 < (387 - 4080\theta_2)/3400$),
 or $(\theta_1 = 139/340$ and $31/340 < \theta_2 \leq 8199/88060$ and $1/12(387/340 - 12\theta_2) \leq \theta_3 < (387 - 4080\theta_2)/3400$),
 or $(\theta_1 = 139/340$ and $8199/88060 < \theta_2 \leq 953/10115$ and $1/12(387/340 - 12\theta_2) \leq \theta_3 < (387 - 4080\theta_2)/3400$),
 or $(\theta_1 = 139/340$ and $953/10115 < \theta_2 \leq 217/2295$ and $1/12(387/340 - 12\theta_2) \leq \theta_3 < (387 - 4080\theta_2)/3400$),
 or $(\theta_1 = 37/90$ and $112/1215 < \theta_2 < 101/1080$ and $1/12(101/90 - 12\theta_2) \leq \theta_3 < 1/900(101 - 1080\theta_2)$),
 or $(\theta_1 = 37/90$ and $4/45 < \theta_2 \leq 2117/23310$ and $1/12(101/90 - 12\theta_2) \leq \theta_3 < 1/900(101 - 1080\theta_2)$),
 or $(\theta_1 = 37/90$ and $2117/23310 < \theta_2 \leq 29/315$ and $1/12(101/90 - 12\theta_2) \leq \theta_3 < 1/900(101 - 1080\theta_2)$),
 or $(\theta_1 = 37/90$ and $29/315 < \theta_2 \leq 112/1215$ and $1/12(101/90 - 12\theta_2) \leq \theta_3 < 1/900(101 - 1080\theta_2)$),

or $(\theta_1 = 127/306 \text{ and } 364/4131 < \theta_2 \leq 1/11 \text{ and } 1/12(335/306 - 12\theta_2) \leq \theta_3 < (335 - 3672\theta_2)/3060)$,
 or $(\theta_1 = 127/306 \text{ and } 1/11 < \theta_2 < 335/3672 \text{ and } 1/12(335/306 - 12\theta_2) \leq \theta_3 < (335 - 3672\theta_2)/3060)$,
 or $(\theta_1 = 127/306 \text{ and } 13/153 < \theta_2 \leq 6887/79254 \text{ and } 1/12(335/306 - 12\theta_2) \leq \theta_3 < (335 - 3672\theta_2)/3060)$,
 or $(\theta_1 = 127/306 \text{ and } 6887/79254 < \theta_2 \leq 364/4131 \text{ and } 1/12(335/306 - 12\theta_2) \leq \theta_3 < (335 - 3672\theta_2)/3060)$,
 or $(\theta_1 = 143/340 \text{ and } 27/340 \leq \theta_2 \leq 7163/88060 \text{ and } 1/12(359/340 - 12\theta_2) \leq \theta_3 < (359 - 4080\theta_2)/3400)$,
 or $(\theta_1 = 143/340 \text{ and } 7163/88060 < \theta_2 \leq 7/85 \text{ and } 1/12(359/340 - 12\theta_2) \leq \theta_3 < (359 - 4080\theta_2)/3400)$,
 or $(\theta_1 = 143/340 \text{ and } 7/85 < \theta_2 < 359/4080 \text{ and } 1/12(359/340 - 12\theta_2) \leq \theta_3 < (359 - 4080\theta_2)/3400)$,
 or $(\theta_1 = 53/126 \text{ and } 5/63 \leq \theta_2 \leq 379/4662 \text{ and } 1/12(19/18 - 12\theta_2) \leq \theta_3 < 1/180(19 - 216\theta_2))$,
 or $(\theta_1 = 53/126 \text{ and } 379/4662 < \theta_2 \leq 20/243 \text{ and } 1/12(19/18 - 12\theta_2) \leq \theta_3 < 1/180(19 - 216\theta_2))$,
 or $(\theta_1 = 53/126 \text{ and } 20/243 < \theta_2 < 19/216 \text{ and } 1/12(19/18 - 12\theta_2) \leq \theta_3 < 1/180(19 - 216\theta_2))$,
 or $(\theta_1 = 17/40 \text{ and } 797/10360 < \theta_2 \leq 7/90 \text{ and } 1/12(41/40 - 12\theta_2) \leq \theta_3 < 1/400(41 - 480\theta_2))$,
 or $(\theta_1 = 17/40 \text{ and } 7/90 < \theta_2 \leq 11/140 \text{ and } 1/12(41/40 - 12\theta_2) \leq \theta_3 < 1/400(41 - 480\theta_2))$,
 or $(\theta_1 = 17/40 \text{ and } 11/140 < \theta_2 < 41/480 \text{ and } 1/12(41/40 - 12\theta_2) \leq \theta_3 < 1/400(41 - 480\theta_2))$,
 or $(\theta_1 = 73/170 \text{ and } 7/85 < \theta_2 < 169/2040 \text{ and } 1/12(169/170 - 12\theta_2) \leq \theta_3 < (169 - 2040\theta_2)/1700)$,
 or $(\theta_1 = 21/50 \text{ and } 56/675 < \theta_2 < 53/600 \text{ and } 1/12(53/50 - 12\theta_2) \leq \theta_3 < 1/500(53 - 600\theta_2))$,
 or $(31/70 < \theta_1 < 4/9 \text{ and } 1/27(14 - 28\theta_1) < \theta_2 < 1/12(4 - 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(\theta_1 = 4/9 \text{ and } 1/18 \leq \theta_2 \leq 134/2331 \text{ and } 1/12(8/9 - 12\theta_2) \leq \theta_3 < 1/45(4 - 54\theta_2))$,
 or $(\theta_1 = 4/9 \text{ and } 134/2331 < \theta_2 \leq 14/243 \text{ and } 1/12(8/9 - 12\theta_2) \leq \theta_3 < 1/45(4 - 54\theta_2))$,
 or $(\theta_1 = 4/9 \text{ and } 14/243 < \theta_2 < 2/27 \text{ and } 1/12(8/9 - 12\theta_2) \leq \theta_3 < 1/45(4 - 54\theta_2))$,
 or $(\theta_1 = 17/40 \text{ and } 327/4760 < \theta_2 < 3/40 \text{ and } 1/12(41/40 - 12\theta_2) \leq \theta_3 < 1/400(41 - 480\theta_2))$,
 or $(\theta_1 = 143/340 \text{ and } 89/1190 < \theta_2 < 27/340 \text{ and } 1/12(359/340 - 12\theta_2) \leq \theta_3 < (359 - 4080\theta_2)/3400)$,
 or $(\theta_1 = 7/17 \text{ and } 19/272 \leq \theta_2 < 3/34 \text{ and } 1/12(19/17 - 12\theta_2) \leq \theta_3 < 1/170(19 - 204\theta_2))$,
 or $(\theta_1 = 7/17 \text{ and } 47/680 \leq \theta_2 < 19/272 \text{ and } 1/12(19/17 - 12\theta_2) \leq \theta_3 < 1/170(19 - 204\theta_2))$,
 or $(\theta_1 = 15/34 \text{ and } 31/544 \leq \theta_2 < 1/17 \text{ and } 1/12(31/34 - 12\theta_2) \leq \theta_3 < 1/340(31 - 408\theta_2))$,
 or $(\theta_1 = 53/126 \text{ and } 163/2520 \leq \theta_2 < 19/288 \text{ and } 1/12(19/18 - 12\theta_2) \leq \theta_3 < 1/180(19 - 216\theta_2))$,
 or $(\theta_1 = 53/126 \text{ and } 19/288 \leq \theta_2 < 167/2499 \text{ and } 1/12(19/18 - 12\theta_2) \leq \theta_3 < 1/180(19 - 216\theta_2))$,
 or $(\theta_1 = 15/34 \text{ and } 37/680 \leq \theta_2 < 1/18 \text{ and } 1/12(31/34 - 12\theta_2) \leq \theta_3 < 1/340(31 - 408\theta_2))$,
 or $(79/180 \leq \theta_1 < 15/34 \text{ and } \theta_2 = 1/18 \text{ and } 1/12(10/3 - 7\theta_1) \leq \theta_3 < 1/30(10 - 21\theta_1))$,
 or $(15/34 < \theta_1 < 31/70 \text{ and } \theta_2 = 1/18 \text{ and } 1/12(10/3 - 7\theta_1) \leq \theta_3 < 1/30(10 - 21\theta_1))$,
 or $(\theta_1 = 5/12 \text{ and } 1/14 < \theta_2 < 1/12 \text{ and } 1/12(13/12 - 12\theta_2) \leq \theta_3 < 1/120(13 - 144\theta_2))$,
 or $(\theta_1 = 5/12 \text{ and } 1/15 \leq \theta_2 < 13/192 \text{ and } 1/12(13/12 - 12\theta_2) \leq \theta_3 < 1/120(13 - 144\theta_2))$,
 or $(\theta_1 = 5/12 \text{ and } 13/192 \leq \theta_2 \leq 1/14 \text{ and } 1/12(13/12 - 12\theta_2) \leq \theta_3 < 1/120(13 - 144\theta_2))$,
 or $(\theta_1 = 11/26 \text{ and } 33/520 \leq \theta_2 < 27/416 \text{ and } 1/12(27/26 - 12\theta_2) \leq \theta_3 < 1/260(27 - 312\theta_2))$,
 or $(\theta_1 = 11/26 \text{ and } 27/416 \leq \theta_2 < 15/221 \text{ and } 1/12(27/26 - 12\theta_2) \leq \theta_3 < 1/260(27 - 312\theta_2))$,
 or $(\theta_1 = 11/26 \text{ and } 15/221 < \theta_2 < 1/13 \text{ and } 1/12(27/26 - 12\theta_2) \leq \theta_3 < 1/260(27 - 312\theta_2))$,
 or $(\theta_1 = 61/140 \text{ and } 2/35 \leq \theta_2 < 19/320 \text{ and } 1/12(19/20 - 12\theta_2) \leq \theta_3 < 1/200(19 - 240\theta_2))$,
 or $(\theta_1 = 61/140 \text{ and } 19/320 \leq \theta_2 < 9/140 \text{ and } 1/12(19/20 - 12\theta_2) \leq \theta_3 < 1/200(19 - 240\theta_2))$,
 or $(\theta_1 = 15/34 \text{ and } 1/18 < \theta_2 < 31/544 \text{ and } 1/12(31/34 - 12\theta_2) \leq \theta_3 < 1/340(31 - 408\theta_2))$,
 or $(\theta_1 = 43/98 \text{ and } 109/1960 \leq \theta_2 < 13/224 \text{ and } 1/12(13/14 - 12\theta_2) \leq \theta_3 < 1/140(13 - 168\theta_2))$,
 or $(\theta_1 = 43/98 \text{ and } 13/224 \leq \theta_2 < 3/49 \text{ and } 1/12(13/14 - 12\theta_2) \leq \theta_3 < 1/140(13 - 168\theta_2))$,
 or $(\theta_1 = 13/32 \text{ and } 23/320 \leq \theta_2 < 37/512 \text{ and } 1/12(37/32 - 12\theta_2) \leq \theta_3 < 1/320(37 - 384\theta_2))$,
 or $(\theta_1 = 13/32 \text{ and } 37/512 \leq \theta_2 < 3/32 \text{ and } 1/12(37/32 - 12\theta_2) \leq \theta_3 < 1/320(37 - 384\theta_2))$,
 or $(\theta_1 = 139/340 \text{ and } 6/85 \leq \theta_2 < 387/5440 \text{ and } 1/12(387/340 - 12\theta_2) \leq \theta_3 < (387 - 4080\theta_2)/3400)$,
 or $(\theta_1 = 139/340 \text{ and } 387/5440 \leq \theta_2 < 31/340 \text{ and } 1/12(387/340 - 12\theta_2) \leq \theta_3 < (387 - 4080\theta_2)/3400)$,
 or $(\theta_1 = 37/90 \text{ and } 5/72 \leq \theta_2 < 101/1440 \text{ and } 1/12(101/90 - 12\theta_2) \leq \theta_3 < 1/900(101 - 1080\theta_2))$,
 or $(\theta_1 = 37/90 \text{ and } 101/1440 \leq \theta_2 < 4/45 \text{ and } 1/12(101/90 - 12\theta_2) \leq \theta_3 < 1/900(101 - 1080\theta_2))$,
 or $(\theta_1 = 127/306 \text{ and } 25/357 < \theta_2 < 23/306 \text{ and } 1/12(335/306 - 12\theta_2) \leq \theta_3 < (335 - 3672\theta_2)/3060)$,
 or $(\theta_1 = 127/306 \text{ and } 23/306 < \theta_2 < 13/153 \text{ and } 1/12(335/306 - 12\theta_2) \leq \theta_3 < (335 - 3672\theta_2)/3060)$,
 or $(\theta_1 = 127/306 \text{ and } 413/6120 \leq \theta_2 < 335/4896 \text{ and } 1/12(335/306 - 12\theta_2) \leq \theta_3 < (335 - 3672\theta_2)/3060)$,

or $(\theta_1 = 127/306 \text{ and } 335/4896 \leq \theta_2 \leq 25/357 \text{ and } 1/12(335/306 - 12\theta_2) \leq \theta_3 < (335 - 3672\theta_2)/3060)$,
 or $(\theta_1 = 143/340 \text{ and } 11/170 \leq \theta_2 < 359/5440 \text{ and } 1/12(359/340 - 12\theta_2) \leq \theta_3 < (359 - 4080\theta_2)/3400)$,
 or $(\theta_1 = 143/340 \text{ and } 359/5440 \leq \theta_2 < 159/2380 \text{ and } 1/12(359/340 - 12\theta_2) \leq \theta_3 < (359 - 4080\theta_2)/3400)$,
 or $(\theta_1 = 143/340 \text{ and } 159/2380 < \theta_2 \leq 89/1190 \text{ and } 1/12(359/340 - 12\theta_2) \leq \theta_3 < (359 - 4080\theta_2)/3400)$,
 or $(\theta_1 = 53/126 \text{ and } 11/147 < \theta_2 < 5/63 \text{ and } 1/12(19/18 - 12\theta_2) \leq \theta_3 < 1/180(19 - 216\theta_2))$,
 or $(\theta_1 = 53/126 \text{ and } 167/2499 < \theta_2 \leq 11/147 \text{ and } 1/12(19/18 - 12\theta_2) \leq \theta_3 < 1/180(19 - 216\theta_2))$,
 or $(\theta_1 = 17/40 \text{ and } 1/16 \leq \theta_2 < 41/640 \text{ and } 1/12(41/40 - 12\theta_2) \leq \theta_3 < 1/400(41 - 480\theta_2))$,
 or $(\theta_1 = 17/40 \text{ and } 41/640 \leq \theta_2 < 327/4760 \text{ and } 1/12(41/40 - 12\theta_2) \leq \theta_3 < 1/400(41 - 480\theta_2))$,
 or $(\theta_1 = 12/29 \text{ and } 2/29 < \theta_2 < 5/58 \text{ and } 1/12(32/29 - 12\theta_2) \leq \theta_3 < 1/145(16 - 174\theta_2))$,
 or $(\theta_1 = 12/29 \text{ and } 79/1160 \leq \theta_2 < 2/29 \text{ and } 1/12(32/29 - 12\theta_2) \leq \theta_3 < 1/145(16 - 174\theta_2))$,
 or $(9/20 < \theta_1 < 73/162 \text{ and } 1/40(11 - 20\theta_1) \leq \theta_2 < 1/24(3 - 4\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(130/289 \leq \theta_1 < 9/20 \text{ and } \theta_2 = 1/204(-28 + 85\theta_1) \text{ and } 1/17(8 - 17\theta_1) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(451/1003 \leq \theta_1 < 9/20 \text{ and } 2/37 < \theta_2 < 1/60(-7 + 25\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(451/1003 \leq \theta_1 < 9/20 \text{ and } 1/70(32 - 63\theta_1) \leq \theta_2 \leq 2/37 \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(451/1003 \leq \theta_1 < 9/20 \text{ and } \theta_2 = 1/60(-7 + 25\theta_1) \text{ and } 1/20(9 - 20\theta_1) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(130/289 \leq \theta_1 < 9/20 \text{ and } 1/204(-28 + 85\theta_1) < \theta_2 < 1/70(32 - 63\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(9/20 < \theta_1 < 73/162 \text{ and } 1/24(3 - 4\theta_1) \leq \theta_2 < 1/204(-28 + 85\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(9/20 < \theta_1 < 73/162 \text{ and } \theta_2 = 1/204(-28 + 85\theta_1) \text{ and } 1/17(8 - 17\theta_1) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(9/20 < \theta_1 < 73/162 \text{ and } 1/204(-28 + 85\theta_1) < \theta_2 < 1/70(32 - 63\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(9/20 < \theta_1 < 73/162 \text{ and } 2/37 < \theta_2 < 1/5(8 - 17\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(9/20 < \theta_1 < 73/162 \text{ and } 1/70(32 - 63\theta_1) \leq \theta_2 \leq 2/37 \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(157/343 \leq \theta_1 < 11/24 \text{ and } 1/40(11 - 20\theta_1) \leq \theta_2 < 1/24(3 - 4\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(16/35 < \theta_1 < 11/24 \text{ and } 1/24(3 - 4\theta_1) \leq \theta_2 < 1/204(-28 + 85\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(1772/3893 < \theta_1 < 11/24 \text{ and } \theta_2 = 1/204(-28 + 85\theta_1) \text{ and } 1/17(8 - 17\theta_1) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(1772/3893 < \theta_1 < 11/24 \text{ and } 1/204(-28 + 85\theta_1) < \theta_2 \leq 2/37 \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(11/24 < \theta_1 \leq 1444/3145 \text{ and } 1/40(11 - 20\theta_1) \leq \theta_2 < 1/24(3 - 4\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(11/24 < \theta_1 \leq 1444/3145 \text{ and } 1/24(3 - 4\theta_1) \leq \theta_2 < 1/204(-28 + 85\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(11/24 < \theta_1 \leq 1444/3145 \text{ and } \theta_2 = 1/204(-28 + 85\theta_1) \text{ and } 1/17(8 - 17\theta_1) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(11/24 < \theta_1 < 1444/3145 \text{ and } 1/204(-28 + 85\theta_1) < \theta_2 \leq 2/37 \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(\theta_1 = 11/24 \text{ and } 2/37 < \theta_2 < 19/288 \text{ and } 1/288(19 - 288\theta_2) \leq \theta_3 < 1/240(19 - 288\theta_2))$,
 or $(\theta_1 = 11/24 \text{ and } 11/240 \leq \theta_2 < 7/144 \text{ and } 1/288(19 - 288\theta_2) \leq \theta_3 < 1/48)$,
 or $(\theta_1 = 11/24 \text{ and } 263/4896 < \theta_2 \leq 2/37 \text{ and } 1/288(19 - 288\theta_2) \leq \theta_3 < 1/240(19 - 288\theta_2))$,
 or $(\theta_1 = 11/24 \text{ and } 7/144 \leq \theta_2 < 263/4896 \text{ and } 1/288(19 - 288\theta_2) \leq \theta_3 < 1/240(19 - 288\theta_2))$,
 or $(87/188 < \theta_1 < 13/28 \text{ and } (1 - \theta_1)/12 < \theta_2 < 1/24(3 - 4\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(87/188 < \theta_1 < 13/28 \text{ and } 1/24(3 - 4\theta_1) \leq \theta_2 < 1/5(-3 + 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(769/1673 \leq \theta_1 < 13/28 \text{ and } \theta_2 = 1/204(-28 + 85\theta_1) \text{ and } 1/17(8 - 17\theta_1) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(769/1673 \leq \theta_1 < 13/28 \text{ and } 2/37 < \theta_2 < 1/204(-28 + 85\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(87/188 \leq \theta_1 < 13/28 \text{ and } 1/5(-3 + 7\theta_1) < \theta_2 \leq 2/37 \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(\theta_1 = 13/28 \text{ and } 5/112 < \theta_2 < 1/21 \text{ and } 1/16(1 - 16\theta_2) \leq \theta_3 < 1/56)$,
 or $(13/28 < \theta_1 < 316/679 \text{ and } 1/28(-2 + 7\theta_1) \leq \theta_2 < 1/24(3 - 4\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(13/28 < \theta_1 < 316/679 \text{ and } (1 - \theta_1)/12 < \theta_2 < 1/28(-2 + 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(23/49 < \theta_1 < 8/17 \text{ and } (1 - \theta_1)/12 < \theta_2 \leq 1/4(3 - 6\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(15/32 < \theta_1 < 8/17 \text{ and } \theta_2 = 1/204(-28 + 85\theta_1) \text{ and } 1/17(8 - 17\theta_1) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(15/32 < \theta_1 < 8/17 \text{ and } 1/5(-3 + 7\theta_1) < \theta_2 < 1/204(-28 + 85\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(226/481 < \theta_1 < 8/17 \text{ and } 1/2(-6 + 13\theta_1) \leq \theta_2 < 1/5(-3 + 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(226/481 < \theta_1 < 8/17 \text{ and } 2/37 < \theta_2 < 1/2(-6 + 13\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1 - 2\theta_1 - \theta_2)$,
 or $(226/481 < \theta_1 < 8/17 \text{ and } 1/28(-2 + 7\theta_1) \leq \theta_2 \leq 2/37 \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1 - 2\theta_1 - \theta_2)$,
 or $(23/49 < \theta_1 < 8/17 \text{ and } 1/4(3 - 6\theta_1) < \theta_2 < 1/28(-2 + 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1 - 2\theta_1 - \theta_2)$,
 or $(\theta_1 = 4/9 \text{ and } 19/360 \leq \theta_2 < 1/18 \text{ and } 1/12(8/9 - 12\theta_2) \leq \theta_3 < 1/45(4 - 54\theta_2))$,

or $(31/70 < \theta_1 < 4/9$ and $\theta_2 = 1/18$ and $1/12(10/3 - 7\theta_1) \leq \theta_3 < 1/30(10 - 21\theta_1)$),

or $(1444/3145 < \theta_1 < 74/161$ and $1/40(11 - 20\theta_1) \leq \theta_2 < 1/24(3 - 4\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),

or $(74/161 < \theta_1 < 769/1673$ and $1/40(11 - 20\theta_1) \leq \theta_2 < 1/24(3 - 4\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),

or $(769/1673 \leq \theta_1 < 23/50$ and $1/40(11 - 20\theta_1) \leq \theta_2 < 1/24(3 - 4\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),

or $(23/50 \leq \theta_1 \leq 41/89$ and $(1 - \theta_1)/12 < \theta_2 < 1/24(3 - 4\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),

or $(177/388 < \theta_1 \leq 53/116$ and $1/5(8 - 17\theta_1) < \theta_2 < 1/24(3 - 4\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),

or $(53/116 < \theta_1 \leq 16/35$ and $1/40(11 - 20\theta_1) \leq \theta_2 < 1/24(3 - 4\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),

or $(279/616 < \theta_1 < 5/11$ and $1/70(32 - 63\theta_1) \leq \theta_2 < 1/24(3 - 4\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),

or $(1772/3893 < \theta_1 \leq 51/112$ and $1/70(32 - 63\theta_1) \leq \theta_2 < 1/24(3 - 4\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),

or $(51/112 < \theta_1 \leq 177/388$ and $1/40(11 - 20\theta_1) \leq \theta_2 < 1/24(3 - 4\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),

or $(16/35 < \theta_1 < 157/343$ and $1/40(11 - 20\theta_1) \leq \theta_2 < 1/24(3 - 4\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),

or $(5/11 \leq \theta_1 < 1772/3893$ and $1/70(32 - 63\theta_1) \leq \theta_2 < 1/24(3 - 4\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),

or $(41/89 < \theta_1 < 87/188$ and $1/5(-3 + 7\theta_1) < \theta_2 < 1/24(3 - 4\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),

or $(316/679 < \theta_1 \leq 121/259$ and $1/28(-2 + 7\theta_1) \leq \theta_2 < 1/24(3 - 4\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),

or $(121/259 < \theta_1 \leq 15/32$ and $1/28(-2 + 7\theta_1) \leq \theta_2 < 1/24(3 - 4\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),

or $(1772/3893 < \theta_1 < 51/112$ and $1/40(11 - 20\theta_1) \leq \theta_2 < 1/70(32 - 63\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),

or $(5/11 \leq \theta_1 < 1772/3893$ and $1/40(11 - 20\theta_1) \leq \theta_2 < 1/70(32 - 63\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),

or $(41/89 < \theta_1 \leq 87/188$ and $(1 - \theta_1)/12 < \theta_2 < 1/5(-3 + 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),

or $(316/679 < \theta_1 \leq 15/32$ and $(1 - \theta_1)/12 < \theta_2 < 1/28(-2 + 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),

or $(15/32 < \theta_1 \leq 23/49$ and $(1 - \theta_1)/12 < \theta_2 < 1/28(-2 + 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),

or $(177/388 < \theta_1 < 53/116$ and $1/40(11 - 20\theta_1) \leq \theta_2 < 1/5(8 - 17\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),

or $(15/32 < \theta_1 < 23/49$ and $1/28(-2 + 7\theta_1) \leq \theta_2 \leq 1/4(3 - 6\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),

or $(4/9 < \theta_1 < 116/259$ and $1/16(4 - 7\theta_1) \leq \theta_2 \leq 1/259(130 - 259\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),

or $(73/162 \leq \theta_1 \leq 4244/9401$ and $1/24(3 - 4\theta_1) \leq \theta_2 < 1/204(-28 + 85\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),

or $(279/616 < \theta_1 < 5/11$ and $1/24(3 - 4\theta_1) \leq \theta_2 < 1/204(-28 + 85\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),

or $(1772/3893 < \theta_1 < 177/388$ and $1/24(3 - 4\theta_1) \leq \theta_2 < 1/5(8 - 17\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),

or $(177/388 < \theta_1 \leq 16/35$ and $1/24(3 - 4\theta_1) \leq \theta_2 < 1/204(-28 + 85\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),

or $(4244/9401 < \theta_1 < 279/616$ and $1/24(3 - 4\theta_1) \leq \theta_2 < 1/70(32 - 63\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),

or $(5/11 \leq \theta_1 < 1772/3893$ and $1/24(3 - 4\theta_1) \leq \theta_2 < 1/204(-28 + 85\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),

or $(769/1673 \leq \theta_1 < 87/188$ and $1/24(3 - 4\theta_1) \leq \theta_2 \leq 2/37$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),

or $(13/28 \leq \theta_1 < 316/679$ and <

or $(116/259 < \theta_1 < 185/413 \text{ and } 2/37 < \theta_2 < 1/60(-7 + 25\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(116/259 < \theta_1 < 185/413 \text{ and } 1/70(32 - 63\theta_1) < \theta_2 \leq 2/37 \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(116/259 < \theta_1 < 185/413 \text{ and } \theta_2 = 1/60(-7 + 25\theta_1) \text{ and } 1/20(9 - 20\theta_1) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(185/413 \leq \theta_1 < 451/1003 \text{ and } 2/37 < \theta_2 < 1/60(-7 + 25\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(185/413 \leq \theta_1 < 451/1003 \text{ and } 1/70(32 - 63\theta_1) \leq \theta_2 \leq 2/37 \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(185/413 \leq \theta_1 < 451/1003 \text{ and } \theta_2 = 1/60(-7 + 25\theta_1) \text{ and } 1/20(9 - 20\theta_1) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(73/162 \leq \theta_1 < 5/11 \text{ and } 2/37 < \theta_2 < 1/5(8 - 17\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(73/162 \leq \theta_1 < 4244/9401 \text{ and } 1/204(-28 + 85\theta_1) < \theta_2 < 1/70(32 - 63\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(73/162 \leq \theta_1 < 4244/9401 \text{ and } 1/70(32 - 63\theta_1) \leq \theta_2 \leq 2/37 \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(4244/9401 \leq \theta_1 < 5/11 \text{ and } 1/204(-28 + 85\theta_1) < \theta_2 \leq 2/37 \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(286/629 \leq \theta_1 < 1772/3893 \text{ and } 1/204(-28 + 85\theta_1) < \theta_2 < 1/5(8 - 17\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(5/11 \leq \theta_1 < 286/629 \text{ and } 1/204(-28 + 85\theta_1) < \theta_2 \leq 2/37 \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(5/11 \leq \theta_1 < 286/629 \text{ and } 2/37 < \theta_2 < 1/5(8 - 17\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(4244/9401 < \theta_1 \leq 279/616 \text{ and } 1/70(32 - 63\theta_1) \leq \theta_2 < 1/204(-28 + 85\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(13/28 \leq \theta_1 < 316/679 \text{ and } 2/37 < \theta_2 < 1/204(-28 + 85\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(316/679 < \theta_1 \leq 121/259 \text{ and } 2/37 < \theta_2 < 1/204(-28 + 85\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(121/259 < \theta_1 \leq 15/32 \text{ and } 1/5(-3 + 7\theta_1) < \theta_2 < 1/204(-28 + 85\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(1444/3145 < \theta_1 \leq 74/161 \text{ and } 2/37 < \theta_2 < 1/204(-28 + 85\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(74/161 < \theta_1 < 769/1673 \text{ and } 2/37 < \theta_2 < 1/204(-28 + 85\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(13/28 \leq \theta_1 < 316/679 \text{ and } 1/5(-3 + 7\theta_1) < \theta_2 \leq 2/37 \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(316/679 < \theta_1 < 121/259 \text{ and } 1/5(-3 + 7\theta_1) < \theta_2 \leq 2/37 \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(121/259 < \theta_1 \leq 15/32 \text{ and } 2/37 < \theta_2 < 1/5(-3 + 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(15/32 < \theta_1 \leq 226/481 \text{ and } 2/37 < \theta_2 < 1/5(-3 + 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(15/32 < \theta_1 < 226/481 \text{ and } 1/2(-6 + 13\theta_1) \leq \theta_2 \leq 2/37 \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(\theta_1 = 116/259 \text{ and } 529/10360 \leq \theta_2 < 27/518 \text{ and } 1/111(8 - 111\theta_2) \leq \theta_3 < 1/185(16 - 222\theta_2))$,
 or $(\theta_1 = 116/259 \text{ and } 2/37 < \theta_2 < 1087/15540 \text{ and } 1/111(8 - 111\theta_2) \leq \theta_3 < 1/185(16 - 222\theta_2))$,
 or $(\theta_1 = 116/259 \text{ and } 27/518 \leq \theta_2 < 2/37 \text{ and } 1/111(8 - 111\theta_2) \leq \theta_3 < 1/185(16 - 222\theta_2))$,
 or $(\theta_1 = 116/259 \text{ and } 1087/15540 < \theta_2 < 8/111 \text{ and } 1/111(8 - 111\theta_2) \leq \theta_3 < 1/185(16 - 222\theta_2))$,
 or $(\theta_1 = 1772/3893 \text{ and } 7383/155720 \leq \theta_2 < 1294/27251 \text{ and } (264 - 3893\theta_2)/3893 \leq \theta_3 < 349/15572)$,
 or $(\theta_1 = 1772/3893 \text{ and } 1294/27251 \leq \theta_2 < 4591/93432 \text{ and } (264 - 3893\theta_2)/3893 \leq \theta_3 < 349/15572)$,
 or $(\theta_1 = 1772/3893 \text{ and } 2/37 < \theta_2 < 264/3893 \text{ and } (264 - 3893\theta_2)/3893 \leq \theta_3 < (1584 - 23358\theta_2)/19465)$,
 or $(\theta_1 = 1772/3893 \text{ and } 12/229 < \theta_2 \leq 2/37 \text{ and } (264 - 3893\theta_2)/3893 \leq \theta_3 < (1584 - 23358\theta_2)/19465)$,
 or $(\theta_1 = 1772/3893 \text{ and } 4591/93432 \leq \theta_2 < 12/229 \text{ and } (264 - 3893\theta_2)/3893 \leq \theta_3 < (1584 - 23358\theta_2)/19465)$,
 or $(\theta_1 = 316/679 \text{ and } 61/1358 \leq \theta_2 < 773/16296 \text{ and } 1/97(6 - 97\theta_2) \leq \theta_3 < 47/2716)$,
 or $(\theta_1 = 316/679 \text{ and } 121/2716 < \theta_2 < 61/1358 \text{ and } 1/97(6 - 97\theta_2) \leq \theta_3 < 47/2716)$,
 or $(\theta_1 = 316/679 \text{ and } 2/37 < \theta_2 < 654/11543 \text{ and } 1/97(6 - 97\theta_2) \leq \theta_3 < 1/485(36 - 582\theta_2))$,
 or $(\theta_1 = 316/679 \text{ and } 5/97 < \theta_2 \leq 2/37 \text{ and } 1/97(6 - 97\theta_2) \leq \theta_3 < 1/485(36 - 582\theta_2))$,
 or $(\theta_1 = 316/679 \text{ and } 773/16296 \leq \theta_2 < 5/97 \text{ and } 1/97(6 - 97\theta_2) \leq \theta_3 < 1/485(36 - 582\theta_2))$,
 or $(\theta_1 = 316/679 \text{ and } 654/11543 < \theta_2 < 6/97 \text{ and } 1/97(6 - 97\theta_2) \leq \theta_3 < 1/485(36 - 582\theta_2))$,
 or $(\theta_1 = 9/20 \text{ and } 2/37 < \theta_2 < 17/240 \text{ and } 1/240(17 - 240\theta_2) \leq \theta_3 < 1/200(17 - 240\theta_2))$,
 or $(\theta_1 = 9/20 \text{ and } 73/1400 \leq \theta_2 \leq 2/37 \text{ and } 1/240(17 - 240\theta_2) \leq \theta_3 < 1/200(17 - 240\theta_2))$,
 or $(\theta_1 = 9/20 \text{ and } 41/816 < \theta_2 < 73/1400 \text{ and } 1/240(17 - 240\theta_2) \leq \theta_3 < 1/200(17 - 240\theta_2))$,
 or $(\theta_1 = 9/20 \text{ and } 1/20 \leq \theta_2 < 41/816 \text{ and } 1/240(17 - 240\theta_2) \leq \theta_3 < 1/200(17 - 240\theta_2))$,
 or $(\theta_1 = 74/161 \text{ and } 291/6440 \leq \theta_2 < 187/3864 \text{ and } 1/46(3 - 46\theta_2) \leq \theta_3 < 13/644)$,
 or $(\theta_1 = 74/161 \text{ and } 187/3864 \leq \theta_2 \leq 2/37 \text{ and } 1/46(3 - 46\theta_2) \leq \theta_3 < 1/115(9 - 138\theta_2))$,
 or $(15/32 < \theta_1 \leq 23/49 \text{ and } 1/4(3 - 6\theta_1) < \theta_2 < 1/2(-6 + 13\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1 - 2\theta_1 - \theta_2)$,
 or $(23/49 < \theta_1 \leq 226/481 \text{ and } 1/28(-2 + 7\theta_1) \leq \theta_2 < 1/2(-6 + 13\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1 - 2\theta_1 - \theta_2)$,
 or $(15/32 < \theta_1 < 8/17 \text{ and } \theta_2 = 1/5(-3 + 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(13/28 < \theta_1 < 316/679 \text{ and } 1/28(-2 + 7\theta_1) \leq \theta_2 < 1/24(3 - 4\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,

or $(13/28 < \theta_1 < 316/679 \text{ and } (1 - \theta_1)/12 < \theta_2 < 1/28(-2 + 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(13/28 < \theta_1 < 316/679 \text{ and } \theta_2 = 1/5(-3 + 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(316/679 < \theta_1 \leq 121/259 \text{ and } \theta_2 = 1/5(-3 + 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(121/259 < \theta_1 \leq 15/32 \text{ and } \theta_2 = 1/5(-3 + 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(9/20 < \theta_1 < 73/162 \text{ and } 1/40(11 - 20\theta_1) \leq \theta_2 < 1/24(3 - 4\theta_1) \text{ and } 1/2(1 - 2\theta_1 - \theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(157/343 \leq \theta_1 < 11/24 \text{ and } ((1/40(11 - 20\theta_1) \leq \theta_2 \leq 1/6(-2 + 5\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(1/6(-2 + 5\theta_1) < \theta_2 < 1/24(3 - 4\theta_1) \text{ and } 1/2(1 - 2\theta_1 - \theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)))$,
 or $(177/388 < \theta_1 \leq 58/127 \text{ and } \theta_2 = 1/5(8 - 17\theta_1) \text{ and } 1/2(1 - 2\theta_1 - \theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(58/127 < \theta_1 \leq 53/116 \text{ and } \theta_2 = 1/5(8 - 17\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(9/20 < \theta_1 < 73/162 \text{ and } 1/24(3 - 4\theta_1) \leq \theta_2 < 1/7(-1 + 3\theta_1) \text{ and } 1/2(1 - 2\theta_1 - \theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(16/35 < \theta_1 < 11/24 \text{ and } 1/24(3 - 4\theta_1) \leq \theta_2 < 1/7(-1 + 3\theta_1) \text{ and } 1/2(1 - 2\theta_1 - \theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(61/134 < \theta_1 \leq 177/388 \text{ and } \theta_2 = 1/5(8 - 17\theta_1) \text{ and } 1/2(1 - 2\theta_1 - \theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(11/24 < \theta_1 \leq 1444/3145 \text{ and } 1/40(11 - 20\theta_1) \leq \theta_2 < 1/24(3 - 4\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(87/188 < \theta_1 < 13/28 \text{ and } (1 - \theta_1)/12 < \theta_2 < 1/24(3 - 4\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(41/89 < \theta_1 < 87/188 \text{ and } \theta_2 = 1/5(-3 + 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
 or $(11/24 < \theta_1 \leq 1444/3145 \text{ and } ((1/24(3 - 4\theta_1) \leq \theta_2 \leq 1/6(-2 + 5\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(1/6(-2 + 5\theta_1) < \theta_2 < 1/7(-1 + 3\theta_1) \text{ and } 1/2(1 - 2\theta_1 - \theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)))$,
 or $(87/188 < \theta_1 < 13/28 \text{ and } 1/24(3 - 4\theta_1) \leq \theta_2 < 1/5(-3 + 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(769/1673 \leq \theta_1 < 13/28 \text{ and } 2/37 < \theta_2 < 1/7(-1 + 3\theta_1) \text{ and } 1/2(1 - 2\theta_1 - \theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(87/188 \leq \theta_1 < 13/28 \text{ and } ((1/5(-3 + 7\theta_1) < \theta_2 \leq 1/6(-2 + 5\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(1/6(-2 + 5\theta_1) < \theta_2 \leq 2/37 \text{ and } 1/2(1 - 2\theta_1 - \theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)))$,
 or $(\theta_1 = 316/679 \text{ and } 61/1358 \leq \theta_2 < 773/16296 \text{ and } 1/97(6 - 97\theta_2) \leq \theta_3 < 47/2716)$,
 or $(\theta_1 = 316/679 \text{ and } 121/2716 < \theta_2 < 61/1358 \text{ and } 1/97(6 - 97\theta_2) \leq \theta_3 < 47/2716)$,
 or $(\theta_1 = 316/679 \text{ and } (2/37 < \theta_2 \leq 37/679 \text{ and } 1/97(6 - 97\theta_2) \leq \theta_3 < 1/485(36 - 582\theta_2))$,
 or $(37/679 < \theta_2 < 269/4753 \text{ and } 1/2(47/679 - \theta_2) \leq \theta_3 < 1/485(36 - 582\theta_2)))$,
 or $(\theta_1 = 316/679 \text{ and } 5/97 < \theta_2 \leq 2/37 \text{ and } 1/97(6 - 97\theta_2) \leq \theta_3 < 1/485(36 - 582\theta_2))$,
 or $(\theta_1 = 316/679 \text{ and } 773/16296 \leq \theta_2 < 5/97 \text{ and } 1/97(6 - 97\theta_2) \leq \theta_3 < 1/485(36 - 582\theta_2))$,
 or $(\theta_1 = 1772/3893 \text{ and } 7383/155720 \leq \theta_2 < 1294/27251 \text{ and } 1/2(349/3893 - \theta_2) \leq \theta_3 < 349/15572)$,
 or $(\theta_1 = 1772/3893 \text{ and } 1294/27251 \leq \theta_2 < 4591/93432 \text{ and } 1/2(349/3893 - \theta_2) \leq \theta_3 < 349/15572)$,
 or $(\theta_1 = 1772/3893 \text{ and } 4591/93432 \leq \theta_2 < 1423/27251 \text{ and } 1/2(349/3893 - \theta_2) \leq \theta_3 < (1584 - 23358\theta_2)/19465)$,
 or $(\theta_1 = 74/161 \text{ and } 291/6440 \leq \theta_2 < 187/3864 \text{ and } 1/46(3 - 46\theta_2) \leq \theta_3 < 13/644)$,
 or $(\theta_1 = 74/161 \text{ and } ((187/3864 \leq \theta_2 \leq 8/161 \text{ and } 1/46(3 - 46\theta_2) \leq \theta_3 < 1/115(9 - 138\theta_2))$,
 or $(8/161 < \theta_2 \leq 2/37 \text{ and } 1/2(13/161 - \theta_2) \leq \theta_3 < 1/115(9 - 138\theta_2)))$,
 or $(2/5 < \theta_1 \leq 272/679 \text{ and } 1/2(1 - 2\theta_1) < \theta_2 < 1/12(4 - 7\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(130/289 < \theta_1 < 9/20 \text{ and } 1/2(1 - 2\theta_1) \leq \theta_2 < 1/204(-28 + 85\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(4/9 < \theta_1 < 116/259 \text{ and } 1/2(1 - 2\theta_1) \leq \theta_2 < 1/16(4 - 7\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(116/259 < \theta_1 < 185/413 \text{ and } 1/2(1 - 2\theta_1) \leq \theta_2 < 1/70(32 - 63\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(116/259 < \theta_1 < 185/413 \text{ and } \theta_2 = 1/70(32 - 63\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(185/413 \leq \theta_1 < 451/1003 \text{ and } 1/2(1 - 2\theta_1) \leq \theta_2 < 1/70(32 - 63\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(451/1003 \leq \theta_1 < 130/289 \text{ and } 1/2(1 - 2\theta_1) \leq \theta_2 < 1/70(32 - 63\theta_1) \text{ and } 1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(272/679 < \theta_1 \leq 1041/2590 \text{ and } 1/2(1 - 2\theta_1) < \theta_2 < 1/12(4 - 7\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(1041/2590 < \theta_1 \leq 524/1295 \text{ and } 1/2(1 - 2\theta_1) < \theta_2 < 1/12(4 - 7\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(524/1295 < \theta_1 < 13/32 \text{ and } 1/259(130 - 259\theta_1) < \theta_2 < 1/12(4 - 7\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(13/32 < \theta_1 \leq 208/511 \text{ and } 1/259(130 - 259\theta_1) < \theta_2 < 1/12(4 - 7\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(20/49 < \theta_1 < 139/340 \text{ and } 1/27(14 - 28\theta_1) < \theta_2 < 1/12(4 - 7\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(139/340 < \theta_1 < 37/90 \text{ and } 1/27(14 - 28\theta_1) < \theta_2 < 1/12(4 - 7\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(208/511 < \theta_1 < 257/630 \text{ and } 1/119(57 - 112\theta_1) < \theta_2 < 1/12(4 - 7\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(257/630 \leq \theta_1 \leq 20/49 \text{ and } 1/119(57 - 112\theta_1) \leq \theta_2 < 1/12(4 - 7\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
 or $(127/306 < \theta_1 < 32/77 \text{ and } 1/11 < \theta_2 < 1/12(4 - 7\theta_1) \text{ and } 0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,

($32/77 \leq \theta_1 < 5/12$ and $1/27(14 - 28\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($127/308 < \theta_1 \leq 12/29$ and $1/11 \leq \theta_2 < 1/12(4 - 7\theta_1)$ and $0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($12/29 < \theta_1 < 29/70$ and $1/11 < \theta_2 < 1/12(4 - 7\theta_1)$ and $0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($29/70 \leq \theta_1 < 127/306$ and $1/11 \leq \theta_2 < 1/12(4 - 7\theta_1)$ and $0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($5/12 < \theta_1 < 68/163$ and $1/27(14 - 28\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($34/81 \leq \theta_1 < 21/50$ and $1/27(14 - 28\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($21/50 < \theta_1 < 143/340$ and $1/27(14 - 28\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($143/340 < \theta_1 < 53/126$ and $1/27(14 - 28\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($53/126 < \theta_1 < 11/26$ and $1/27(14 - 28\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($11/26 < \theta_1 < 76/179$ and $1/27(14 - 28\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($76/179 \leq \theta_1 < 17/40$ and $1/7(-2 + 6\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($17/40 < \theta_1 < 73/170$ and $1/7(-2 + 6\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($52/121 \leq \theta_1 < 31/72$ and $1/27(14 - 28\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($31/72 \leq \theta_1 \leq 4061/9418$ and $1/27(14 - 28\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($4061/9418 < \theta_1 < 61/140$ and $1/238(-27 + 102\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($61/140 < \theta_1 < 43/98$ and $1/238(-27 + 102\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($43/98 < \theta_1 \leq 29/66$ and $1/238(-27 + 102\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($29/66 < \theta_1 < 15/34$ and $1/238(-27 + 102\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($15/34 < \theta_1 < 638/1445$ and $1/238(-27 + 102\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($638/1445 < \theta_1 < 4/9$ and $1/27(14 - 28\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($68/163 \leq \theta_1 < 28/67$ and $-2 + 5\theta_1 < \theta_2 < 1/12(4 - 7\theta_1)$ and $0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($73/170 < \theta_1 < 52/121$ and $1/7(-2 + 6\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($23/55 < \theta_1 < 34/81$ and $1/27(14 - 28\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($28/67 < \theta_1 \leq 23/55$ and $1/27(14 - 28\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ and $0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($524/1295 < \theta_1 < 13/32$ and $1/2(1 - 2\theta_1) < \theta_2 \leq 1/259(130 - 259\theta_1)$ and $0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($13/32 < \theta_1 < 139/340$ and $1/2(1 - 2\theta_1) < \theta_2 \leq 1/259(130 - 259\theta_1)$ and $0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($139/340 < \theta_1 < 37/90$ and $1/2(1 - 2\theta_1) < \theta_2 \leq 1/259(130 - 259\theta_1)$ and $0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($127/306 < \theta_1 < 5/12$ and $1/2(1 - 2\theta_1) < \theta_2 \leq 1/259(130 - 259\theta_1)$ and $0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($12/29 < \theta_1 < 127/306$ and $1/2(1 - 2\theta_1) < \theta_2 \leq 1/259(130 - 259\theta_1)$ and $0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($257/630 \leq \theta_1 < 139/340$ and $1/259(130 - 259\theta_1) < \theta_2 < 1/119(57 - 112\theta_1)$ and $0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($20/49 < \theta_1 < 139/340$ and $1/119(57 - 112\theta_1) \leq \theta_2 \leq 1/27(14 - 28\theta_1)$ and $0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($139/340 < \theta_1 < 37/90$ and $1/259(130 - 259\theta_1) < \theta_2 \leq 1/119(57 - 112\theta_1)$ and $0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($139/340 < \theta_1 < 37/90$ and $1/119(57 - 112\theta_1) < \theta_2 \leq 1/27(14 - 28\theta_1)$ and $0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($208/511 < \theta_1 < 257/630$ and $1/259(130 - 259\theta_1) < \theta_2 \leq 1/119(57 - 112\theta_1)$ and $0 < \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),

(11/26 < θ_1 < 204/481 and 1/7(-2 + 6 θ_1) < $\theta_2 \leq 1/259(130 - 259\theta_1)$ and 0 < θ_3 < 1/10(4 - 7 θ_1 - 12 θ_2)),
or (11/26 < $\theta_1 \leq 204/481$ and 1/259(130 - 259 θ_1) < $\theta_2 \leq 1/27(14 - 28\theta_1)$ and 0 < θ_3 < 1/10(4 - 7 θ_1 - 12 θ_2)),
or (76/179 < θ_1 < 17/40 and 1/27(14 - 28 θ_1) < $\theta_2 \leq 1/7(-2 + 6\theta_1)$ and 0 < θ_3 < 1/10(4 - 7 θ_1 - 12 θ_2)),
or (76/179 < θ_1 < 17/40 and 1/259(130 - 259 θ_1) < $\theta_2 \leq 1/27(14 - 28\theta_1)$ and 0 < θ_3 < 1/10(4 - 7 θ_1 - 12 θ_2)),
or (17/40 < θ_1 < 73/170 and 1/27(14 - 28 θ_1) < $\theta_2 \leq 1/7(-2 + 6\theta_1)$ and 0 < θ_3 < 1/10(4 - 7 θ_1 - 12 θ_2)),
or (17/40 < θ_1 < 73/170 and 1/259(130 - 259 θ_1) < $\theta_2 \leq 1/27(14 - 28\theta_1)$ and 0 < θ_3 < 1/10(4 - 7 θ_1 - 12 θ_2)),
or (73/170 $\leq \theta_1$ < 31/72 and 1/238(-27 + 102 θ_1) < $\theta_2 \leq 1/259(130 - 259\theta_1)$ and 0 < θ_3 < 1/10(4 - 7 θ_1 - 12 θ_2)),
or (73/170 $\leq \theta_1$ < 31/72 and 1/259(130 - 259 θ_1) < $\theta_2 \leq 1/27(14 - 28\theta_1)$ and 0 < θ_3 < 1/10(4 - 7 θ_1 - 12 θ_2)),
or (73/170 $\leq \theta_1$ < 52/121 and 1/27(14 - 28 θ_1) < $\theta_2 \leq 1/7(-2 + 6\theta_1)$ and 0 < θ_3 < 1/10(4 - 7 θ_1 - 12 θ_2)),
or (31/72 $\leq \theta_1$ < 5419/12580 and 1/238(-27 + 102 θ_1) < $\theta_2 \leq 1/259(130 - 259\theta_1)$ and 0 < θ_3 < 1/10(4 - 7 θ_1 - 12 θ_2)),
or (31/72 $\leq \theta_1 \leq 5419/12580$ and 1/259(130 - 259 θ_1) < $\theta_2 \leq 1/27(14 - 28\theta_1)$ and 0 < θ_3 < 1/10(4 - 7 θ_1 - 12 θ_2)),
or (4061/9418 < θ_1 < 61/140 and 1/27(14 - 28 θ_1) < θ_2 < 1/238(-27 + 102 θ_1) and 0 < θ_3 < 1/10(4 - 7 θ_1 - 12 θ_2)),
or (4061/9418 < θ_1 < 61/140 and 1/259(130 - 259 θ_1) < $\theta_2 \leq 1/27(14 - 28\theta_1)$ and 0 < θ_3 < 1/10(4 - 7 θ_1 - 12 θ_2)),
or (61/140 < θ_1 < 43/98 and 1/27(14 - 28 θ_1) < θ_2 < 1/238(-27 + 102 θ_1) and 0 < θ_3 < 1/10(4 - 7 θ_1 - 12 θ_2)),
or (61/140 < θ_1 < 43/98 and 1/259(130 - 259 θ_1) < $\theta_2 \leq 1/27(14 - 28\theta_1)$ and 0 < θ_3 < 1/10(4 - 7 θ_1 - 12 θ_2)),
or (43/98 < θ_1 < 15/34 and 1/27(14 - 28 θ_1) < θ_2 < 1/238(-27 + 102 θ_1) and 0 < θ_3 < 1/10(4 - 7 θ_1 - 12 θ_2)),
or (43/98 < θ_1 < 15/34 and 1/259(130 - 259 θ_1) < $\theta_2 \leq 1/27(14 - 28\theta_1)$ and 0 < θ_3 < 1/10(4 - 7 θ_1 - 12 θ_2)),
or (15/34 < θ_1 < 4/9 and 1/259(130 - 259 θ_1) < $\theta_2 \leq 1/27(14 - 28\theta_1)$ and 0 < θ_3 < 1/10(4 - 7 θ_1 - 12 θ_2)),
or (15/34 < $\theta_1 \leq 638/1445$ and 1/27(14 - 28 θ_1) < θ_2 < 1/238(-27 + 102 θ_1) and 0 < θ_3 < 1/10(4 - 7 θ_1 - 12 θ_2)),
or (108/259 < θ_1 < 68/163 and -2 + 5 θ_1 < $\theta_2 \leq 1/27(14 - 28\theta_1)$ and 0 < θ_3 < 1/10(4 - 7 θ_1 - 12 θ_2)),
or (204/481 < $\theta_1 \leq 76/179$ and 1/259(130 - 259 θ_1) < $\theta_2 \leq 1/7(-2 + 6\theta_1)$ and 0 < θ_3 < 1/10(4 - 7 θ_1 - 12 θ_2)),
or (5419/12580 < $\theta_1 \leq 4061/9418$ and 1/259(130 - 259 θ_1) < θ_2 < 1/238(-27 + 102 θ_1) and 0 < θ_3 < 1/10(4 - 7 θ_1 - 12 θ_2)),
or (5419/12580 < θ_1 < 4061/9418 and 1/238(-27 + 102 θ_1) < $\theta_2 \leq 1/27(14 - 28\theta_1)$ and 0 < θ_3 < 1/10(4 - 7 θ_1 - 12 θ_2)),
or (204/481 < θ_1 < 76/179 and 1/7(-2 + 6 θ_1) < $\theta_2 \leq 1/27(14 - 28\theta_1)$ and 0 < θ_3 < 1/10(4 - 7 θ_1 - 12 θ_2)),
or (108/259 < $\theta_1 \leq 68/163$ and 1/259(130 - 259 θ_1) < θ_2 < -2 + 5 θ_1 and 0 < θ_3 < 1/10(4 - 7 θ_1 - 12 θ_2)),
or (68/163 < $\theta_1 \leq 28/67$ and 1/27(14 - 28 θ_1) < θ_2 < -2 + 5 θ_1 and 0 < θ_3 < 1/10(4 - 7 θ_1 - 12 θ_2)),
or (4/9 < θ_1 < 116/259 and $\theta_2 = 1/60(-7 + 25\theta_1)$ and 1/20(9 - 20 θ_1) $\leq \theta_3$ < 1/10(4 - 7 θ_1 - 12 θ_2)),
or (116/259 < θ_1 < 185/413 and $\theta_2 = 1/60(-7 + 25\theta_1)$ and 1/20(9 - 20 θ_1) $\leq \theta_3$ < 1/10(4 - 7 θ_1 - 12 θ_2)),
or (185/413 $\leq \theta_1$ < 451/1003 and $\theta_2 = 1/60(-7 + 25\theta_1)$ and 1/20(9 - 20 θ_1) $\leq \theta_3$ < 1/10(4 - 7 θ_1 - 12 θ_2)),
or ($\theta_1 = 116/259$ and 2/37 < θ_2 < 1087/15540 and 1/111(8 - 111 θ_2)

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or $(7/17 < \theta_1 \leq 12/29$ and $1/24(3 - 4\theta_1) \leq \theta_2 < 1/17(8 - 17\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),

or $(7/17 < \theta_1 \leq 2049/4964$ and $1/28(5 - 8\theta_1) \leq \theta_2 < 1/238(-27 + 102\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),

or $(17/40 < \theta_1 < 73/170$ and $1/20 < \theta_2 < 1/19(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),

or $(73/170 \leq \theta_1 < 31/72$ and $1/20 < \theta_2 < 1/19(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),

or $(31/72 \leq \theta_1 < 61/140$ and $1/20 < \theta_2 < 1/19(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),

or $(5/12 < \theta_1 < 21/50$ and $1/18 < \theta_2 < 1/19(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),

or $(21/50 \leq \theta_1 < 143/340$ and $1/18 < \theta_2 < 1/19(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),

or $(143/340 < \theta_1 < 53/126$ and $1/18 < \theta_2 < 1/19(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),

or $(272/679 < \theta_1 \leq 45/112$ and $1/112(29 - 56\theta_1) < \theta_2 < 1/19(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),

or $(45/112 < \theta_1 < 13/32$ and $1/24(3 - 4\theta_1) \leq \theta_2 < 1/19(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),

or $(13/32 < \theta_1 < 139/340$ and $1/24(3 - 4\theta_1) \leq \theta_2 < 1/19(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),

or $(139/340 < \theta_1 < 37/90$ and $1/24(3 - 4\theta_1) \leq \theta_2 < 1/19(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),

or $(53/126 < \theta_1 < 11/26$ and $1/24(3 - 4\theta_1) \leq \theta_2 < 1/19(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),

or $(11/26 < \theta_1 < 39/92$ and $1/24(3 - 4\theta_1) \leq \theta_2 < 1/19(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),

or $(205/504 < \theta_1 < 139/340$ and $1/112(29 - 56\theta_1) < \theta_2 < 1/18$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),

or $(139/340 < \theta_1 < 37/90$ and $1/112(29 - 56\theta_1) < \theta_2 < 1/18$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),

or $(12/29 < \theta_1 < 127/306$ and $1/112(29 - 56\theta_1) < \theta_2 < 1/18$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),

or $(127/306 < \theta_1 < 5/12$ and $1/112(29 - 56\theta_1) < \theta_2 < 1/17(8 - 17\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),

or $(43/98 < \theta_1 < 37/84$ and $1/20 < \theta_2 < 1/35(11 - 21\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),

or $(75/172 < \theta_1 < 43/98$ and $1/20 < \theta_2 < 1/23(6 - 11\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),

or $(37/84 < \theta_1 < 97/220$ and $1/20 < \theta_2 < 1/23(6 - 11\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),

or $(5/12 < \theta_1 \leq 117/280$ and $1/112(29 - 56\theta_1) < \theta_2 < 1/17(8 - 17\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),

or $(117/280 < \theta_1 < 21/50$ and $1/20 < \theta_2 < 1/17(8 - 17\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),

or $(21/50 \leq \theta_1 < 143/340$ and $1/20 < \theta_2 < 1/17(8 - 17\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),

or $(61/140 < \theta_1 \leq 75/172$ and $1/20 < \theta_2 < 1/24(3 - 4\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),

or $(97/220 < \theta_1 < 15/34$ and $1/20 < \theta_2 < 1/24(3 - 4\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),

or $(15/34 < \theta_1 < 31/70$ and $1/20 < \theta_2 < 1/24(3 - 4\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),

or $(85/198 < \theta_1 < 73/170$ and $1/23(6 - 11\theta_1) < \theta_2 < 1/18$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),

or $(73/170 \leq \theta_1 < 31/72$ and $1/23(6 - 11\theta_1) < \theta_2 < 1/18$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),

or $(163/378 < \theta_1 \leq 19/44$ and $1/35(11 - 21\theta_1) < \theta_2 < 1/18$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),

or $(5/12 < \theta_1 < 21/50$ and $1/24(3 - 4\theta_1) \leq \theta_2 < 1/18$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),

or $(21/50 \leq \theta_1 < 143/3$

or $(139/340 < \theta_1 < 37/90$ **and** $1/18 < \theta_2 < 1/24(3 - 4\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
or $(12/29 < \theta_1 \leq 141/340$ **and** $1/18 < \theta_2 < 1/24(3 - 4\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
or $(159/364 < \theta_1 < 43/98$ **and** $1/23(6 - 11\theta_1) < \theta_2 < 1/35(11 - 21\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
or $(43/98 < \theta_1 \leq 97/220$ **and** $1/23(6 - 11\theta_1) < \theta_2 < 1/24(3 - 4\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
or $(75/172 < \theta_1 \leq 159/364$ **and** $1/23(6 - 11\theta_1) < \theta_2 < 1/24(3 - 4\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
or $(43/98 < \theta_1 \leq 37/84$ **and** $1/35(11 - 21\theta_1) < \theta_2 < 1/23(6 - 11\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
or $(97/220 < \theta_1 < 15/34$ **and** $1/35(11 - 21\theta_1) < \theta_2 < 1/23(6 - 11\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
or $(15/34 < \theta_1 < 31/70$ **and** $1/35(11 - 21\theta_1) < \theta_2 < 1/23(6 - 11\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
or $(159/364 < \theta_1 < 43/98$ **and** $1/35(11 - 21\theta_1) < \theta_2 < 1/24(3 - 4\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1))$,
or $(127/306 < \theta_1 < 5/12$ **and** $1/28(5 - 8\theta_1) \leq \theta_2 < 1/23(6 - 11\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
or $(127/306 < \theta_1 \leq 509/1224$ **and** $1/23(6 - 11\theta_1) < \theta_2 < 1/238(-27 + 102\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
or $(127/306 < \theta_1 < 509/1224$ **and** $1/238(-27 + 102\theta_1) < \theta_2 < 1/35(11 - 21\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
or $(127/306 < \theta_1 \leq 509/1224$ **and** $1/35(11 - 21\theta_1) < \theta_2 < 1/40(11 - 20\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
or $(509/1224 < \theta_1 < 5/12$ **and** $1/23(6 - 11\theta_1) < \theta_2 < 1/35(11 - 21\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
or $(509/1224 < \theta_1 < 5/12$ **and** $1/35(11 - 21\theta_1) < \theta_2 < 1/238(-27 + 102\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
or $(509/1224 < \theta_1 < 5/12$ **and** $1/238(-27 + 102\theta_1) < \theta_2 < 1/40(11 - 20\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
or $(53/126 < \theta_1 < 11/26$ **and** $1/18 < \theta_2 < 1/28(5 - 8\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
or $(11/26 < \theta_1 < 17/40$ **and** $1/18 < \theta_2 < 1/28(5 - 8\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
or $(17/40 < \theta_1 \leq 53/124$ **and** $1/18 < \theta_2 < 1/28(5 - 8\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
or $(85/198 < \theta_1 < 73/170$ **and** $1/18 < \theta_2 < 1/28(5 - 8\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
or $(73/170 \leq \theta_1 < 31/72$ **and** $1/18 < \theta_2 < 1/28(5 - 8\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
or $(31/72 \leq \theta_1 < 163/378$ **and** $1/18 < \theta_2 < 1/35(11 - 21\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
or $(53/124 < \theta_1 < 85/198$ **and** $1/18 < \theta_2 < 1/23(6 - 11\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
or $(163/378 < \theta_1 < 61/140$ **and** $1/18 < \theta_2 < 1/40(11 - 20\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
or $(61/140 < \theta_1 < 43/98$ **and** $1/18 < \theta_2 < 1/40(11 - 20\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
or $(43/98 < \theta_1 < 79/180$ **and** $1/18 < \theta_2 < 1/40(11 - 20\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2))$,
or $(31/72 \leq \theta_1 \leq 19/44$ **and** $1/23(6 - 11\theta_1) < \theta_2 < 1/28(5 - 8\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4$

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or ($19/45 < \theta_1 < 11/26$ and $1/40(11 - 20\theta_1) \leq \theta_2 < 1/16(4 - 7\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($11/26 < \theta_1 < 17/40$ and $1/40(11 - 20\theta_1) \leq \theta_2 < 1/16(4 - 7\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($17/40 < \theta_1 < 73/170$ and $1/40(11 - 20\theta_1) \leq \theta_2 < 1/16(4 - 7\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($73/170 \leq \theta_1 < 31/72$ and $1/40(11 - 20\theta_1) \leq \theta_2 < 1/16(4 - 7\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($31/72 \leq \theta_1 \leq 163/378$ and $1/40(11 - 20\theta_1) \leq \theta_2 < 1/16(4 - 7\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($692/1649 \leq \theta_1 < 21/50$ and $1/8(6 - 13\theta_1) < \theta_2 \leq 1/7(-2 + 6\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($21/50 \leq \theta_1 \leq 822/1955$ and $1/8(6 - 13\theta_1) < \theta_2 \leq 1/7(-2 + 6\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($822/1955 < \theta_1 < 143/340$ and $1/238(-27 + 102\theta_1) < \theta_2 \leq 1/7(-2 + 6\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($143/340 < \theta_1 < 53/126$ and $1/238(-27 + 102\theta_1) < \theta_2 \leq 1/7(-2 + 6\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($53/126 < \theta_1 < 11/26$ and $1/238(-27 + 102\theta_1) < \theta_2 \leq 1/7(-2 + 6\theta_1)$ and $1 - 2\theta_1 - 2\theta_2 \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($37/90 < \theta_1 < 7/17$ and $1/40(11 - 20\theta_1) \leq \theta_2 < 1/16(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($\theta_1 = 157/380$ and $\theta_2 = 13/190$ and $109/4560 \leq \theta_3 < 109/3800$),
or ($157/380 < \theta_1 < 12/29$ and $1/40(11 - 20\theta_1) \leq \theta_2 \leq 1/7(-2 + 6\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($7/17 < \theta_1 < 157/380$ and $1/40(11 - 20\theta_1) \leq \theta_2 < 1/16(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($157/380 \leq \theta_1 < 12/29$ and $1/7(-2 + 6\theta_1) < \theta_2 < 1/16(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($79/180 < \theta_1 < 15/34$ and $1/40(11 - 20\theta_1) \leq \theta_2 < 1/18$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($15/34 < \theta_1 < 31/70$ and $1/40(11 - 20\theta_1) \leq \theta_2 < 1/18$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($127/306 < \theta_1 < 5/12$ and $1/16(4 - 7\theta_1) \leq \theta_2 \leq 1/7(-2 + 6\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($127/306 < \theta_1 < 5/12$ and $1/7(-2 + 6\theta_1) < \theta_2 < -2 + 5\theta_1$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($127/306 < \theta_1 \leq 509/1224$ and $1/40(11 - 20\theta_1) \leq \theta_2 < 1/16(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($509/1224 < \theta_1 < 5/12$ and $1/40(11 - 20\theta_1) \leq \theta_2 < 1/16(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($163/378 < \theta_1 < 61/140$ and $1/40(11 - 20\theta_1) \leq \theta_2 < 1/16(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($61/140 < \theta_1 < 43/98$ and $1/40(11 - 20\theta_1) \leq \theta_2 < 1/16(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($43/98 < \theta_1 < 79/180$ and $1/40(11 - 20\theta_1) \leq \theta_2 < 1/16(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($79/180 \leq \theta_1 < 15/34$ and $1/18 < \theta_2 < 1/16(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($15/34 < \theta_1 < 31/70$ and $1/18 < \theta_2 < 1/16(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($12/29 < \theta_1 < 127/306$ and $1/16(4 - 7\theta_1) \leq \theta_2 \leq 1/7(-2 + 6\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($5/12 < \theta_1 < 692/1649$ and $1/16(4 - 7\theta_1) \leq \theta_2 \leq 1/7(-2 + 6\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($692/1649 < \theta_1 < 21/50$ and $1/16(4 - 7\theta_1) \leq \theta_2 < 1/238(-27 + 102\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($21/5$

or $(12/29 < \theta_1 < 127/306$ and $1/7(-2 + 6\theta_1) < \theta_2 < -2 + 5\theta_1$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or $(692/1649 \leq \theta_1 < 21/50$ and $1/238(-27 + 102\theta_1) < \theta_2 \leq 1/7(-2 + 6\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or $(21/50 \leq \theta_1 < 143/340$ and $1/238(-27 + 102\theta_1) < \theta_2 \leq 1/7(-2 + 6\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or $(143/340 < \theta_1 < 53/126$ and $1/238(-27 + 102\theta_1) < \theta_2 \leq 1/7(-2 + 6\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or $(53/126 < \theta_1 < 11/26$ and $1/238(-27 + 102\theta_1) < \theta_2 \leq 1/7(-2 + 6\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or $(31/70 < \theta_1 < 4/9$ and $1/40(11 - 20\theta_1) \leq \theta_2 < 1/18$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or $(31/70 < \theta_1 < 4/9$ and $1/18 < \theta_2 < 1/16(4 - 7\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or $(130/289 < \theta_1 < 9/20$ and $1/28(5 - 8\theta_1) \leq \theta_2 < 1/40(11 - 20\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or $(4/9 < \theta_1 < 116/259$ and $1/28(5 - 8\theta_1) \leq \theta_2 < 1/40(11 - 20\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or $(116/259 < \theta_1 < 185/413$ and $1/28(5 - 8\theta_1) \leq \theta_2 < 1/40(11 - 20\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or $(185/413 \leq \theta_1 < 451/1003$ and $1/28(5 - 8\theta_1) \leq \theta_2 < 1/40(11 - 20\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or $(451/1003 \leq \theta_1 < 841/1870$ and $1/204(-28 + 85\theta_1) < \theta_2 < 1/40(11 - 20\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or $(53/116 < \theta_1 \leq 16/35$ and $1/5(8 - 17\theta_1) < \theta_2 < 1/40(11 - 20\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
or $(51/112 < \theta_1 \leq 177/388$ and $1/70(32 - 63\theta_1) \leq \theta_2 < 1/40(11 - 20\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
or $(16/35 < \theta_1 < 157/343$ and $1/70(32 - 63\theta_1) \leq \theta_2 < 1/40(11 - 20\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
or $(1772/3893 < \theta_1 \leq 51/112$ and $(1 - \theta_1)/12 < \theta_2 < 1/40(11 - 20\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
or $(51/112 < \theta_1 \leq 16/35$ and $(1 - \theta_1)/12 < \theta_2 < 1/70(32 - 63\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
or $(5/11 \leq \theta_1 < 1772/3893$ and $(1 - \theta_1)/12 < \theta_2 < 1/40(11 - 20\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
or $(91/199 < \theta_1 < 157/343$ and $(1 - \theta_1)/12 < \theta_2 < 1/70(32 - 63\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
or $(16/35 < \theta_1 < 91/199$ and $(1 - \theta_1)/12 < \theta_2 < 1/5(8 - 17\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
or $(16/35 < \theta_1 \leq 91/199$ and $1/5(8 - 17\theta_1) < \theta_2 < 1/70(32 - 63\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
or $(177/388 < \theta_1 \leq 53/116$ and $1/70(32 - 63\theta_1) \leq \theta_2 < 1/40(11 - 20\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
or $(53/116 < \theta_1 < 16/35$ and $1/70(32 - 63\theta_1) \leq \theta_2 < 1/5(8 - 17\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
or $(73/162 \leq \theta_1 \leq 4244/9401$ and $1/2(1 - 2\theta_1) < \theta_2 < 1/40(11 - 20\theta_1)$ and $1/2(1 - 2\theta_1 - \theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
or $(4244/9401 < \theta_1 \leq 279/616$ and $1/2(1 - 2\theta_1) < \theta_2 < 1/40(11 - 20\theta_1)$ and $1/2(1 - 2\theta_1 - \theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
or $(279/616 < \theta_1 < 5/11$ and $1/2(1 - 2\theta_1) < \theta_2 < 1/40(11 - 20\theta_1)$ and $1/2(1 - 2\theta_1 - \theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
or $(11/24 < \theta_1 \leq 1444/3145$ and $(1 - \theta_1)/12 < \theta_2 < 1/40(11 - 20\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
or $(1444/3145 < \theta_1 < 74/161$ and $(1 - \theta_1)/12 < \theta_2 < 1/40(11 - 20\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
or $(74/161 < \theta_1 < 769/1673$ and $(1 - \theta_1)/12 < \theta_2 < 1/40(11 - 20\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
or $(769/1673 \leq \theta_1 < 23/50$ and $(1 - \theta_1)/12 < \theta_2 < 1/40(11 - 20\theta_1)$ and $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
or $(53/116 < \theta_1 \leq 16/35$

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or ($121/259 < \theta_1 \leq 15/32$ **and** $1/28(-2 + 7\theta_1) \leq \theta_2 < 1/24(3 - 4\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
or ($316/679 < \theta_1 \leq 15/32$ **and** $(1 - \theta_1)/12 < \theta_2 < 1/28(-2 + 7\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
or ($15/32 < \theta_1 \leq 23/49$ **and** $(1 - \theta_1)/12 < \theta_2 < 1/28(-2 + 7\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
or ($15/32 < \theta_1 < 23/49$ **and** $1/28(-2 + 7\theta_1) \leq \theta_2 \leq 1/4(3 - 6\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
or ($13/28 < \theta_1 < 316/679$ **and** $1/24(3 - 4\theta_1) \leq \theta_2 < 1/5(-3 + 7\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($316/679 < \theta_1 \leq 121/259$ **and** $1/24(3 - 4\theta_1) \leq \theta_2 < 1/5(-3 + 7\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($121/259 < \theta_1 \leq 15/32$ **and** $1/24(3 - 4\theta_1) \leq \theta_2 \leq 2/37$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($13/28 < \theta_1 \leq 86/185$ **and** $2/37 < \theta_2 < 1/7(-1 + 3\theta_1)$ **and** $1/2(1 - 2\theta_1 - \theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($86/185 < \theta_1 < 316/679$ **and** $((2/37 < \theta_2 \leq 1/6(-2 + 5\theta_1))$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$)),
or ($1/6(-2 + 5\theta_1) < \theta_2 < 1/7(-1 + 3\theta_1)$ **and** $1/2(1 - 2\theta_1 - \theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$))),
or ($316/679 < \theta_1 \leq 121/259$ **and** $((2/37 < \theta_2 \leq 1/6(-2 + 5\theta_1))$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($1/6(-2 + 5\theta_1) < \theta_2 < 1/7(-1 + 3\theta_1)$ **and** $1/2(1 - 2\theta_1 - \theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$))),
or ($121/259 < \theta_1 \leq 15/32$ **and** $((1/5(-3 + 7\theta_1) < \theta_2 \leq 1/6(-2 + 5\theta_1))$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($1/6(-2 + 5\theta_1) < \theta_2 < 1/7(-1 + 3\theta_1)$ **and** $1/2(1 - 2\theta_1 - \theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$))),
or ($13/28 < \theta_1 \leq 86/185$ **and** $((1/5(-3 + 7\theta_1) < \theta_2 \leq 1/6(-2 + 5\theta_1))$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($1/6(-2 + 5\theta_1) < \theta_2 \leq 2/37$ **and** $1/2(1 - 2\theta_1 - \theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$))),
or ($86/185 < \theta_1 < 316/679$ **and** $1/5(-3 + 7\theta_1) < \theta_2 \leq 2/37$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($316/679 < \theta_1 < 121/259$ **and** $1/5(-3 + 7\theta_1) < \theta_2 \leq 2/37$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($121/259 < \theta_1 \leq 15/32$ **and** $2/37 < \theta_2 < 1/5(-3 + 7\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($15/32 < \theta_1 \leq 226/481$ **and** $2/37 < \theta_2 < 1/5(-3 + 7\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($15/32 < \theta_1 < 226/481$ **and** $1/2(-6 + 13\theta_1) \leq \theta_2 \leq 2/37$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or ($15/32 < \theta_1 \leq 23/49$ **and** $1/4(3 - 6\theta_1) < \theta_2 < 1/2(-6 + 13\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1 - 2\theta_1 - \theta_2$),
or ($23/49 < \theta_1 \leq 226/481$ **and** $1/28(-2 + 7\theta_1) \leq \theta_2 < 1/2(-6 + 13\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1 - 2\theta_1 - \theta_2$),
or ($177/388 < \theta_1 \leq 58/127$ **and** $1/5(8 - 17\theta_1) < \theta_2 < 1/24(3 - 4\theta_1)$ **and** $1/2(1 - 2\theta_1 - \theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
or ($58/127 < \theta_1 \leq 53/116$ **and** $((1/5(8 - 17\theta_1) < \theta_2 \leq 1/6(-2 + 5\theta_1))$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
or ($1/6(-2 + 5\theta_1) < \theta_2 < 1/24(3 - 4\theta_1)$ **and** $1/2(1 - 2\theta_1 - \theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$))),
or ($53/116 < \theta_1 \leq 16/35$ **and** $((1/40(11 - 20\theta_1) \leq \theta_2 \leq 1/6(-2 + 5\theta_1))$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
or ($1/6(-2 + 5\theta_1) < \theta_2 < 1/24(3 - 4\theta_1)$ **and** $1/2(1 - 2\theta_1 - \theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$))),
or ($279/616 < \theta_1 < 5/11$ **and** $1/70(32 - 63\theta_1) \leq \theta_2 < 1/24(3 - 4\theta_1)$ **and** $1/2(1 - 2\theta_1 - \theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1$

or $(14/31 < \theta_1 \leq 279/616$ **and** $1/70(32 - 63\theta_1) \leq \theta_2 < 1/7(-1 + 3\theta_1)$ **and** $1/2(1 - 2\theta_1 - \theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or $(1444/3145 < \theta_1 < 74/161$ **and** $1/40(11 - 20\theta_1) \leq \theta_2 < 1/24(3 - 4\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
or $(74/161 < \theta_1 < 769/1673$ **and** $1/40(11 - 20\theta_1) \leq \theta_2 < 1/24(3 - 4\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
or $(769/1673 \leq \theta_1 < 23/50$ **and** $1/40(11 - 20\theta_1) \leq \theta_2 < 1/24(3 - 4\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
or $(23/50 \leq \theta_1 \leq 41/89$ **and** $(1 - \theta_1)/12 < \theta_2 < 1/24(3 - 4\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
or $(41/89 < \theta_1 < 87/188$ **and** $1/5(-3 + 7\theta_1) < \theta_2 < 1/24(3 - 4\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
or $(41/89 < \theta_1 \leq 87/188$ **and** $(1 - \theta_1)/12 < \theta_2 < 1/5(-3 + 7\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/4(1 - 2\theta_1)$),
or $(769/1673 \leq \theta_1 < 87/188$ **and** $((1/24(3 - 4\theta_1) \leq \theta_2 \leq 1/6(-2 + 5\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or $(1/6(-2 + 5\theta_1) < \theta_2 \leq 2/37$ **and** $1/2(1 - 2\theta_1 - \theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$))),
or $(1444/3145 < \theta_1 \leq 17/37$ **and** $((1/24(3 - 4\theta_1) \leq \theta_2 \leq 1/6(-2 + 5\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or $(1/6(-2 + 5\theta_1) < \theta_2 < 1/7(-1 + 3\theta_1)$ **and** $1/2(1 - 2\theta_1 - \theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$))),
or $(17/37 < \theta_1 < 74/161$ **and** $((1/24(3 - 4\theta_1) \leq \theta_2 \leq 1/6(-2 + 5\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or $(1/6(-2 + 5\theta_1) < \theta_2 \leq 2/37$ **and** $1/2(1 - 2\theta_1 - \theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$))),
or $(74/161 < \theta_1 < 769/1673$ **and** $((1/24(3 - 4\theta_1) \leq \theta_2 \leq 1/6(-2 + 5\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or $(1/6(-2 + 5\theta_1) < \theta_2 \leq 2/37$ **and** $1/2(1 - 2\theta_1 - \theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$))),
or $(17/37 < \theta_1 \leq 74/161$ **and** $2/37 < \theta_2 < 1/7(-1 + 3\theta_1)$ **and** $1/2(1 - 2\theta_1 - \theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or $(74/161 < \theta_1 < 769/1673$ **and** $2/37 < \theta_2 < 1/7(-1 + 3\theta_1)$ **and** $1/2(1 - 2\theta_1 - \theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or $(451/1003 \leq \theta_1 < 9/20$ **and** $1/60(-7 + 25\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or $(4/9 < \theta_1 < 116/259$ **and** $1/60(-7 + 25\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or $(116/259 < \theta_1 < 185/413$ **and** $1/60(-7 + 25\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or $(185/413 \leq \theta_1 < 451/1003$ **and** $1/60(-7 + 25\theta_1) < \theta_2 < 1/12(4 - 7\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or $(451/1003 \leq \theta_1 < 9/20$ **and** $2/37 < \theta_2 < 1/60(-7 + 25\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or $(451/1003 \leq \theta_1 < 9/20$ **and** $1/70(32 - 63\theta_1) \leq \theta_2 \leq 2/37$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or $(130/289 \leq \theta_1 < 9/20$ **and** $1/204(-28 + 85\theta_1) < \theta_2 < 1/70(32 - 63\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or $(4/9 < \theta_1 < 116/259$ **and** $1/16(4 - 7\theta_1) \leq \theta_2 \leq 1/259(130 - 259\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or $(4/9 < \theta_1 < 116/259$ **and** $1/259(130 - 259\theta_1) < \theta_2 \leq 1/27(14 - 28\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or $(4/9 < \theta_1 < 116/259$ **and** $1/27(14 - 28\theta_1) < \theta_2 < 1/60(-7 + 25\theta_1)$ **and** $1/12(4 - 7\theta_1 - 12\theta_2) \leq \theta_3 < 1/10(4 - 7\theta_1 - 12\theta_2)$),
or $(116/259 < \theta_1 < 185/413$ **and** $2/37 < \theta_2 < 1/60(-7 + 25\theta_1)$ **and** $1/12(4 - 7\theta_1 -$